



Participant Handbook

Sector

Green Jobs

Sub Sector

Other Green Jobs

Occupation

Green Logistics

Reference Id

SGJ/N4601, Version 1.0; NSQF Level 4



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Green Logistics Practices

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Shri Narendra Modi
Prime Minister of India

“

Skilling is building a better India.
If we have to move India towards
development then Skill Development
should be our mission.

”



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This participant's handbook meant for promoting green logistics practices in logistics sector and attempt to ensure the availability of all the relevant information to the existing and prospective job holders in green logistics practices. We have compiled the content with inputs from the relevant Subject Matter Experts (SMEs) and industry members to ensure it is the latest and authentic. We express our sincere gratitude to all the SMEs and industry members who have made invaluable contributions to the completion of this participant's handbook.

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This handbook will help deliver skill-based training with a hope that it will benefit all the stakeholders, such as participants, trainers, and evaluators. We have made all efforts to ensure the publication meets the current quality standards for the successful delivery of QP/NOS-based training programs. We welcome and appreciate any suggestions for future improvements to this handbook.



About this Book

This participant handbook has been designed to serve as a guide for participants who aim to obtain the required knowledge and skills to undertake various activities promoting green logistics practices at warehouse, supply chain and transportations. Its content has been aligned with the National Occupation Standard. With a qualified trainer's guidance, the participants will be equipped with the following for working efficiently in the job role:

- **Knowledge and Understanding:** The relevant operational knowledge and understanding to perform the required tasks.
- **Performance Criteria:** The essential skills through hands-on training to perform the required operations to the applicable quality standards.
- **Professional Skills:** The Ability to make appropriate operational decisions about the field of work.

After studying this handbook, job holders will be adequately skilled in carrying out their duties according to the applicable quality standards. The handbook is aligned with the following Modules detailed in the latest and approved version of Green Logistics Practices NOS (SGJ /N4601):

Module 1: Basics of Climate Change, Environment Concerns and Renewable Energy

Module 2: Introduction to Green Logistics

Module 3: Steps required for Greening the Logistics Sector

Module 4: Identifying Green Supply and Demand networks

Module 5: Health and Safety in Green Logistics.

The handbook has been divided into an appropriate number of units and sub-units based on the content of the relevant NOS. We hope it will facilitate easy and structured learning for the participants, allowing them to obtain enhanced knowledge and skills.

Symbols Used



**Key Learning
Outcomes**



Steps



Exercise



Activity



Notes



**Unit
Objectives**

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1. Basics of Climate Change, Environmental Concerns and Renewable Energy



Unit **1.1** Introduction to Sustainable Development and Sustainable Development Goals (SDG's)

Unit **1.2** Overview of Greenhouse Gas Emissions and its sources in Logistics

Unit **1.3** Introduction to renewable energy

Unit **1.4** Uses of renewable energy in logistics sector



Chapter Highlights

After reading this chapter, you will be able to: %

- 1.** Know about the concept of Sustainable Development and SDG's
- 2.** Understand about Green House Gas emissions (GHG)
- 3.** Identify the sources of GHG emissions in Logistics
- 4.** Identify the environmental impact of Logistics
- 5.** Understand the concept of Circular Economy and its relevance to Logistics Sector
- 6.** Know the key benefits of circular economy

UNIT 1.1: Introduction to Sustainable Development and SDG

Unit objectives

At the end of this unit, you will be able to:

- 1- Understand What is Sustainable Development
- 2- Know about the pillars of sustainable development
- 3- Understand about UN Sustainable Development Goals (SDG's 7 and 13)
- 4- Understanding how one goal leads to simultaneous advancements in multiple areas

Introduction to Sustainable Development

1.1.1 What is Sustainability

The name sustainability is derived from the Latin word "sustinere" i.e., to hold; sub, under. Sustain can mean "maintain", "support", or "endure". Thus, sustainability is essentially about holding up human existence by meeting the needs of the present without compromising the ability of future generations to meet their own needs. Originally, sustainability means making only such use of natural, renewable resources that people can continue to rely on their yields in the long term.

Sustainability encompasses three dimensions namely:

1. Environmental Sustainability
2. Economic Sustainability
3. Social Sustainability

Sustainability is a framework that acknowledges the interconnectedness of environmental, social, and economic systems, emphasizing the importance of balance and long-term thinking.

1.1.2 Sustainable Development

The industrial revolution is connected to the rise of the idea of sustainable development. From the second half of the 19th century, Western societies started to discover that their economic and industrial activities had a significant impact on the environment and the social balance. Several ecological and social crises took place in the world and rose awareness that a more sustainable model was needed.

In 1980, the International Union for the Conservation of Nature published a world conservation strategy that included one of the first references to sustainable development as a global priority and introduced

the term "sustainable development". The concept of sustainable development arises from a worldwide which sees the survival, progress and continued maintenance of human community.

Today's framework on sustainable development is quite strong although there is still a huge way to go. There are many actors working with different audiences in different areas of sustainability. They share the same goal – to raise awareness on this topic and to create conditions for it to grow and develop. One of the main players is the United Nations, where different teams actively work on multiple campaigns.

According to United Nations (UN), sustainable development has been defined as development that meets the needs of the present without compromising the ability of future generations to meet their own needs and calls for concerted efforts towards building an inclusive, sustainable and resilient future for people and planet.

Sustainable development is based on the three pillars of sustainability: **economic**, **environmental** and **social** sustainability. It is only achieved when there is balance or a trade-off between these three aspects:



Fig:1.1.2

The above figure shows how each dimension is dependent on the next. The economy is dependent on society, and both are dependent on the environment.

1. **Environmental Sustainability:** occurs when humanity's rate of consumption does not exceed nature's rate of replenishment and when humanity's rate of generating pollution and emitting greenhouse gases does not exceed nature's rate of restoration.
2. **Economic Sustainability:** is the ability of human communities around the world to maintain their independence and have access to the resources required to meet their needs, meaning that secure sources of livelihood are available to everyone
3. **Social Sustainability:** is the ability of a society to uphold universal human rights and meet people's basic needs, such as healthcare, education, and transportation. Healthy communities

ensure personal, labour, and cultural rights are respected and all people are protected from discrimination.

1.1.3 Roadmap from Millennium Development Goals (MDGs) to Sustainable Development Goals (SDGs)

In September 2000, leaders of 189 countries at the United Nations headquarters signed the historic Millennium Declaration, in which they committed to achieving a set of eight measurable goals that range from halving extreme poverty and hunger to promoting gender equality and reducing child mortality, by the target date of 2015.

For 15 years, the MDGs drove progress in several important areas: reducing income poverty, providing much needed access to water and sanitation, driving down child mortality and drastically improving maternal health. They also kick-started a global movement for free primary education, inspiring countries to invest in their future generations. Most significantly, the MDGs made huge strides in combatting HIV/AIDS and other treatable diseases such as malaria and tuberculosis.

Key MDG achievements

- More than 1 billion people have been lifted out of extreme poverty (since 1990)
- Child mortality dropped by more than half (since 1990)
- The number of out of school children has dropped by more than half (since 1990)
- HIV/AIDS infections fell by almost 40 percent (since 2000)

Substantial progress has been made regarding the MDGs. The world has already realized the first MDG of halving the extreme poverty rate by 2015. However, the achievements have been uneven. The MDGs were set to expire in 2015 and the discussion of a post-2015 agenda continued. The focus was then on building a sustainable world where environmental sustainability, social inclusion, and economic development were equally valued."

To support the declaration, United Nations established 8 accompanying objectives. These objectives (given in the figure 1.1.2) were set with a deadline of 2015 and became known as the Millennium Development Goals (MDGs).



Fig. 1.1.3

1.1.4 Sustainable Development Goals (SDGs)

Unlike Millennium Development Goals (MDGs), the Sustainable Development Goals (SDGs) were uniformly applicable to all countries of the world.

The Sustainable Development Goals (SDGs) were born at the United Nations Conference on Sustainable Development in Rio de Janeiro in 2012. The objective was to produce a set of universal goals that meet the urgent environmental, political and economic challenges facing our world.

The SDGs replace the Millennium Development Goals (MDGs), which started a global effort in 2000 to tackle the indignity of poverty. The United Nations defines sustainable development as the pathway to meet the needs of the present without compromising the ability of future generations to meet their own needs. It's an overarching framework that integrates economic, social, and environmental dimensions to foster prosperity for people today while safeguarding resources and opportunities for the future.

The Sustainable Development Goals (SDGs) or Global Goals are a collection of seventeen interlinked objectives designed to serve as a "shared blueprint for peace and prosperity for people and the planet, now and into the future".

The SDGs coincided with another historic agreement reached in 2015 at the COP21 Paris Climate Conference:

About Conference of Parties (COP)

The Conference of the Parties, commonly abbreviated as COP, is a key decision-making body of the United Nations Framework Convention on Climate Change (UNFCCC). It's an annual gathering where countries come together to discuss and negotiate global responses to climate change.

During these conferences, countries discuss and negotiate various measures, commitments, and agreements aimed at reducing greenhouse gas emissions, adapting to the impacts of climate change, providing financial and technological support to developing countries, and addressing other pertinent issues related to climate action.

One of the most significant COP meetings was COP21 in 2015, where the Paris Agreement was adopted. This agreement set forth ambitious goals to limit global warming and establish a framework for international cooperation on climate change mitigation and adaptation.

The SDGs have been developed through an unprecedented consultative process that brought national governments and millions of citizens from the globe together to negotiate and adopt this ambitious agenda

In July 2014, the UN General Assembly Open Working Group (OWG) proposed a document containing 17 goals to be put forward for the General Assembly's approval in September 2015. This document set the ground for the new SDGs and the global development agenda spanning from 2015-2030.



1.1.4

Goal 1: No Poverty- End poverty in all its forms everywhere

Goal 2: Zero Hunger- End hunger, achieve food security, improve nutrition, and promote sustainable agriculture.

Goal 3: Good Health and Well-being- Ensure healthy lives and promote well-being for all at all ages.

Goal 4: Quality Education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

Goal 5: Gender Equality: Achieve gender equality and empower all women and girls.

Goal 6: Clean Water and Sanitation: Ensure availability and sustainable management of water and sanitation for all.

Goal 7: Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable, and modern energy for all.

Goal 8: Decent Work and Economic Growth: Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.

Goal 9: Industry, Innovation, and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

Goal 10: Reduced Inequalities: Reduce inequality within and among countries.

Goal 11: Sustainable Cities and Communities: Make cities and human settlements inclusive, safe, resilient, and sustainable.

Goal 12: Responsible Consumption and Production: Ensure sustainable consumption and production patterns.

Goal 13: Climate Action: Take urgent action to combat climate change and its impacts.

Goal 14: Life Below Water: Conserve and sustainably use the oceans, seas, and marine resources.

Goal 15: Life on Land: Protect, restore, and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, halt and reverse land degradation, and halt biodiversity loss.

Goal 16: Peace, Justice, and Strong Institutions: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all, and build effective, accountable, and inclusive institutions at all levels.

Goal 17: Partnerships for the Goals: Strengthen the means of implementation and revitalize the global partnership for sustainable development.

Each goal has specific targets and indicators, and achieving these goals requires collaborative efforts, innovative solutions, and commitment from governments, businesses, civil society, and individuals worldwide.

The 17 Sustainable Development Goals (SDG's) are structured around 5 pillars of the 2030 Agenda: People, Planet, Prosperity, Peace and Partnership. These 5 Ps highlight how the SDGs are an intertwined framework instead of a group of solo goals.



Fig: 1.1.4

1.1.5 Interlinkage between SDG 7 and SDG 13

Sustainable Development Goal 7 (SDG 7) and Sustainable Development Goal 13 (SDG 13) are interconnected and complementary in addressing global challenges related to sustainable development, particularly regarding energy and climate action.

SDG 7 focuses on ensuring access to affordable, reliable, sustainable, and modern energy for all. This goal emphasizes the significance of clean energy, energy efficiency, and renewable energy sources to meet growing energy demands while mitigating environmental impact.

On the other hand, SDG 13 aims to take urgent action to combat climate change and its impacts. It highlights the need to strengthen resilience and adaptive capacity to climate-related hazards and natural disasters.

SDG 7 ensure Access to Affordable, Reliable, Sustainable, and Modern Energy for all. This goal emphasizes on:

- **Energy Access:** Providing universal access to modern energy services is essential for socio-economic development.
- **Renewable Energy:** Encouraging the use of renewable energy sources to reduce reliance on fossil fuels.

- **Energy Efficiency:** Promoting energy efficiency practices across sectors to minimize waste and optimize energy usage.

While SDG 13 take Urgent Action to Combat Climate Change and Its impacts. This goal focuses on:

- **Mitigating Climate Change:** Reducing greenhouse gas emissions to limit global temperature rise.
- **Adaptation and Resilience:** Strengthening resilience and adaptive capacities to climate-related hazards.

SDG 7 (Affordable and Clean Energy) and SDG 13 (Climate Action) are deeply interconnected and mutually reinforcing in several ways:

- **Renewable Energy and Climate Mitigation:** SDG 7 promotes the use of renewable energy sources such as solar, wind, hydro, and geothermal power. Transitioning from fossil fuels to renewable energy significantly reduces greenhouse gas emissions, aligning with the objectives of SDG 13 by mitigating climate change.
- **Energy Efficiency and Emission Reduction:** Improving energy efficiency, a key aspect of SDG 7, directly contributes to reducing emissions. Using energy more efficiently in industries, transportation, and buildings helps mitigate climate change impacts outlined in SDG 13.
- **Climate Resilience through Energy Access:** SDG 7 aims to provide universal access to affordable, reliable, and modern energy services. Access to sustainable energy sources is crucial for building resilience to climate change impacts, particularly in vulnerable communities, supporting the aims of both goals simultaneously.
- **Policy Integration:** Coordinated policies are essential for achieving both SDG 7 and SDG 13. Integrated approaches that promote renewable energy deployment, energy efficiency, and climate-resilient infrastructure contribute to achieving targets across both goals.
- **Global Collaboration:** Both goals require international cooperation. Collaborative efforts in research, technology transfer, and capacity-building initiatives are crucial for advancing renewable energy technologies, implementing sustainable practices, and mitigating climate change globally.
- **Co-benefits and Synergies:** Actions taken to achieve one goal often have positive effects on the other. For instance, investing in renewable energy not only reduces emissions but also improves energy access and creates employment opportunities, aligning with both SDGs.

Ultimately, the success of one goal significantly contributes to the achievement of the other. Addressing the global challenges of energy access and climate change requires a comprehensive and interconnected approach that considers the synergy between SDG 7 and SDG 13.

UNIT 1.2 Overview of Greenhouse Gas Emissions – its sources in Logistics

Unit objectives



After reading this unit, you will be able to understand the following points:

- 1- What are Greenhouse Gases (GHG)
- 2- What are Greenhouse Gas emissions
- 3- The sources of Greenhouse Gases in Logistics
- 4- Examples of sources of GHG emissions in logistics
- 5- Environmental impacts of logistics

1.2.1 What are Greenhouse Gases

Greenhouse gases (also known as GHGs) are gases such as Carbon dioxide (CO₂), Methane (CH₄), Water vapour (H₂O), Nitrous oxide (N₂O), Ozone (O₃) in the earth's atmosphere that can trap heat. These gases create a natural "blanket" around the Earth, allowing sunlight to enter and warm the surface. When the Earth's surface releases this heat energy, some of it is absorbed and re-emitted by greenhouse gases, trapping heat and preventing it from escaping into space. This process helps maintain a stable temperature, enabling life to thrive.

Nonetheless, energy demand has risen dramatically in recent decades as global populations have grown and economies have developed. The increased combustion of fossil fuels to meet this demand has seen global greenhouse gas (GHG) emissions soar to more than 50 billion metric tons of carbon dioxide equivalent (GtCO₂e) annually. GHGs - which can stay in the atmosphere for centuries - trap heat within Earth's atmosphere by absorbing the sun's energy and preventing it from escaping into space.

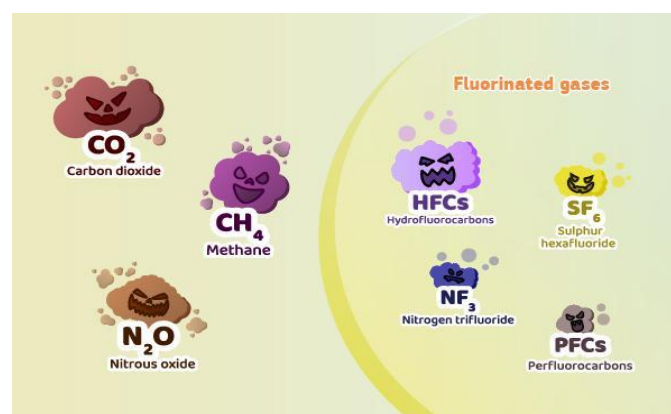


Fig: 1.2.1

Image Courtesy: [European Commission](https://climate.ec.europa.eu/climate-change/causes-climate-change_en#greenhouse-gases) (https://climate.ec.europa.eu/climate-change/causes-climate-change_en#greenhouse-gases)

Statista, Published by [Ian Tiseo](#), Aug 31, 2023

What is Greenhouse effect:

In a greenhouse, sunlight enters, and heat is retained. The surface of the Earth absorbs just under half of the sun's energy, while the atmosphere absorbs 23 per cent, and the rest is reflected back into space. Natural processes ensure that the amount of incoming and outgoing energy is equal, keeping the planet's temperature stable.

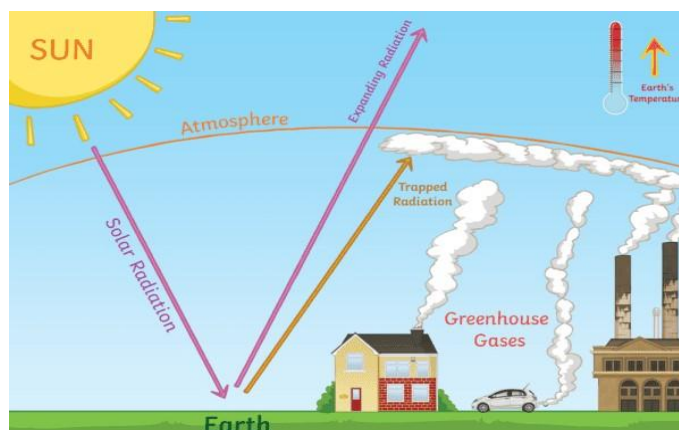


Image Courtesy: [What are Greenhouse Gases? - Answered - Twinkl Teaching Wiki](#)

However, human activity has resulted in the increased emission of so-called greenhouse gases (GHGs) which, unlike other atmospheric gases such as oxygen and nitrogen, becomes trapped in the atmosphere, unable to escape the planet. This energy returns to the surface, where it is reabsorbed. Because more energy enters than exits the planet, surface temperatures increase until a new balance is achieved.

1.2.2 The Greenhouse Gas Emissions

The release of greenhouse gases associated with human activities and climate change is referred to as greenhouse gas emissions, or climate pollution

Since the start of the Industrial Revolution and the advent of coal-powered steam engines, human activities have supersized the volume of greenhouse gases emitted into the atmosphere. It is estimated that between 1750 and 2019, atmospheric concentrations of carbon dioxide increased by 47 percent, methane by 156 percent, and nitrous oxide by 23 percent. In the late 1920s, the man-made fluorinated gases like chlorofluorocarbons to the mix. Of all the human-driven emissions of carbon dioxide, approximately half were generated in the last 30 years alone.

How Greenhouse Gases Warm Our Planet

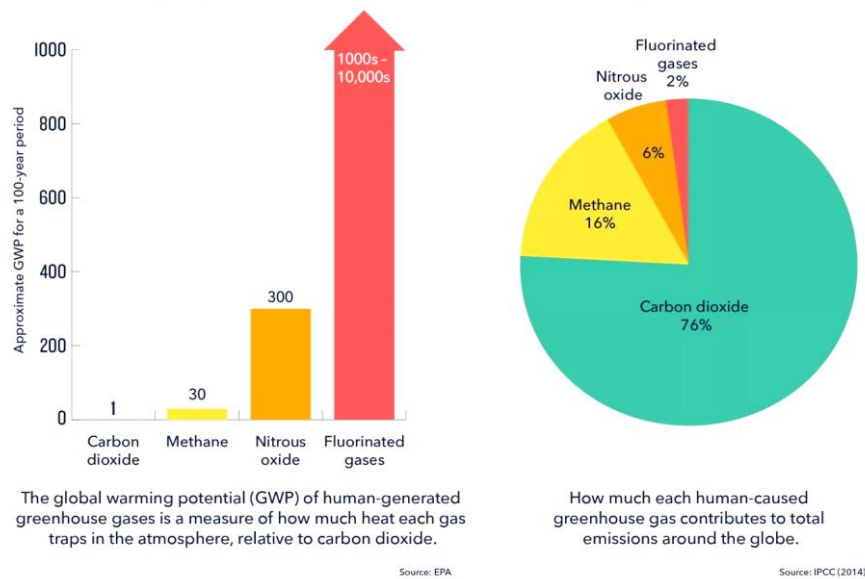


Fig: 1.2.2

[United Nations, NRDC.org story](https://www.nrdc.org/stories/greenhouse-effect-101#gases) (<https://www.nrdc.org/stories/greenhouse-effect-101#gases>)

What is Global Warming?

Global warming is the long-term heating of Earth's climate system observed since the pre-industrial period (since 1850) due to the anthropogenic activities, primarily fossil fuel burning, which increases heat-trapping greenhouse gas (GHGs) levels in Earth's atmosphere. The term is interchangeably used with climate change, though the latter refers to both human- and naturally produced warming and the effects it has on our planet. It is most commonly measured as the average increase in Earth's global surface temperature.

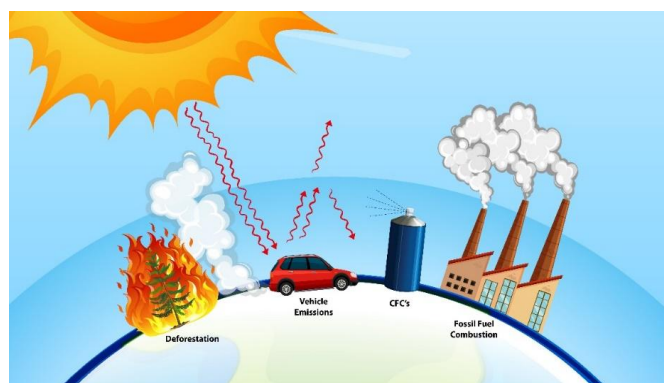


Image Courtesy: Designed by Freepik (<https://www.freepik.com/vectors/poster>)>Poster vector created by brgfx - www.freepik.com)

How Greenhouse Gas emissions causes Climate Change

The main driver of climate change is the **greenhouse effect**. Some gases in the Earth's atmosphere act a bit like the glass in a greenhouse, trapping the sun's heat and stopping it from leaking back into space and causing global warming.

Many of these greenhouse gases occur naturally, but human activities are increasing the concentrations of some of them in the atmosphere, these are:

- carbon dioxide (CO₂)
- methane (CH₄)
- nitrous oxide (N₂O)
- fluorinated gases (Fluorocarbons)

CO₂ produced by human activities is the largest contributor to global warming. By 2020, its concentration in the atmosphere had risen to 48% above its pre-industrial level (before 1750).

Other greenhouse gases are emitted by human activities in smaller quantities. Methane is a more powerful greenhouse gas than CO₂, but has a shorter atmospheric lifetime. Nitrous oxide, like CO₂, is a long-lived greenhouse gas that accumulates in the atmosphere over decades to centuries. Non-greenhouse gas pollutants, including aerosols like soot, have different warming and cooling effects and are also associated with other issues such as poor air quality. Natural causes, such as changes in solar radiation or volcanic activity are estimated to have contributed less than plus or minus 0.1°C to total warming between 1890 and 2010.

Causes for rising emissions

While CO₂ is the most talked about GHG, annual emissions of methane and often-overlooked nitrous oxide have also reached record highs. These two GHGs have a greater global warming potential than CO₂, and the growth in their emissions is of great concern to climate scientists.



Fig: 1.2.2

by [Ian Tiseo](#), Aug 31, 2023 (Greenhouse gas emissions worldwide - Statistics & Facts | Statista), [European Commission](#)

- **Burning coal, oil and gas:** produces carbon dioxide and nitrous oxide.
- **Cutting down forests (deforestation):** Trees help to regulate the climate by absorbing CO₂ from the atmosphere. When they are cut down, that beneficial effect is lost and the carbon stored in the trees is released into the atmosphere, adding to the greenhouse effect.
- **Increasing livestock farming:** Cows and sheep produce large amounts of methane when they digest their food.
- **Fertilisers containing nitrogen:** produce nitrous oxide emissions.
- **Fluorinated gases:** emitted from equipment and products that use these gases. Such emissions have a very strong warming effect, up to 23 000 times greater than CO₂. Emitted from a variety of manufacturing and industrial processes, fluorinated gases are man-made. There are four main categories: hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulfur hexafluoride (SF₆), and nitrogen trifluoride (NF₃).

As every tonne of CO₂ emitted contributes to global warming, all emissions reductions contribute to slowing it down. In order to stop global warming completely, CO₂ emissions have to reach net zero worldwide. In addition, reducing emissions of other greenhouse gases, such as methane, can also have a powerful effect on slowing global warming – especially in the short term.

Indian Scenario on increasing GHG emissions in Logistics Sector

India is among the fastest growing economies globally and will soon become the world's most populous country. While road transport expansion and improvement typically serve as a catalyst for socio-economic development, as has happened in many countries, it has unleashed several negative environmental problems in India, namely, burgeoning emissions of CO₂ as well as air pollutants such as nitrogen oxides (NO_x) and fine particulate matter (PM_{2.5}).

Every year, India moves around 4.6 billion tonnes of freight, which creates a need for transport that covers 2.2 trillion tonne-kilometres, emitting 2.3 gigatonnes of carbon dioxide (CO₂). Experts predict that this demand will go up in the future, leading to a surge in road freight transportation to 9.6 trillion tonne-km by 2050. Today, India is third in the world for CO₂ emissions, preceded by China and the United States.

1.2.3 Sources of Greenhouse gas emissions in Logistics Sector are as follows:

In the logistics sector, greenhouse gas (GHG) emissions primarily arise from various activities associated with transportation, warehousing, and supply chain management. Some significant sources include:

→ **Transportation Fuel:**

- **Road Transport:** Emissions from trucks, vans, and cars used for transporting goods.
- **Maritime Transport:** Shipping vessels using fossil fuels.
- **Aviation:** Emissions from air freight and cargo planes.

→ **Warehousing and Distribution:**

- **Energy Use:** Heating, cooling, and lighting of warehouses.
- **Material Handling Equipment:** Emissions from forklifts, cranes, and other machinery.

→ **Supply Chain Operations:**

- **Inventory Management:** Emissions linked to storing excessive inventory or inefficient inventory practices.
- **Packaging:** Emissions related to packaging materials and waste disposal.

→ **Last-Mile Delivery:**

- **Urban Delivery Services:** Emissions from final delivery to consumers in urban areas.
- **Delivery Vehicles:** Emissions from vans and trucks for last-mile delivery.

→ **Reverse Logistics:**

- **Product Returns:** Emissions from managing and transporting returned goods.

1.2.4 Key Environmental impacts of the Logistics Sector include:

Logistics is one of the most important basic industries for any economic growth as it is the management of the flow of products from the place of their origin to the place of their consumption, thus the industry also involves the integration of material handling, warehousing, packaging, transportation, shipping security, inventory management, supply chain management, procurement, and customs service.

However, the logistics industry has been a major contributor to climate change, accounting for 9% of global greenhouse gas emissions. This is due to the industry's heavy reliance on fossil fuels through various mechanisms, primarily due to its reliance on transportation, infrastructure, and energy consumption around the world.

The logistics sector, while crucial for global trade and commerce, also has significant environmental impacts.

→ **Greenhouse Gas Emissions:**

- **CO₂ Emissions:** Arise from fossil fuel combustion in transportation (trucks, ships, planes), warehouses, and distribution centers.
- **Methane and Nitrous Oxide:** Emissions from transportation, especially in the case of inefficient vehicles or outdated technologies.

→ **Air Pollution:**

- **Particulate Matter (PM):** Emitted by diesel engines in trucks, leading to respiratory issues and air quality degradation.
- **Nitrogen Oxides (NOx) and Sulfur Oxides (SOx):** Emitted from burning fossil fuels, contributing to air pollution and acid rain.

→ **Resource Consumption:**

- **Energy Use:** High energy consumption in warehouses, distribution centers, and transportation.
- **Water Use:** Large-scale logistics operations can have significant water demands for various purposes.

→ **Waste Generation:**

- **Packaging Waste:** Large volumes of packaging materials lead to increased waste.
- **Obsolete Equipment:** Disposal of old or outdated vehicles and machinery generates waste.

→ **Land Use and Habitat Disruption:**

- **Infrastructure Development:** Construction of transport infrastructure and logistics facilities can disrupt natural habitats and ecosystems.
- **Land Degradation:** Increased traffic and development can contribute to soil erosion and habitat fragmentation.

→ **Noise Pollution:**

- **Transportation and Warehousing Activities:** Noise from transportation vehicles, loading/unloading, and operations in warehouses.



Fig: 1.2.4

Probable areas to adopt sustainable practices in Logistics Sector

In the logistics sector, several areas offer opportunities to adopt sustainable practices, reducing environmental impact and promoting efficiency. Some probable areas include:

Transportation: Implementing eco-friendly vehicles, such as electric or hybrid trucks, and optimizing routes to reduce fuel consumption and emissions. Also, exploring alternative fuel sources, like biofuels or hydrogen-powered vehicles, can significantly reduce carbon footprints.

Packaging: Adopting sustainable packaging materials and optimizing packaging designs to reduce waste. Using recyclable, biodegradable, or reusable packaging can significantly minimize environmental impact.

Warehouse Operations: Implementing energy-efficient practices within warehouses and distribution centers, such as using LED lighting, optimizing heating and cooling systems, and employing renewable energy sources like solar panels.

Reverse Logistics: Establishing efficient systems for product returns, repairs, and recycling to reduce waste sent to landfills. Refurbishing or recycling returned products contributes to a circular economy and minimizes waste generation.

Technology and Data Utilization: Utilizing technology like IoT (Internet of Things), AI (Artificial Intelligence), and data analytics to optimize logistics operations. This includes route optimization, inventory management, and energy consumption tracking to enhance efficiency and reduce environmental impact.

Collaborative Logistics: Exploring collaborative logistics models, such as sharing transportation or warehouse facilities, to optimize asset utilization and reduce overall environmental impact through resource sharing.

Supply Chain Transparency and Sustainable Sourcing: Ensuring transparency throughout the supply chain and collaborating with suppliers who adhere to sustainable practices. Ethical sourcing and environmentally conscious suppliers help minimize environmental and social impacts.

Employee Awareness and Training: Educating and training employees on sustainable practices and environmental responsibility within the logistics sector can significantly contribute to the adoption and success of sustainable initiatives.

By focusing on these areas, the logistics sector can make significant strides toward sustainability, reducing its environmental footprint and contributing to a more environmentally friendly future.

Unit 1.3 Introduction to Renewable Energy & its Sources

Unit objectives



After reading this unit, you will be able to understand the following points:

1. Growing significance of renewable energy in today's world
2. Meaning of renewable energy and its sources
3. Difference between conventional and renewable energy sources.
4. Importance & benefits of renewable energy

1.3.1 Growing significance of renewable energy in today's world

It's hard to imagine the way we live and work without energy. But our energy system continues to rely on the burning of fossil fuels, emitting huge amounts of greenhouse gases that are radically warming and destabilising our climate. We're seeing more extreme weather events. Previously reliable rains are failing and sea levels are rising.

The sources of electricity production such as coal, oil, and natural gas have contributed to one-third of global greenhouse gas emissions. It is essential to raise the standard of living by providing cleaner and more reliable electricity

Promisingly, the world is responding - from the large-scale deployment of renewables like wind and solar and exciting innovations in battery storage and grid flexibility to the emergence of new business models that are fair to consumers and communities.

According to the International Energy Agency (IEA), world electricity demand will have increased by 70% by 2040 - its share of final energy use rising from 18 to 24% during the same period – driven mainly by the emerging economies of India, China, Africa, the Middle East and South-East Asia.

Clean energy development is vital for combating climate change and limiting its most devastating effects. 2022 was the fifth warmest year since records have been kept and the eighth consecutive year in which global temperatures have increased by at least 1 degree Celsius compared to pre-industrial levels.

https://www.acciona.com/renewable-energy/?_adin=02021864894

Note:

The global surface temperature rose to roughly 1.09°C higher in the period between 2011 and 2020 compared to 1850 and 1900 and that projected global emission reductions by 2030, if countries follow their targets agreed upon prior to COP26.

The 1.5° Celsius target is the global climate target that aims to limit warming to said level by 2100, in order to prevent the planet from slipping into further climate crises.

As per World Meteorological Organisation (WMO), the annual mean global surface temperature between 2023 and 2027 will be 1.1-1.8 degree Celsius higher than the baseline temperature of 1850-1900 or pre-industrial levels. In 2022, it was 1.15 degrees above the baseline, and by 2027, the average will exceed 1.5 degrees, a critical point beyond which there may be no return.

In 2010, at the Cancun COP16, countries agreed to limit the global average warming to below 2° Celsius. In 2015, the parties to the Paris Agreement pledged to limit the average temperature rise to below 2°

https://www.chathamhouse.org/2023/03/what-ipcc-report-means-global-action?qclid=Cj0KCQiAm4WsBhCiARIsAEJIEzV2rrEqYG0yqEEoQPTurtxG_AOeSiau-W5mvSc6-T6IJNNR-3MKkaAtQ4EALw_wcB

<https://www.thehindu.com/sci-tech/energy-and-environment/explained-why-is-the-15-degree-celsius-target-critical/article66905617.ece>

The growing significance of renewable energy in today's world stems from its multifaceted impact across several critical areas:

→ **Climate Change Mitigation:**

- **Emission Reduction:** Renewable energy serves as a key tool in combating climate change by significantly reducing greenhouse gas emissions compared to fossil fuel-based energy sources.
- **Sustainability:** Its ability to generate power without depleting finite resources aligns with global efforts to transition to cleaner energy systems and meet emission reduction goals outlined in international agreements.

→ **Energy Security and Independence:**

- **Diversification of Energy Sources:** Embracing renewable energy diversifies energy sources, reducing dependence on volatile global markets for fossil fuels and enhancing energy security.
- **Domestic Resources:** Harnessing renewable resources domestically minimizes reliance on imported energy, fostering greater energy independence for nations.

→ **Economic Advantages:**

- **Cost Competitiveness:** Advances in renewable technology have driven down costs, making solar, wind, and other renewables increasingly competitive with conventional energy sources.
- **Job Creation and Industry Growth:** The renewable energy sector generates employment opportunities and stimulates economic growth, fostering innovation and investment in clean energy technologies.

→ **Technological Progress and Innovation:**

- **Advancements:** Continuous technological advancements enhance efficiency, storage capabilities, and scalability of renewable energy systems.
- **Research and Development:** Increased investment in research and development drives innovation, making renewables more accessible, efficient, and adaptable.

→ **Policy and Societal Shifts:**

- **Governmental Support:** Policies promoting renewable energy adoption through incentives, targets, and subsidies accelerate its integration into national energy systems.
- **Public Awareness:** Growing environmental consciousness among populations fosters demand for cleaner, sustainable energy sources, encouraging the adoption of renewables.

→ **Global Energy Transition:**

- **Shifting Energy Mix:** A global transition towards renewables is underway, evident in rising investment, expanding capacity, and decreasing reliance on fossil fuels in energy portfolios.
- **Leadership and Commitment:** Many nations, industries, and organizations are committing to ambitious renewable energy targets, driving the transition toward a more sustainable energy future.

1.3.2 Meaning of renewable energy and its sources

As per Department of Energy (DOE), Renewable energy is energy produced from sources like the sun and wind that are naturally replenished and do not run out. Renewable energy can be used for electricity generation, space and water heating and cooling, and transportation.

Renewable energy represents a crucial pathway towards a sustainable and low-carbon energy future. Its utilization not only addresses energy needs but also contributes significantly to mitigating climate change, fostering economic growth, enhancing energy security, and promoting environmental sustainability on a global scale.

Few common sources of renewable energy are as under:

<https://www.vedantu.com/physics/renewable-energy>, <https://www.nsenergybusiness.com/features/types-of-renewable-energy/>

Solar energy

Solar energy involves converting energy from the sun into thermal or electrical energy using one of the cleanest and most abundant renewable energy sources. The radiant light and heat energy are harnessed by solar collectors. The sun's energy can be captured to generate electricity or heat through a system of panels or mirrors. Solar or photovoltaic cells convert sunlight directly into electricity while solar thermal collectors use heat-absorbing panels and a series of attached circulation tubes to heat water or buildings.

Wind energy

Wind energy is one of the fastest-growing renewable energy sources. The process involves using wind to produce electricity through the use of the kinetic energy created by air in motion. This is transformed into electrical energy using wind turbines or wind energy conversion systems. The technology can be deployed either onshore – which is by far the largest market – or offshore, using either fixed-bottom turbines anchored to the sea floor or, on a much smaller scale, floating structures that can be based in deeper waters.

Hydro energy

Hydroelectricity is produced by harnessing the gravitational force of flowing water. The technology is broadly classified into four categories – conventional (dams), pumped-storage, run-of-the-river and offshore marine (tidal). Compared to fossil fuel-powered energy plants, hydropower plants emit fewer greenhouse gases

Biomass energy

A collective term for all plant and animal material, biomass can take many forms – from plants and wood, to animal and agricultural waste. In power generation it is commonly used as wood pellets that are harvested from forests and burned to release energy. It utilizes organic matter such as wood, agricultural residues, and waste to produce heat or electricity through combustion, fermentation, or conversion to biofuels.

Tidal energy

Tidal energy is generated from converting energy from the force tides into power and its production is considered more predictable compared to wind energy and solar power. Although the world's first large-scale plant of its type became operational in 1966, tidal power is still not widely used

Geothermal energy

Geothermal energy harnesses natural heat energy generated beneath the earth's crust, with heat pumps extracting steam or hot water to surface level. For the purpose of processing the raw energy, there are three types of geothermal power stations that currently exist – dry steam plants, flash steam plants and binary cycle plants.

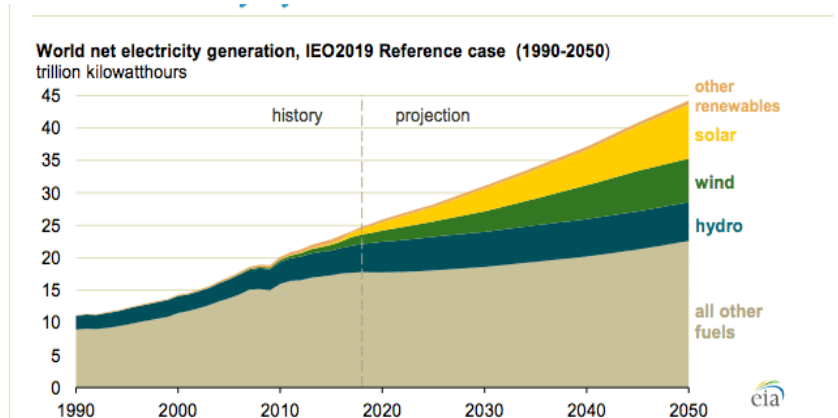


Fig. 2.1.2

Image courtesy: [Energy Information Administration \(EIA\)](https://www.eia.gov/)

Renewable energy will account for half of global power generation by 2050

1.3.3 Difference between conventional and renewable energy sources

Energy is a key component for most of the consumption activities and production activities and hence plays a vital role in a country's economic growth and development, without the energy sector, no industry would exist. It is essential for industries and is now widely utilised in agriculture and related fields such as the manufacture and delivery of fertilisers, insecticides, and farm equipment. It is also required in homes for cooking, lighting, and warmth. Based on the sources from which energy is derived, it can be classified into two categories; viz., conventional energy and renewable Energy.

What is conventional Energy?

The energy derived from traditional sources that have been widely used for a long time. This includes non-renewable resources like coal, oil, natural gas, and nuclear power. These sources are considered conventional due to their widespread use and historical dominance in powering various industries, transportation, and households. They are typically mined or extracted from the earth and processed to generate electricity or produce heat through methods like burning or nuclear reactions. However, these sources are finite and contribute significantly to environmental issues such as pollution and climate change due to their high carbon emissions and other pollutants when used for energy generation.

Renewable energy

Renewable or clean energy is a type of energy that comes from natural sources or processes that are constantly replenished. Natural resources are sunlight, wind, or tides, etc. It is expected that renewable energy will become the main solution addressing climate change. For this reason, most countries are heavily investing in different types of renewable energy, and they try to substitute non-renewable with the renewable energy as fast as economically feasible.

Key difference between conventional energy and renewable energy

Basis of Difference	Conventional Energy Sources	Non-Conventional Energy Sources (Renewable Energy Sources)
Definition	The sources of energy that have been used all around the world for a long time are called conventional energy sources.	Those energy sources whose evolution has been done the recent years are known as non-conventional energy sources.
Existence	Conventional energy sources exist for many years.	The non-conventional energy sources are relatively new sources of energy.
Renewability	Conventional sources of energy are generally non-renewable sources.	Non-conventional sources are renewable sources of energy.
Availability	Conventional sources of energy available in limited quantity in nature.	Since non-conventional sources are renewable, hence they available in abundance in nature.
Formation	Conventional energy sources require many years (or long period of time) for formation.	Non-conventional energy sources can be formed in short period of time.
Exhaustible	Conventional energy sources shall exhaust one day.	Non-conventional energy sources shall not exhaust because they are renewable.
Examples	The common examples of conventional sources of energy are coal, petroleum, natural gas, firewood, etc.	The examples of non-conventional energy sources are sun, wind, tides, water, biofuels, etc.
Safety concerns	There are many safety concerns surrounding the conventional sources such as fire in coal mines, explosion on oil platforms, burst of gas cylinders, etc.	With non-conventional energy sources, no such safety concerns associated.
Energy density	Conventional sources of energy have high energy density.	The energy density of non-conventional energy sources is less.
Obtained from	Conventional sources are generally obtained from the earth crust.	Non-conventional energy sources are present in the environment such sun, wind, water, biomass, etc.
Environmental impact	Most of the conventional energy sources pollute the environment and cause global warming.	Non-conventional energy sources are environment friendly and do not cause pollution.
Cost	Conventional energy sources are expensive.	Non-conventional are relatively less expensive.
Applications	Conventional energy sources are mainly used at large scale in commercial and industrial applications such in thermal power plants, fuel in vehicles, etc.	Non-conventional sources are used at small scale in domestic and specific commercial and industrial applications such as solar power for houses, offices, etc.

<https://www.tutorialspoint.com/difference-between-conventional-and-non-conventional-sources-of-energy>

1.3.4 Key Benefits of renewable energy

Like any human activity, all energy sources have an impact on our environment. Renewable energy is no exception, and each source has its own trade-offs. However, the advantages over the devastating impacts of fossil fuels are undeniable: from the reduction of water and land use, less air and water pollution, less wildlife and habitat loss, to no or lower greenhouse gas emissions.

Renewable energy emits no or low air pollutants

Worldwide increases in fossil fuel-based road transport, industrial activity, and power generation (as well as the open burning of waste in many cities) contribute to elevated levels of air pollution.

Renewable energy does not impact air quality during operations. Replacing the current fossil fuel-based energy system to a renewables-based system is the most urgent and efficient way to tackle air pollution.

Renewable energy prices are more stable and secure.

Evolving energy markets and geopolitical uncertainty have moved energy security and energy infrastructure resilience to the forefront of many national energy strategies. Since renewable energy is produced locally and reduces the need for energy imports, it is less affected by geopolitical crisis, price spikes, or sudden disruptions in the supply chain, thereby enhancing regional energy security

Renewable energy comes at lower costs

In most parts of the world, renewable energy technologies, and particularly wind and solar energy, remain more cost-effective than fossil fuels.

Renewable energy creates jobs and can address gender inequality.

Deployment of renewables creates employment opportunities and contributes to growth in gross domestic product, largely through the expansion of renewable energy manufacturing and deployment. The energy transition is projected to result in a global net gain in employment, with more jobs created by ramping up renewable capacity than lost by phasing out fossil energy. Providing further hope is that an estimated 70% of jobs in the oil and gas industry overlap with the skills needed for the energy transition.

Renewables can also play a key role in reducing gender inequality, for instance by creating jobs and entrepreneurial opportunities for women.

India's position in renewable energy

India stands 4th globally in Renewable Energy Installed Capacity (including Large Hydro), 4th in Wind Power capacity & 4th in Solar Power capacity (as per REN21 Renewables 2022 Global Status Report). India has set an enhanced target at the COP26 of 500 GW of non-fossil fuel-based energy by 2030. This has been a key pledge under the Panchamrit. This is the world's largest expansion plan in renewable energy.

India's installed non-fossil fuel capacity has increased 396% in the last 8.5 years and stands at more than 176.49 GW (including large Hydro and nuclear), about 43% of the country's total capacity (as of July 2023). India saw the highest year-on-year growth in renewable energy additions of 9.83% in 2022. The installed solar energy capacity has increased by 30 times in the last 9 years and stands at 70.10

<https://www.energy.gov/eere/renewable-energy>

GW as of July 2023. The installed Renewable energy capacity (including large hydro) has increased by around 128% since 2014.

Industry Scenario

Renewable energy sources have a combined installed capacity of 150+ GW. As of July 2023, Renewable energy sources, including large hydropower, have a combined installed capacity of 176.49 GW.

The following is the installed capacity for Renewable till 2023:

- Wind power: 44.5 GW
- Solar Power: 72.3 GW
- Biomass/Co-generation: 10.2 GW
- Small Hydro Power: 4.98 GW
- Waste To Energy: 0.57 GW
- Large Hydro: 46.88 GW

Unit 1.4 Uses of Renewable Energy in Logistics Sector

Unit objectives



After reading this unit, you will be able to understand the following points:

1. Overview of Logistics Sector and Energy needs
2. Current energy landscape in logistics sector
3. Various renewable energy sources suitable for logistics
4. Concept of circular economy
5. Case study on circular economy in logistics sector
6. Example of calculating the electric load of the workplace and identifying solar plant capacity to be installed

1.4.1 Overview of logistics sector in energy consumption

Logistics refers to the overall process of managing how resources are acquired, stored, and transported to their final destination. Logistics management involves identifying prospective distributors and suppliers and determining their effectiveness and accessibility.

Since India has made the commitment to achieve net zero emissions by 2070. The government's support for the idea and business initiatives centered on sustainable practices has seen a noticeable improvement in recent years. It is important to recognize that transportation is one of the major contributors to the overall pollution in the nation at a time when both small and large businesses are looking for ways to reduce their overall emissions by switching to alternative fuels.

the logistics sector is responsible for more than 11% of the Nation's total air pollution. According to a different study, the contribution might be much higher.

The logistics sector, often referred to as the backbone of any economy, plays a pivotal role in facilitating the movement of goods and connecting producers with consumers. In India, this sector contributes an impressive 14.4% to the nation's GDP, making it a significant player in the country's overall economic success. However, recent reports and statistics shed light on the challenges that the Indian logistics sector faces, particularly in terms of sustainability and high costs.

The World Bank's report highlights that logistics costs in India are among the highest globally, accounting for nearly 14% of the country's GDP, significantly surpassing the global average of around 8%. These exorbitant costs not only hinder trade and competitiveness but also burden the economy as a whole.

Therefore, it is imperative to note that only sustainable practices can control a business's contribution to pollutant emissions and carbon emissions, regardless of the type and scale of the enterprise.

1.4.2 Current energy landscape in logistics sector

The global energy logistics market was valued at \$351.2 billion in 2021, and is projected to reach \$1,383.7 billion by 2031, growing at a CAGR of 14.7% from 2022 to 2031. Logistics is widely known as the process of coordinating and moving resources, such as equipment, food, liquids, inventory, materials, and people, from one location to the storage of the desired destination. It is the management of the flow of goods from one point of origin to the point of consumption, to meet the requirement of customers. Energy logistics includes transportation and distribution of energy products within the energy industry by efficiently utilizing labour, infrastructure, and equipment. Products, such as freight shipping, parcel, and document shipping, come under the services provided in the energy logistics market.

The energy in logistics services make an important contribution to the market integration of renewables and, at the same time, improve customer profits.

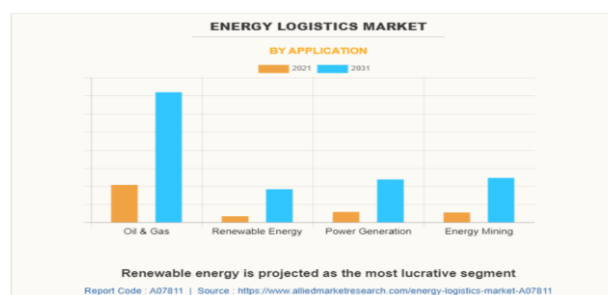


Fig. 1.4.2

Image courtesy: [allied market research](https://www.alliedmarketresearch.com)

<https://www.tvsscs.com/the-evolving-landscape-of-logistics-services-in-india-trends-and-outlook/>,
<https://www.alliedmarketresearch.com/energy-logistics-market-A07811>

Currently, increasing Internet of things (IoT) penetration in the energy logistics sector enables freight companies to directly access the company network via the internet. In addition, some logistics companies worldwide are increasing their spending and using technologically advanced systems for logistics enhancement, which are further expected to propel the growth of the energy in the logistics market.

The factors such as rise in trade-related agreements, rise of tech-driven energy logistics services and the growing adoption of IoT-enabled connected devices, and increasing wind energy production capabilities supplement the growth of the energy logistics market. However, poor infrastructure & higher logistics costs and lack of control of manufacturers on logistics service are the factors expected to hamper the growth of the energy in the logistics market.

As one of the fastest-growing major economies in the world, India's logistics sector has witnessed a remarkable transformation in recent years. The liberalization of the Indian economy in the early 1990s opened doors for foreign investments and paved the way for the integration of the Indian logistics industry with global supply chains. In recent years, the rapid growth of e-commerce, manufacturing, retail sectors and the government's focus on infrastructure development has further accelerated the evolution of logistics services in the country. As per the India Infrastructure Report released in September 2022, the Indian logistics industry has demonstrated a consistent upward trajectory in growth in recent years. Despite the challenges posed by the pandemic, this sector has exhibited remarkable resilience and remains on a robust growth path. Projections indicate that the logistics market is poised to achieve an impressive Compound Annual Growth Rate (CAGR) of 14.8%, ultimately reaching a market size of \$400 billion by Fiscal Year 2026. This represents a significant expansion from the \$230 billion figure reported in Fiscal Year 2022.

One of the key drivers of this growth has been the government's emphasis on infrastructure development. The introduction of the Goods and Services Tax (GST) in 2017 streamlined tax processes and reduced transit times, resulting in a more efficient movement of goods across the country. Initiatives like the **"Make in India"** campaign and the **"Sagarmala Project"** for port development have also played a crucial role in boosting the logistics sector.

Key initiative taken by India in boosting logistics sector

About Sagarmala Project

The concept of Sagarmala was approved by the Union Cabinet on 25th March 2015. As part of the programme, a National Perspective Plan (NPP) for the comprehensive development of India's 7,500 km coastline, 14,500 km of potentially navigable waterways and maritime sector has been prepared which was released by the Hon'ble Prime Minister, on 14th April, 2016 at the Maritime India Summit 2016.

India is one of the fastest growing large economies in the world with a GDP growth rate of 7.3% in 2018-19 and ports play an important role in the overall economic development of the country. Approximately 95 % of India's merchandise trade (by volume) passes through sea ports.

However, Indian ports still have to address infrastructural and operational challenges before they graduate to the next level. For example, operational efficiency of Indian ports has improved over the years but still lags behind the global average. Turnaround time (TAT) at major ports was approximately 2.5 days in 2018-19, whereas global average benchmark is 1-2 days. Some of the private sector ports in India like Mundra and Gangavaram, have been able to achieve a turnaround time of around 2 days.

Secondly, last mile connectivity to the ports is one of the major constraints in smooth movement of cargo to/from the hinterland. Around 87% of Indian freight uses either road or rail for transportation of goods. A significant share of this cargo experiences “idle time” during its transit to the ports due to capacity constraints on highways and railway lines connecting ports to production and consumption centers.

However, more than 90% of coal currently moves via railways. The constraints on connectivity and sub-optimal modal mix results in higher logistics cost thereby affecting the manufacturing sector and export competitiveness

The third factor is the location of industries / manufacturing centres vis-à-vis the ports. While cost differential between India and China is not significant on a per tonne km basis, China still has a lower container exporting cost, than the cost in India, due to lower lead distances. Presence of major manufacturing and industrial zones in coastal regions in China, which were developed as part of the Port-Led Policy of the government is the main reason for lower lead distances.

Vision of the Sagarmala Programme is to reduce logistics cost for EXIM and domestic trade with minimal infrastructure investment. This includes:

- Reducing cost of transporting domestic cargo through optimizing modal mix
- Lowering logistics cost of bulk commodities by locating future industrial capacities near the coast
- Improving export competitiveness by developing port proximate discrete manufacturing clusters
- Optimizing time/cost of EXIM container movement

<https://sagarmala.gov.in/about-sagarmala/background>

1.4.3 Various renewable energy sources suitable for logistics

As the world grapples with the pressing challenges of climate change, industries across the spectrum are reevaluating their carbon footprints. The logistics industry, historically dependent on fossil fuels, is at the forefront of this green shift. This transformation is not just about environmental responsibility; it's about pioneering a sustainable future for global trade and transportation.

The Rise of Renewable Energy in Logistics: The logistics sector is vast, encompassing everything from shipping to trucking. Each segment is finding innovative ways to incorporate renewable energy:

- **Solar Power:** Warehouses and storage facilities are installing solar panels, converting sunlight directly into electricity. This not only powers the facilities but also reduces reliance on the grid.
- **Wind Energy:** Some shipping companies are revisiting ancient maritime traditions by integrating modern sails or "kites" to harness wind energy, supplementing traditional fuel.
- **Biofuels:** Derived from organic materials, biofuels are becoming a viable alternative for trucks and ships, offering a cleaner burn than conventional fuels.
- **Investment in Green Infrastructure:** Logistics companies are investing in infrastructure that supports renewable energy adoption, such as building EV charging stations, establishing renewable energy partnerships, and integrating sustainability into their supply chains.
- **Innovation and Research:** Ongoing research and development in renewable energy technologies, including advancements in battery storage, hydrogen fuel cells, and renewable energy generation, are paving the way for more efficient and scalable solutions for the logistics industry.

Benefits of renewable energy in logistics Sector

- **Carbon Reduction:** The most evident benefit is the significant reduction in CO2 emissions, helping companies meet environmental targets and reduce their carbon footprint.
- **Cost Savings:** While the initial investment might be significant, renewable energy sources often lead to long-term savings, especially with rising fossil fuel prices.
- **Brand Reputation:** In an era where consumers are environmentally conscious, companies that adopt green practices enhance their brand image and appeal to a broader audience.

Conclusion: The green shift in the logistics industry is not just a fleeting trend; it's a necessary evolution. As the world moves towards a more sustainable future, the logistics sector's role is pivotal. By embracing renewable energy, the industry is not only reducing its environmental impact but also setting the stage for a more sustainable, efficient, and cost-effective future.

1.4.4 Concept of Circular Economy

The concept of a “circular economy” was devised with the aim of minimizing waste and pollution and making the most of resources by keeping products and materials in use as much as possible, and by recovering and regenerating products and materials at the end of each service life.

The Circular Economy Model: Less raw material, less waste, fewer emissions

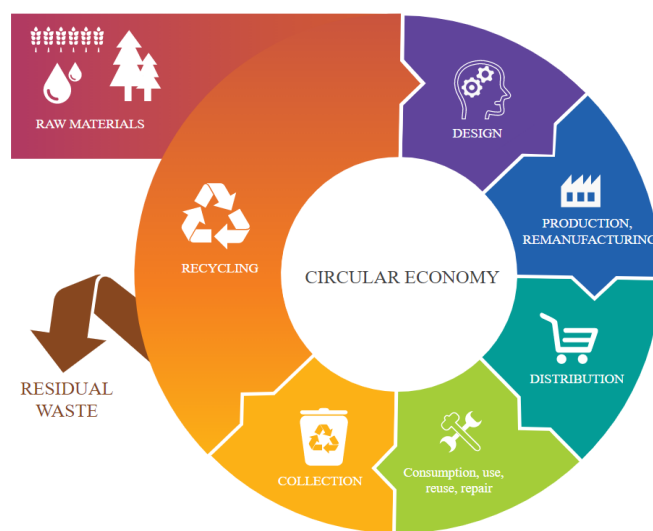


Fig. 1.4.4

Source: European Parliament Research Service

The circular economy encourages businesses to shift from a linear model to a more sustainable approach. It promotes innovations in product design, material sourcing, waste reduction, and encourages collaboration across industries to create closed-loop systems where materials are continuously recycled and reused.

The transition from linear economy to circular economy can put economic growth on a sustainable pathway, by reducing finite resources consumption and minimizing waste and environmental impacts. According to the Ellen MacArthur Foundation, the implementation of circular economy is guided by three principles:

1. **Waste equals Food** – Redefining the purpose of end-of-life products can extend their durability and reduce the environmental impacts of manufacturing new products. Within a closed loop, appropriate maintenance, reuse, refurbishment and recycling can extend the life cycle of products.
2. **Use renewable resources** – By increasing the utilization of renewable or waste-derived resource and energy, circular economy model could create new types of jobs and reduce environmental impacts, including carbon emissions.
3. **Build resilience through diversity** – In order to achieve the reduction of virgin material consumption and waste generation, supply chains need to be developed to reorient products from one manufacturing process into another. Therefore, designing a circular economy model requires bringing together various companies and stakeholders, which serve different functions within a circular economy system.

Application of Circular Economy in Aviation Sector

Circular economy has potential to reshape the whole supply chain from product design to end-of-life management, and aviation already utilises some of the concepts associated with circular economy.

For the aviation sector, circular economy is an emerging concept and while its application is still not widespread, the utilization of circular economy concepts could provide valuable learning opportunities for the future. Aviation is a sector expecting substantial growth, with the annual world air traffic expected to double by 2035, with an average annual growth rate of 4.4 per cent.

According to Boeing and Airbus, the projection of new delivered aircraft by 2034 would be 38,050 and 32,585, respectively. All these estimations indicate a potential increase in resource consumption, waste and emissions generation in global aviation. The transition from linear economy to circular economy could contribute to the reduction of the adverse environmental impacts and associated economic costs.

The application of the circular economy principles to the aviation sector would primarily focus on two elements:

1. **Aircraft**
2. **Airports.**

For aircraft, the circular economy model can be applied into aircraft operations and for the management of aircraft end-of-life. Since 2015, Air France-KLM Group adopted circular economy strategy in their flight operations. The strategy entails four components of the circular economy as follows:

Redesign: Redesigning the catering services to separate the waste correctly

Reduce: Reducing the mass of food packaging and switching manuals from hard copy to digital

Reuse: Reusing the seats and on-board entertainment systems in other systems

Recycle: Recycling the reusable equipment, including trays, drawers, blankets and trolleys, etc.

To handle end-of-life aircraft, Boeing and Airbus have developed their individual management approaches. Airbus launched the PAMELA project (Process for Advanced Management of End-of-Life Aircraft), which includes three stages: decommissioning, disassembly, and smart and selective dismantling. It demonstrated that 85 per cent of an Airbus A300 aircraft weight can be recycled, reused or recovered as secondary raw materials. On the other side, Boeing co-founded the industry association AFRA (Aircraft Fleet Recycling Association), which aims to set-up a new standard for an environmentally responsible management of end-of-life aircraft.

At airports, the application of circular economy has also demonstrated great potential for environmental and economic benefits.

Schiphol Airport (also known as Amsterdam Airport Schiphol, located at Amsterdam, Netherlands) and Philips developed a partnership and provided a circular lighting solution for airport. In this light service solution, Philips remains the owner of the lamps and fittings. It is possible to replace separate components with ease, thus extending the service life of the lighting fixtures. When lamps reach the end of their service life, Philips will collect and recycle them. This circular solution not only reduces 50 per cent energy consumption by energy-efficient LED lighting and extends 75 per cent service life of the fittings, but also reduces maintenance costs and raw material consumption.

By: [*International Civil Association Organisation \(ICAO\)*](#)

Circular Economy in Logistics Sector

In the context of the circular economy, logistics plays a crucial role in reshaping traditional linear supply chains into more sustainable and resource-efficient systems. The current take-make-waste economy is designed for high volume output of materials and products that lead to enormous landfills. Combined with a growing world population and ever-growing consumption patterns, this take-make-waste model has led to severe degradation of natural systems and anthropogenic climate change.

To help overcome these challenges, the transition to a circular economy has been set in motion. As the economy transitions from linear to circular, logistics will change as well. One obvious change is the growing importance of reverse logistics which concerns the activities related to products, components and materials returning into the value chain.

At the core of circular logistics is sustainable packaging. It's not just about wrapping products; it's about rethinking how to package them. Eco-friendly materials that are efficient for transportation and kind to the environment are the new standard. These materials are lightweight, durable, and recyclable—minimising waste and optimising transportation. They are crucial in reducing carbon footprints and contributing to climate neutrality.

In a circular economy, logistics providers are not just delivering goods but charting efficient transportation routes by identifying the most efficient transportation routes through data analytics and technology, cutting fuel consumption and emissions. The goal is transporting goods while leaving the lightest possible ecological footprint which is a crucial step in achieving climate neutrality.

One of the secrets to circular logistics success is intelligent inventory management. It's about walking the tightrope between meeting demand and avoiding overstocking. By closely monitoring the market and adjusting supply accordingly, logistics companies minimise excess inventory, reducing costs and waste. This balance helps streamline operations and boosts overall efficiency.

The circular economy principles applied in the logistics sector involve innovative strategies and practices to minimize waste, optimize resources, and reduce the environmental impact of transportation and supply chain activities. Here's how circular economy concepts are integrated into the logistics sector:

- **Reverse Logistics:** Logistics operations are instrumental in managing the reverse flow of products and materials, enabling activities like product returns, recycling, refurbishment, and remanufacturing. Efficient reverse logistics reduce waste and extend the lifecycle of products.
- **Resource Optimization:** Logistics strategies contribute to resource efficiency by optimizing transportation routes, minimizing empty haulage, and maximizing vehicle or container capacity utilization. This reduces fuel consumption and environmental impact.
- **Circular Supply Chains:** Logistics facilitates the creation of closed-loop supply chains where materials are recycled, reused, or remanufactured. Effective logistics management ensures seamless integration of recovered materials back into production processes.
- **Packaging Innovation:** Logistics professionals contribute to sustainable packaging solutions that reduce waste and enhance recyclability. Optimized packaging designs and materials lessen the environmental footprint of products during transportation and storage.
- **Collaborative Networks:** Logistics providers collaborate with suppliers, manufacturers, and customers to design and implement circular solutions. Shared logistics infrastructure and collaborative transportation efforts contribute to resource optimization and waste reduction.
- **Technology Integration:** Logistics leverage innovative technologies such as IoT, AI, and data analytics to optimize resource use, track and trace materials, predict demand, and streamline operations. These advancements contribute to increased efficiency and reduced environmental impact.
- **Embracing Circular Business Models:** Logistics companies are increasingly adopting circular business models, aligning their operations with circular economy principles. This involves shifting from a linear 'take-make-dispose' model to circularity, emphasizing resource efficiency and waste reduction.

By integrating these circular economy principles into logistics operations, companies can significantly reduce waste, conserve resources, cut emissions, and enhance overall sustainability while maintaining or even improving operational efficiency. This approach fosters resilience and responsiveness to environmental challenges while driving innovation within the logistics sector.

1.4.5 Case Studies for the Circular Economy

Tata Motors Limited, a USD \$42 billion organization, is a leading automobile manufacturer practicing sustainability and the spirit of “giving back to society” as a core philosophy. One of the company’s highly successful circular economy initiatives is Tata Prolife, a pioneering after-market product support strategy for Tata Motors' customers. Use of Tata Motors Prolife aggregate ensures original equipment-like-performance of the vehicle even after the first life cycle. Tata Motors conducts its reconditioning at Prolife plants in Lucknow and Coimbatore. The ambition is to minimize the product life cycle cost, allowing the company to offer high quality products at a reduced price. Tata Motors’ Prolife business has a wide variety of reconditioned products, from engine long blocks, gear boxes, turbo chargers and air compressors to electrical components such as starter motors and alternators. Tata prolife business reconditions about 23,000 equivalent engines in a year.

Renault is a circular economy pioneer. Renault has implemented circular strategies at all stages of the vehicle life cycle, including dematerialization through design as well as end-of-life vehicle collection and dismantling, reuse, remanufacture and recycling. In 2008, Renault entered a joint venture with SUEZ to acquire 100% of INDRA vehicle recycling services. 8 The objective of the partnership was to recover 95% of all end-of-life vehicles in France, addressing the European Community ELV Directive 2000/53. 9 INDRA’s dismantling network processed 400,000 end of life vehicles in 2016. Renault supported INDRA’s development to supply its own plant and foundries (plastics, aluminium, platinoids or copper), as well as second hand part premium services for repair in its dealer network (offering more affordable solutions for repair to its customers). Simultaneously, Renault is increasing recycled content in its vehicles, achieving over 32% of new vehicles by weight in EU in 2016.10 The large amount of recycled plastics reintroduced into the system saves between 10% and 15% of the cost virgin material.

Philips is transforming its business model, moving from selling equipment alone to providing solutions to hospitals and other care providers, sharing risk for results through long-term partnerships. Through this new approach, Philips can directly manage when to upgrade equipment, refurbish systems, reuse parts and recycle materials. This allows Philips to reduce the total lifecycle costs of equipment and extend their lifetimes. The model enables care providers to also reduce costs and access state-of-the-art technology at a more predictable cost. By moving away from a purely transactional sales approach, Philips has achieved 50-90% material reuse (depending on the product) through its refurbishing activities, including its reuse of 940 tons of refurbished medical imaging equipment in 2016.

<https://www.copper8.com/wp-content/uploads/2020/06/Whitepaper-Logistics-in-the-transition-to-a-circular-economy.pdf>

<https://www.logisticsandscm.com/logistics/circular-economy-in-logistics-reducing-waste-maximising-efficiency>

<https://www.cbd.int/business/case-studies/wbcsd-8-business-cases-for-the-circular-economy-en.pdf>

1.4.6 Example of calculating the electric load of the workplace and identifying solar plant capacity to be installed

Through load calculation, one is able to analyse the solar plant capacity you need to install to meet your energy demands

Step 1: Make a list of electrical equipment:

The first step is to list all the workplace electrical appliances accurately. For this, one can follow the table. List every appliance individually to understand the power consumption.

Electrical Appliance	Power Rating (Watt)	Run Time (Hour)	Total Consumption (Watt/Hour)
A1			
A2			
A3			
C1			
C2			
C3			

Load Calculation

- The sheet has 4 columns, the first one is the appliance.
- The second one is the power rating of the appliance, which means, how much power that appliance consumes in 1 hour.
- The third column is of total running time in hours, which means how long the appliance has been running in a 24-hour time cycle.
- The last column is of total power consumption which can be calculated by multiplying the 2nd and 3rd columns.

For example, if you have 3 fans each with a 50-watt rating, that run for 10 hours in a day. Then your power consumption would be $3 \times 50 \times 10 = 1500\text{Wh}$



Step 2: Accommodate the power Losses

Add the 4th column and multiply it by 1.25 to accommodate the power losses. In every electrical device, there is some loss of power during transmission which gets converted into heat. As a result, there is a difference between the input and the output of a device.

Appliance	Power Rating (Watt)	Run Time (Hour)	Total Consumption (wh)
3 Lights	15W (each)	9	$3*15*9= 405$
1 Fridge	500W	24	$1*500*24= 12000$
1 PC	70W	4	$1*70*4= 280$
3 Fans	40W (each)	10	$3*40*10= 1200$
	Total		13,885

Total per day consumption: $13,885*1.25= 14,185\text{Wh}$ or **14.18kWh.**

This means that in a single day, the consumption is around 14.18 units (which roughly translates to 425 units in a month).

Generally, 1 kW solar system generates average 4.5 units per day, therefore, plant capacity can be calculated as: total consumption divided by total generation per day i.e., 3.15 kW (plant capacity)

<https://ornatesolar.com/blog/load-calculation-analysis>

Exercise

1. What is sustainable development and sustainable development goals?

2. Explain the interlinkage between SDG7 & SDG13?

3. What are greenhouse gas (GHG) emissions?

4. What are the sources of GHG emissions in logistics sector?

5. Identify the areas to adopt sustainable practices in logistics sector?

6. What are renewable energy and its sources?

7. Identify the difference between conventional energy and renewable energy?

8. What are the various renewable sources that can be used in logistics?

9. Explain circular economy?

10. Explain the best practices where circular economy concept can be integrated in logistics sector?

2. Introduction to Green Logistics



Unit 2.1 Brief concept of Green Logistics

Unit 2.2 Overview of green supply chain management

Unit 2.3 Introduction to carbon footprint



Chapter Highlights

After reading this chapter, you will be able to: %

1. Know the basic concept of green logistics
2. Understand about sustainable logistics practices
3. Identify the difference between conventional logistics and green logistics
4. Know the key initiatives taken for green logistics in India
5. Identify the list of components of green logistics
6. Establish the drivers and key levers for green logistics in Indian context
7. Understand the meaning of green supply chain management, warehouse management, transport/ shipping and their impact on environment
8. Understand about the supply chain performance metrics
9. Establish the components of measuring carbon footprint of a shipping/ transport agency

Unit 2.1 Concept of Green Logistics

Unit objectives

After reading this unit, you will be able to understand the following points:

1. Overview of green logistics and sustainable logistics
2. Sustainable logistics practices
3. Difference between conventional logistics and green logistics
4. Key drivers and levers for green logistics in Indian context
5. Green logistics initiatives **taken by India**

Introduction to Green Logistics

2.1.1 Overview of Green Logistics and Sustainable Logistics

What are Logistics

Logistics refers to the overall process of managing how resources are acquired, stored, and transported to their final destination. Logistics management involves identifying prospective distributors and suppliers and determining their effectiveness and accessibility. Logistics managers are referred to as logisticians.

"**Logistics**" was initially a military-based term used in reference to how military personnel obtained, stored, and moved equipment and supplies. The term is now used widely in the business sector, particularly by companies in the manufacturing sectors, to refer to how resources are handled and moved along the supply chain.

For example, in the natural gas industry, logistics involves managing the pipelines, trucks, storage facilities, and distribution centers that handle oil as it is transformed along the supply chain.

The concept of business logistics has been transformed since the 1960s. The increasing intricacy of supplying companies with the materials and resources they need, along with the global expansion of supply chains, has led to a need for specialists known as supply chain logisticians.

Meaning of Green Logistics

Green logistics is the process of reducing the environmental impact of delivery and logistics processes. As more customers become more environmentally conscious and place emphasis on choosing green companies, shippers must decrease their carbon dioxide emissions, deal with waste disposal and overall waste management, use recyclable materials, and more.

Green logistics includes the carbon footprint of the supply chain, dealing with carbon emissions from waste management and waste disposal, packaging, recycling, reducing energy consumption, etc. Between different national and global entities encouraging or enforcing more sustainability among companies, and more consumers prioritizing green consumption, more companies are pledging zero-net targets in an effort to go as green as possible.

What are the components of green logistics

Green logistics consists in using more eco-friendly and sustainable processes in order to reduce the environmental impact of logistics. This approach covers the entire life cycle of the product: manufacturing, storage, transport, marketing, use and disposal.

Green logistics encompasses various components aimed at reducing the environmental impact of logistics operations. Some key components include:

- **Transportation Optimization:** Utilizing eco-friendly modes of transportation, such as electric or hybrid vehicles, to reduce emissions. Optimizing transport routes to minimize mileage, fuel consumption, and environmental impact.
- **Efficient Packaging:** Implementing sustainable packaging solutions using recyclable, biodegradable, or reusable materials. Reducing packaging waste and adopting innovative designs to minimize materials used.
- **Energy-Efficient Warehousing:** Incorporating energy-efficient practices in warehouses, such as LED lighting, motion sensors, and smart HVAC systems. Adopting green building designs to minimize energy consumption.
- **Waste Reduction and Recycling:** Developing strategies to minimize waste generation throughout the supply chain. Promoting recycling programs and encouraging the reuse of materials to reduce environmental impact.
- **Technology Integration:** Utilizing advanced technologies like IoT, AI, and data analytics for optimized inventory management, route planning, and resource utilization. Implementing real-time tracking and monitoring for efficiency.
- **Supplier Collaboration:** Engaging suppliers committed to sustainable practices and ethical sourcing. Encouraging suppliers to adopt environmentally friendly practices throughout the supply chain.
- **Regulatory Compliance and Certifications:** Adhering to environmental regulations and obtaining certifications (e.g., ISO 14001) to demonstrate commitment to sustainable logistics practices.
- **Employee Engagement:** Providing training programs for employees on sustainability practices. Fostering a culture of environmental responsibility within the workforce.

These components collectively contribute to the implementation of green logistics practices, aiming to strike a balance between operational efficiency and environmental responsibility within logistics operations. Each

component focuses on reducing the carbon footprint, minimizing waste, and adopting sustainable practices throughout the supply chain.

Meaning of Sustainable Logistics

“Sustainable logistics” refers to the practices and processes aimed at improving the sustainability of supply-chain activities, ranging from the supply of raw materials to the transformation processes, the storage, the packaging, the distribution and the management of the end of the lifecycle of products.

This concept is not limited to environmental footprint, but also involves the economic and social sustainability of companies’ **logistics** activities.

Sustainable logistics is becoming more and more relevant in the transition from a linear economic model (based on extraction, transformation, distribution and consumption cycles) to a circular model of industrial economy, whose main goal is to improve the quality of raw materials and final products.

At its core, sustainable logistics recognizes the interconnectedness of environmental, social, and economic factors within the logistics industry. By adopting this approach, businesses seek to reduce their carbon footprint, promote ethical practices, and enhance operational efficiency throughout the supply chain.



Fig. 2.1.2

Source: [Go Green](#)

2.1.2 Sustainable logistics practices

Sustainable practices within green logistics encompass a range of approaches aimed at reducing environmental impact while maintaining efficient supply chain operations. Here are some key sustainable practices in green logistics:

1. The 4R's Strategy

Reduce, Reuse, Recycle and Recover. Aiming for sustainability is a crucial problem for the production model and the supply chain. This approach maximizes the use of resources and eliminates those not required. These include decreasing waste, reusing when practical, and recovering to prevent resource waste. Doing so generates savings that raise the profitability of the distribution process in addition to helping the environment.

2. Explore the Use of Alternative Vehicle and Fuel Technologies

Many delivery truck fleets rely on diesel fuel due to its affordability. Diesel engines are not necessarily better for the environment, but compared to gas engines, diesel trucks emit 13% more CO₂.

To achieve ecological and social sustainability, companies should replace outdated trucks with eco-friendly vehicles and use lower-carbon fuels like biofuel. Many eco-friendly products available on the market improve on previously established practices. More industries employ green transportation options like hybrid or electric cars, eco-friendly train systems, and sea cargo.

3. Using Data for Greater Operational Visibility

Businesses may operate more effectively and efficiently by putting data at the core of supply chain operations. An open logistics ecosystem enables corporate partners to manage logistics operations better. Constituents and staff can monitor real-time transit methods and select the best choices. This improves freight coordination and material and goods tracking. Additionally, by filling trucks, businesses may reduce the danger of empty miles, which is advantageous for both the environment and their bottom line.

4. Smart Warehouse Network

With the rise of last-mile delivery emissions, logistics companies constantly look for ways to lower their carbon footprints. This is one of the primary indicators of genuinely sustainable supply chain logistics.

Central distribution warehouses are a great way to cut last-mile emissions since they bring businesses closer to customers. Businesses benefit from reduced delivery costs and time and increased flexibility to lessen supply chain disruptions.

5. Optimized Planning and Practices

Doing business with other green companies guarantees that every link in the supply chain is sustainable and green. Reducing paper consumption in the workplace and going digital can lessen the environmental impact. The first step in achieving sustainable goals is understanding how much pollution a company's logistics efforts produce. There are specialized software solutions that calculate a fleet's carbon footprint. This data makes organizing transport routes and loads easier to make them more efficient and sustainable.

Companies can explore waste reduction and the use of clean energy sources. It can also utilize sustainable materials, the use of eco-friendly technologies, and more. To develop a sustainable supply chain and achieve carbon neutrality, one of the key building blocks is the implementation of environmentally friendly logistical solutions.

<https://zhenhub.com/blog/sustainable-logistics/#five-sustainable-logistics-practices>

6. Use of renewable energy:

Installation of solar panels or wind turbines to generate renewable energy for powering warehouses, distribution centers, or operational facilities. Utilization of geothermal systems or other renewable energy sources for heating, cooling, and powering logistics operations. The integration of renewable energy sources in sustainable logistics not only helps in reducing greenhouse gas emissions but also contributes to long-term cost savings, energy independence, and a more environmentally friendly supply chain.

2.1.3 Difference between conventional logistics and green logistics

Logistics is the term widely used to describe the transport, storage and handling of products as they move from raw material source, through the production system to their final point of sale or consumption. The logistical activities involve freight transport, storage, inventory management, materials handling and all the related information processing with the main objective to co-ordinate these activities in a way that meets customer requirements at minimum cost.

Conventional logistics refers to the traditional practices and methodologies used in the management and movement of goods and services within a supply chain. It involves the planning, implementation, and control of the flow of materials, products, and information from the point of origin to the point of consumption.

Whereas, on the other hand, green logistics is a form of logistics which is calculated to be environmentally and often socially friendly in addition to economically functional. It describes all attempts to measure and minimize the ecological impact of logistics activities. This includes all activities of the forward and reverse flows of products, information and services between the point of origin and the point of consumption with the aim to create a sustainable company value using a balance of economic and environmental efficiency

The key difference between conventional logistics and green logistics are explained below:

Areas	Conventional Logistics	Green Logistics
Focus	Primarily emphasizes efficiency, cost reduction, and timely delivery of goods.	Focuses on reducing the environmental impact of logistics operations.
Areas of Emphasis	• Optimization of transportation routes and modes • Cost-effective storage and warehousing solutions. • Meeting customer demands efficiently. • Operational streamlining for cost reduction.	• Minimizing carbon emissions and pollution from transportation. • Using eco-friendly transportation modes (e.g., electric vehicles). • Sustainable packaging materials and waste reduction. • Optimizing routes to reduce fuel consumption and emissions.
Environmental Considerations	Environmental concerns aren't the primary focus. Practices may prioritize efficiency	Prioritizes reducing environmental impact through eco-friendly practices

	over sustainability.	and technologies.
Operational Approach	Efficiency-Centric: Focuses on optimizing operations for cost reduction and speed. Technology Use: Utilizes basic technologies for tracking and inventory management.	Efficiency with Sustainability: Balances operational efficiency with environmentally friendly practices. Technology Emphasis: Incorporates eco-friendly technologies, such as alternative fuel vehicles and advanced tracking systems.
Perspective Approach	Short term focus: Emphasizes immediate operational efficiency without long-term environmental considerations.	Sustainability focus: Considers long-term effects, aiming for sustainable practices that minimize environmental impact over time



Fig. 2.1.3

Image courtesy: [TruckGuru](https://www.truckguru.com/)

Fig. 2.1.3

Source: [Semantic Scholar](https://www.semanticscholar.org/)

2.1.4 Key drivers and levers for green logistics in Indian context

Over the years, the importance of logistics has changed from being a mere support function to obtaining strategic advantage. Logistics includes all those activities which take products from the source of raw

materials to final consumer, including recycling and disposal activities. On the inbound side, timely availability and superior quality of raw materials ensures uninterrupted production to fulfil orders. On the outbound side, efficient logistics activities lead to increase in customer reach and satisfaction. Irrespective of business size, logistics activities not only help in getting the product to customer on time but also get a clear idea of requirements of different customer segments. In today's globalized economy, having a robust and flexible logistics system can be a differentiating factor leading to success or failure of business activities.

Drivers for Green Logistics

However, consistent focus on meeting customer requirements in minimum time at minimum cost, has led to a neglect of social and environmental impact of logistics operations. With growing government and public concerns, companies are faced with challenge of developing businesses with lower environmental footprint. This is driving an increased need for companies to “green” their logistics operations

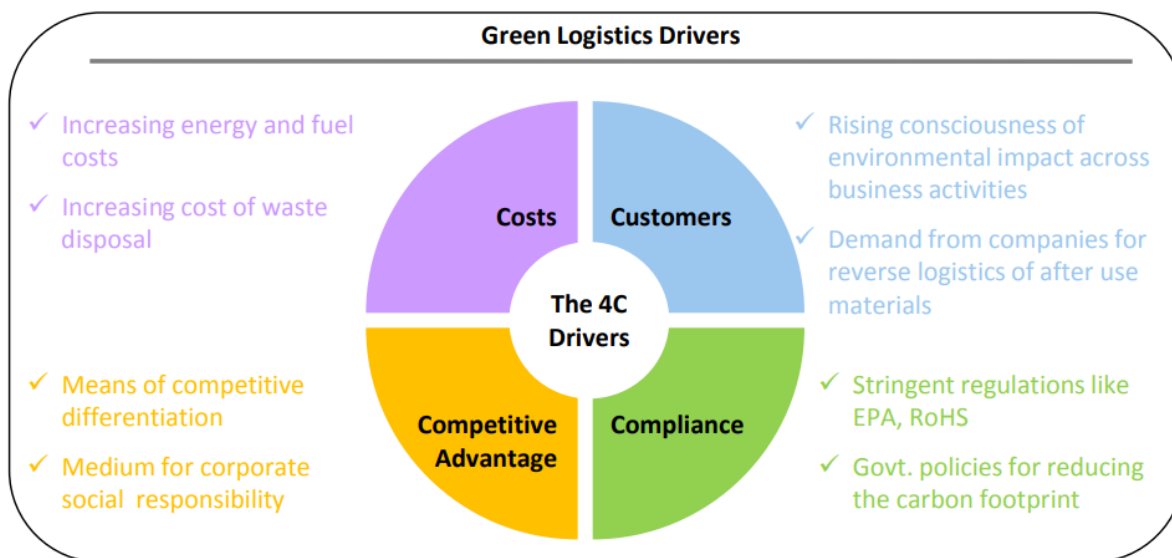


Fig. 2.1.4

1. **Costs:** Increasing power and fuel costs, together with the cost of related raw materials used in infrastructure building and functioning has led to chances for looking into green alternatives that can significantly lead to a reduction in the price.
2. **Customers:** Consumers have become conscious about the environment. They want that the process by which the products are manufactured and delivered to them should create minimum impact on the environment. In addition, the demand of environmentally friendly products is increasing.
3. **Competitive advantage:** Corporations across the globe are now adopting green policies, practices, and procedures to remain competitive in the market and also be acceptable by the consumers. Consumer forums are compelling corporation to be green in all their activities, processes, and practices. Adoption of green principles and best practices like recycling, reducing energy consumption, etc., can significantly reduce costs and improve operational efficiency.

4. **Compliance:** Most of the countries have evolved policies and enacted laws to prohibit products and processes harming the environment. The destructive materials and polluting processes are banned. In this era of globalization, it has become important for the companies to comply with the regulations of the concerned countries.

Green Logistics can therefore be defined as supply chain management practices and strategies that reduce environmental and energy footprint across various stages of supply chain right from procurement of raw materials to finally delivering end product to the customer.

Levers for Green Logistics

Logistics network (including warehouse locations & capacities, modal mix, distribution network), warehousing operations, packaging activities and material handling activities form the backbone of successful logistics operations. Hence, implementation of sustainable green practices across these four levers can help companies create a positive impact on their ecological footprint at optimum costs.

The 4 levers for Green Logistics are:

- Network Optimization
- Green Warehousing
- Green packaging
- Standardization

Network Optimization

Increasing globalization of supply chain activities across industries is leading to highly complex logistics networks. This results in long distance product movements leading to high energy costs and carbon-dioxide emissions. On the other hand, high service level expectations from customers and increasing push for just-in-time practices are leading to expedited orders and multiple small shipments. This causes logistics inefficiencies resulting in aggravated fuel costs and higher emissions per distance travelled. Poor infrastructure in the country further leads to sub-optimal speeds and capacity utilization resulting in higher environment footprint per unit load. This highlights the need for network optimization for successful implementation of green practices.

It is estimated that network restructuring can help companies reduce costs by 10-15% and carbon dioxide emissions up to 10%. However, in order to gain supply chain efficiencies and customer acceptance, companies often take decisions which lead to detrimental impact on environment and pose a question of sustainability. A balancing act in collaboration with supply chain partners is therefore required for companies to restructure their supply chain network for reducing their ecological footprint.

Following strategies for designing green networks:

Warehouse network design: Optimal warehouse numbers, capacities and locations based on number of customers, their geographic distribution & order size, service level expectations and number & locations

<https://ciischooloflogistics.com/wp-content/uploads/2022/02/GreenLog.pdf>

of plants can help companies optimize primary and secondary transportation distances & time leading to cost reduction and lower carbon-dioxide emissions.

Transportation optimization: Transportation constitutes almost 35% of overall logistics costs in India. It is estimated that transportation activities result in 10-15% of overall emissions. Transportation redesigning through following strategies can help companies reduce their environmental footprint while ensuring supply chain efficiencies

- **Distribution & Route Planning:** Once the warehouse locations are finalized, companies can revisit their existing distribution planning system. Selecting economical distance path (not necessarily least distance path) and optimizing order fulfilment process (through differentiated replenishment model) can help companies reduce their fuel consumption.
- **Modal Mix:** Indian logistics sector is heavily dependent on road network (one of the least energy efficient modes) contributing to almost 65% of total freight transport. This is because of lack of transport infrastructure and need for compressing costs and time involved to meet customer demand. While building infrastructure is cost and time consuming, companies can redesign their current modal mix within existing limitations to reduce environmental impact.
- **Green Fleet:** While trucks constitute 5% of vehicle population in India, it is estimated that they are responsible for 63% of carbon-dioxide emissions, 59% of particulate matter emissions and 46% of fuel consumption, thereby highlighting the need of green fleet management. This requires a strategic approach involving a combination of technological improvements and policy initiatives. Technological changes involve use of alternate fuels (CNG/LPG/HCNG) and improved vehicle technology (e.g. aerodynamics, weight reduction, idling reduction, reduced tire rolling resistance, improved fuel injection systems) while policy initiatives involve Government intervention to provide enforcement and incentives with focus on technology adoption and fleet modernization including retirement of vehicles, replacement of powertrains and retrofitting.
- **Green pallets:** Apart from fleet, pallets play a critical role in transportation activities. Traditionally wooden pallets are used for transporting goods as they are cheap and easy to manufacture. However, wooden pallets are fragile, non-recyclable and heavy. This results in frequent replacement & disposal of damaged pallets, and lower fuel efficiencies during transportation. On the other hand, plastic pallets are more resistant to wear & tear, have longer life, can be easily recycled thereby eliminating the need of disposal and are lighter as compared to wooden pallets leading to lower fuel consumption. This also highlights that even non-natural products can bring in sustainability and help reduce environmental footprint.

Supply Chain visibility: An understanding between all the nodes of Supply Chain is critical in any firm. It is a critical to optimize both Cost and Service. This sharing of information via various systems and methodologies will help companies do a better job of planning at all nodes.

Green Warehousing

Warehousing (including inventory management) is one of the most critical supply chain functions, accounting for 20-25% of logistics costs. Traditionally warehousing operations involved receipt, storage and delivery of materials (raw materials, WIP, finished goods and packaging) across supply chain. However, with rising costs and increasing customer demands, the role of warehousing has shifted to

providing value added services. Today warehousing operations involve break-bulk, product mixing, reverse logistics, cross-docking, late product differentiation and just in time & customized deliveries, thereby helping companies to achieve operational efficiency and high levels of customer satisfaction

Lighting, HVAC system and material handling equipments are the largest energy users in warehouse operations. It is estimated that best warehouse energy management practices can save 40-50% of energy use without significant capital investment. Companies can adopt following means for reducing their environment impact without significant cost burden

Daylighting: Natural lighting involves installation of skylights and clerestory windows which provide natural illumination in warehouses with minimum additional construction. As natural light intensity varies across the day, a hybrid setup involving natural means and artificial lights can be used. Natural lighting, thus helps companies to reduce electricity usage, reduce carbon dioxide emissions and improve environment quality for warehouse employees

Lighting systems: Often warehouses are over lit, resulting in unnecessary energy wastage. This can be overcome by using motion sensors, especially in low traffic or non-crucial areas, which ensures that lights are turned on only when the area is in use

Warehouse Management: Proper warehouse management can eliminate unnecessary and repetitive operations which can reduce energy and fuel consumption. For instance, high density areas due to large product quantity in small spaces result in multiple use of forklifts. Lack of streamlined operations result in multiple handling of products making the material flow critical for ensuring energy management. Use of inventory tracking techniques like RFID can help eliminate unnecessary material handling activities, resulting in savings on energy and fuel front. Vertical warehousing (racking & storage) and vertical equipment like vertical carousels & conveyors can help companies reduce land requirement, reduce construction footprint and achieve better facility planning & higher space utilization. This helps companies in preserving natural resources, develop energy efficient buildings, reduce energy costs and overall carbon footprint.

Material handling equipment: Energy efficient equipment like battery-operated or fuel cell forklifts eliminate emissions and produce less noise during operations as compared to fuel operated ones. Automation systems can be developed so that parts of system shut down automatically when not in operation.

Green Envelope: Green envelope involves building warehouses with energy efficient walls and roofing systems. Energy efficiency of walls can be improved up to two times by replacement of traditional batt insulation with loose-fill or sprayed foam.

Temperature Control: Optimum warehouse temperature is critical for ensuring product maintenance in industries like pharmaceuticals, agro-products etc.. Apart from product quality it also helps in regularizing energy requirement. It is estimated that a 1 oC decrease in internal warehouse temperature can lead upto 10% energy consumption savings

Green warehousing provides significant opportunities for companies to decrease their operational costs, while simultaneously reducing the environmental and social impact of their operations, thereby building a sustainable ecosystem. In a long term, companies should target for net-zero warehouses which would involve generating as much energy as much is consumed.

Standardization

Typical logistics activities involve supplier packing raw materials, loading them on trucks and transporting them to manufacturer. At manufacturer's location, raw materials are unloaded for production activities. Finished goods are then packed, loaded on trucks and then transported to warehouses where boxes are unloaded and then stacked. The goods are then loaded on trucks and delivered to retailer, where they are unloaded, re-arranged and stacked. All these activities involve multiple permutations and combinations. Goods can be transported through various modes & capacities as individual boxes or palletized loads, handled (loading and unloading) manually or through automated equipment and can be stored either on ground or on racks. These complexities due to lack of standardization, combined with multiple supply chain stages lead to operational inefficiencies, ineffective use of manpower & space, lower truck turnaround time and safety risks leading to cost escalations and bottlenecks across the supply chain.

Standardization across Racking-Handling-Trucking-Palletizing (RHTP) activities helps suppliers unitize raw materials on standard pallets and load them on standard dimension trucks using standard material handling equipments. With standardization across supply chain in place, manufacturer can easily unload the pallets with minimum product damage. As pallets are of standard dimensions, they need not be returned back to supplier. Post manufacturing, finished goods can be loaded on same pallets and transported again in standard trucks to warehouses. The palletized loads can easily be stacked on standard racks with minimum man handling. The same pallets can then be used to transfer goods to retailer

Standardization across RHTP activities would eliminate the need of empty pallet returns and reduce reverse logistics across various supply chain stages, thereby reducing transportation costs and emissions. As pallets and trucks are standardized, same trucks can be used to transfer multiple products. After unloading goods at a particular warehouse, a truck can be used to transport goods of some other manufacturer (or supplier) located in warehouse' vicinity, thereby overcoming the challenge of empty truck returns. Standardization also brings in efficiencies in terms of reduction in manual & multiple material handling, reduction of loading & unloading time, improvement in space utilization, reduces chances of product damage and creates potential for using information system at a broader scale for supply chain visibility. This standardization can help companies reduce inefficiencies associated with energy & fuel consumption and greenhouse emissions.

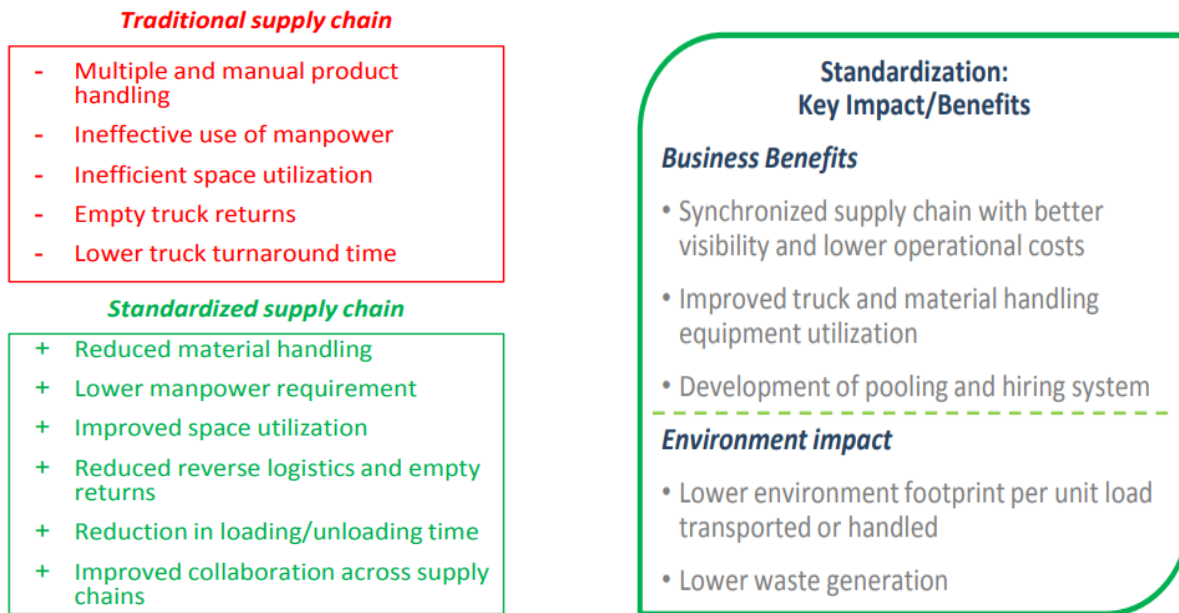


Fig. 2.1.4

Green Packaging

Traditionally packaging functions involved protection & preservation, transportation, material handling, warehousing and storage. However, over the years, with emphasis on sales promotion, customer attention, brand communication and product differentiation packaging operations have gained prominence. Packaging thus plays an integral role in ensuring smooth and successful logistics and marketing activities. Packaging also is a significant cost driver accounting for 5-12% of overall costs across supply chain of various industry sectors.

However, packaging results in significant environment problems. Various kinds of materials like wood, glass, metals, paper & board and plastics are used for packaging applications with varied degree of recyclability. This results in waste generation across various stages of supply chain leading to reverse logistics and waste disposal. It is estimated that packaging contributes to 4-8% of total carbon emissions across supply chain of various industry sectors. This highlights the importance of developing green packaging solutions for achieving cost efficiencies and reducing environmental footprint.

Green Packaging involves developing environment friendly packaging solutions. It also takes into consideration ecological footprint of a product across its life cycle. Green packaging thus provides companies opportunities to increase their revenues, reduce costs, streamline their supply chain and achieve intangible benefits like brand recognition and customer preference.

In order to develop green solutions, companies need to explore options which are inherently green and sustainable. This involves a 3R strategy involving – Reduce, Reuse and Recycle

Reduce: This involves using light weighing, less toxic & thin packaging material, eliminating excess/unnecessary packaging and bulk packaging in primary distribution, while ensuring product safety

during various stages of distribution. This would result in reduction in packaging and transportation expenses, reduction in waste generated & disposal challenges across supply chain, better cube utilization and less generation of greenhouse emissions due to weight reduction.

Reuse: This involves reusing packaging material resulting in improved life cycle performance, reduction in packaging inventory & improved warehouse utilization and reduction in waste management costs

Recycle: This involves deploying recyclable packaging materials like paperboard, corrugated cartons, glass and bio-polymers resulting in reduction of carbon footprint, energy savings and reduction in use of natural virgin resources while upholding requisite packaging standards with no harmful impact on health and safety standards

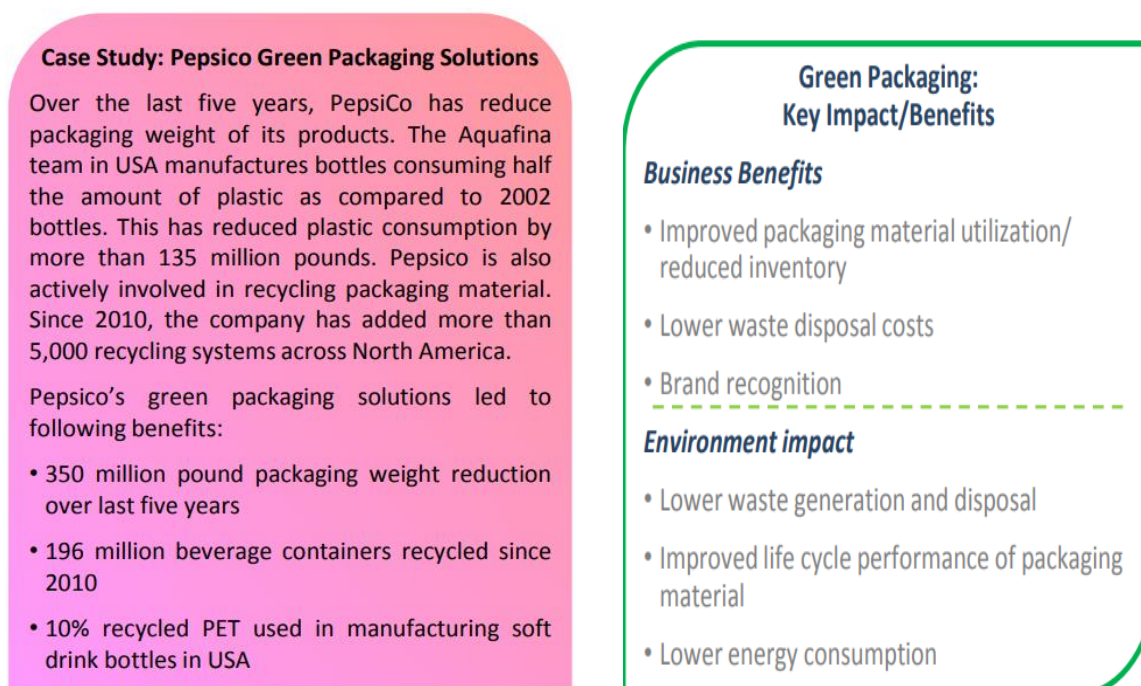


Fig. 2.1.4

2.1.5 Green logistics initiatives in India

National Logistics Policy (2022)

National Logistics Policy is a significant step in creating an efficient and cost-effective logistics system that is indispensable for the transition of India from a developing economy to a developed economy.

Logistics efficiency is a function of infrastructure, services (digital systems/processes/ regulatory framework) and human resource. The PM GatiShakti National Master Plan (NMP) for multimodal connectivity infrastructure to various economic zones, has been launched. PM GatiShakti National Master Plan is a transformative approach for improving logistics efficiency and reducing logistics cost, with focus on integration of existing and proposed infrastructure development initiatives of different agencies, to ensure first and last mile connectivity, for seamless movement of people and goods.

PM GatiShakti is a transformative approach for economic growth and sustainable development. The approach is driven by 7 engines, namely:

- Railways
- Roads

- Ports
- Waterways
- Airports
- Mass Transport
- Logistics Infrastructure

The Gati Shakti aims to create roads linked to railway lines that will further feed into major and minor ports, leading to the efficient movement of goods, especially from the frontiers.

While integrating all the multimodal connectivity projects, the NMP will optimise the efficiency of the movement of goods and people.

The Ministry of Road Transport and Highways (MoRTH) and Shipping plan to connect all state capitals in the northeast by expanding the national highway (NH) network to 2 lakh km by 2025. It will reduce the turnaround time and enhance efficiency in the supply chain.

The National Logistics Policy (NLP) was announced by the Finance Minister in the 2022-23 Union Budget. It was formulated by the Union Ministry of Commerce and Industry.

It is a comprehensive effort to address high logistics costs with an interdisciplinary framework for the development of the country's \$150-billion logistics sector.

The new policy is driven by the vision "To develop a technologically enabled, integrated, cost-efficient, resilient, sustainable and trusted logistics ecosystem in the country for accelerated and inclusive growth."

The policy has four features that will be implemented through the **Comprehensive Logistics Action Plan (CLAP)**:

Integration of Digital System (IDS): For the digital integration of different systems of seven various departments (like road transport, railways, aviation, commerce ministries and foreign trade).

Unified Logistics Interface Platform (ULIP): This ensures shorter and smoother cargo movement and enables the exchange of information confidentially on a real-time basis. This National Industrial Corridor Development Corporation (NICDC) Logistics Data Bank Project has been leveraged.

Ease of Logistics (ELOG): This will enable and ensure the ease of logistics business through transparency and accessibility.

System Improvement Group (SIG): This will monitor all logistics-related projects regularly.

The e-handbook on the policy gives insight into the standardisation of the warehousing sector.

CLAP comprises integrated digital logistics systems, standardisation of physical assets, benchmarking service standards, human resource development, capacity building, development of logistics parks, etc.

For the expansion of the sector, the government will also provide managerial training in logistics and supply chain. Multi-modal logistics parks (MMLPs) can be established in key markets to enhance first- and last-mile connectivity.

<https://byjus.com/free-ias-prep/national-logistics-policy-upsc-notes/>



Fig. 2.1.5

Image courtesy: [moneyControl](https://www.moneycontrol.com)

Supporting policies to make NLP more robust:

- The PM GatiShakti launched in October 2021 was a pioneering step in this direction. PM GatiShakti will get further boost and complementarity with the launch of the National Logistics Policy.
- The introduction of GST has brought efficiency to the logistics and transportation sector. It has reduced the turnaround time of trucks by over 20%.
- Faceless assessment in customs and e-way bills and FASTag are bringing efficiency to the logistics sector.
- **The Sagarmala project** that connects ports and dedicated freight corridors has improved logistics connectivity and systematic infrastructure development work.
- Village Storage Scheme through women's self-help groups shall provide backward linkages for seeds thereby reducing logistics costs. Financial assistance under MUDRA loans and NABARD shall be provided for this purpose.
- Other initiatives to improve physical infrastructure such as **Bharatmala Yojana**, dedicated freight corridors, waterways development and multi-modal transport hubs among others

Conclusion: The National Logistics Policy will facilitate a unified policy and regulatory environment for end-to-end logistics services and an overarching institutional framework that will govern the logistics sector and enhance its competitiveness. Both State and central governments would prepare a master plan for logistics that will cover projects planned by different ministries to ensure multi-modality, optimal modal mix and addressing first and last mile connectivity.

IKEA Model

IKEA Model IKEA is a global furniture store that provides “cheap furniture that as many as possible can afford”. Ikeas company view of green logistics is firstly to remove the wooden pallets from the entire supply chain. Instead of wooden pallets they are using Paper/cardboard pallets and so called ledges. Underneath these ledges there are a plastic leg underneath the goods. Because of this approach they are dramatically decreasing transportation, Co2 emissions, PP/LL can be packed with less space than with normal wooden pallets. Also IKEAs lway monitoring system steer transport partners to use low Co2 equipments. The lway monitoring system is IKEAS own measuring system.

The goals for IKEA within green logistics are as the following:

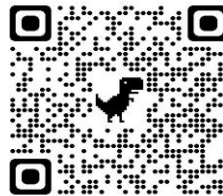
- 2012 60% total flow integrated to non-wooden pallets
- 2014 100% of total flow integrated to non-wooden pallets
- 2015 all transport partners are fulfilling the lway standards

DHL – Go Green Solutions

- DHL Global Forwarding solutions provide a reliable way of accounting for and managing the supply chain CO2 emissions, with best-in-class carbon reporting and all-inclusive carbon offsetting services.
- DHL Global Forwarding has developed a carbon reporting methodology with precision and reliability in mind. Over a number of years, its experience in carbon reporting has provided the expertise to produce accurate CO2 calculations for the customers
- By considering each individual shipment, actual operational data (mode, fleet, age, trade lane, weight/volume, actual distance travelled) and credible sources for emission factors, DHL is able to give the best-in-class CO2 calculation
- Their methodology follows the principles of the Greenhouse Gas Protocol and the World Economic Forum (WEF) Guideline for Consignment-level Carbon Report
- They will create a carbon report that provides the confidence to design own carbon reduction strategy



Understanding the essentials of green logistics



Company like DHL going green

Unit 2.2 overview of green supply chain management

Unit objectives

After reading this unit, you will be able to understand the following points:

1. Introduction to green supply chain management
2. Impact of supply chain management, warehouse management, transport/ shipping on environment
3. Components of green supply chain performance metrics

2.2.1 Introduction to green supply chain management

Since last couple of decades, environmental issues have been increasing and traveling faster than forest fire, country to region, region to world level territory, which is a serious cause of climate change and global warming. While, the green supply chain concept occurs to mitigate environmental degradations and control air, water and waste pollution through the adoption of green practices in business operations. Undeniably, the basic ideology behind green concept is to enhanced environmental sustainability. Because green supply chain can reduce the environmental pollution and production costs and it also can spur economic growth, create competitive advantage in terms of greater customer satisfaction, positive image and reputation and provide better opportunity to export their products in pro-environmental countries

The definition of green idea is expanding with new innovations and techniques to protect environmental sustainability, which can be recognized by corporate social responsibility, green manufacturing, waste reduction, recycling and remanufacturing sustainable/ environmentally friendly supply chain, green supply chain, etc.

The term sustainable or green supply chain refers to the idea of integrating sustainable environmental processes into the traditional supply chain. This can include processes such as supplier selection and purchasing material, product design, product manufacturing and assembling, distribution and end-of-life management. Instead of mitigating harmful impact of business and supply chain operations, green supply chain involves value addition and/or value creation through the operations of whole chain.

2.2.2 Environmental impact of supply chain management, Warehouse management and transport/ shipping

➤ Environmental impact of supply chain management

Supply chain management practices often have several environmental implications due to their focus on efficiency and cost reduction without prioritizing sustainability. Here are some key practices and their environmental impacts:

Global supply chains are growing and becoming more complex as consumer demand increases. This greater need for products, ingredients and raw materials directly translates into increased environmental impact.

The impact of the supply chain on the environment is primarily negative. According to a report from the McKinsey institute, over 90 percent of the environmental damage caused by companies that package consumer goods (cpg) come from their supply chains. This greatly outweighs the damage done by all other forms of business operations. There are two specific areas and effects of the supply chain that are concerning: the greenhouse gas emissions released during the shipping process and international travel through oceans.

Greenhouse Gases in Shipping

Greenhouse gases have long been at the forefront of conversations about climate change, and they are still an issue in the context of supply chain sustainability. The report from the McKinsey Institute notes that the supply chains related to CPG organizations account for more than 80 percent of a given company's greenhouse gas emissions.

International Travel Through Oceans

Ocean travel is one of the primary ways to transport goods internationally; in fact, it is responsible for facilitating about 90 percent of world trade. This popular mode of transportation has contributed to the pollution and decline of our world's oceans. Scientists estimate that only 13 percent of the world's oceans have not been damaged by human activity. In addition to climate change, they attribute this deterioration to global trade, as well as polluted land runoff and commercial fishing practices.

Supply chains play a significant role in shaping environmental impact across various stages of production, transportation, and distribution.

Some key environmental impacts created by supply chains include:

- **Transportation:** Movement of goods through various modes of transportation, primarily fuelled by fossil fuels, contributes to significant carbon dioxide (CO₂) emissions, leading to climate change and air pollution.
- **Manufacturing:** Industrial processes within the supply chain, such as energy-intensive manufacturing, also contribute to greenhouse gas emissions.
- **Resource Depletion:** Extraction of raw materials and resources required for production, coupled with inefficient usage, leads to resource depletion, impacting ecosystems and biodiversity.
- **Waste Generation:** Overproduction, excessive packaging, and inefficient management of returns and end-of-life products contribute to increased waste generation, leading to landfill accumulation and environmental pollution.

- **Water and Soil Pollution:** Chemical runoff from manufacturing processes, improper waste disposal, and agricultural practices in the supply chain can lead to water pollution, contaminating water bodies, and affecting soil quality.
- **Deforestation and Habitat Loss:** Sourcing raw materials like timber, palm oil, and paper can contribute to deforestation, leading to habitat loss, species extinction, and disruption of ecosystems.
- **Energy Consumption:** Operations within warehouses, manufacturing plants, and transportation sectors consume vast amounts of energy, often derived from non-renewable sources, contributing to energy depletion and environmental strain.
- **Supply Chain Complexity:** Complex supply chains with multiple intermediaries, long transportation routes, and global sourcing contribute to increased energy usage, emissions, and the overall environmental footprint.
- **Supplier Practices:** Suppliers' activities and practices upstream in the supply chain can significantly impact environmental sustainability. Suppliers with poor environmental practices contribute to environmental degradation.
- **Social and Ethical Implications:** Environmental impact within the supply chain can also have social and ethical implications, affecting communities, workers' rights, and public health in areas of production and manufacturing.

Understanding and mitigating these environmental impacts within the supply chain are crucial for fostering sustainable practices, reducing emissions, conserving resources, and promoting environmental stewardship across industries. Adopting green and circular economy principles, promoting responsible sourcing, and enhancing transparency throughout the supply chain are essential steps toward minimizing environmental harm.

The impact of supply chain management on Environment



Fig. 2.2.2

Image courtesy: [Shiprocket](#)

➤ **Environmental impact of Warehouse management**

The environmental impact of warehousing has received only little attention by research although significant amount of carbon dioxide emissions logistic-related is caused by storage and material handling processes in warehouses, prior research mostly focused on the transport elements. A warehouse, also referred to as distribution centre, storage facility and logistics service centre. They are used for different operations such as product distribution, cross-docking and composite storage. Nevertheless, carbon emissions caused by material handling activities in logistics buildings, constituting warehouses and separation facilities, are important and account for 13% of overall supply chain emissions (World Economic Forum, 2009).

Warehousing facilities generates important need of use of energy, water and land. They cause various impacts on nature and people, which are associated with impact of land use, atmospheric emissions, waste management, traffic and congestion, public transport, visual intrusion and ecology. (Peter Baker and Clive Marchant, 2015).

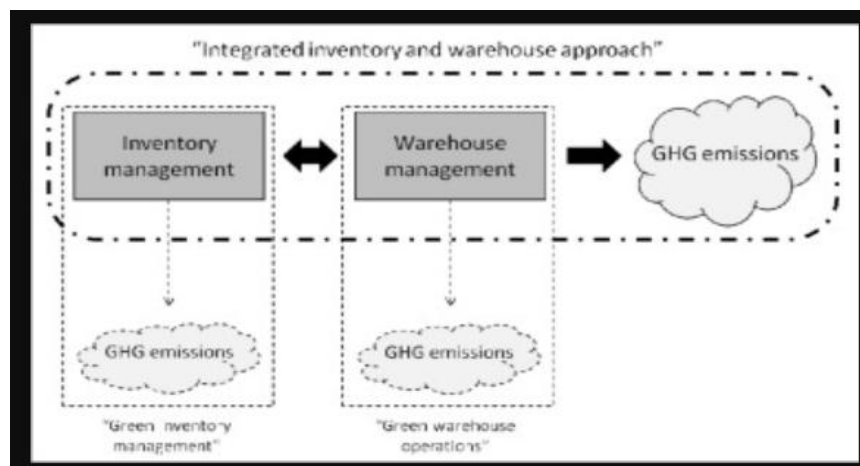


Fig. 2.2.2

Image courtesy: [ScienceDirect](#)

Warehouse practices can significantly impact the environment through various operational activities and resource consumption. Some key environmental impacts of warehouse practices include:

➤ **Energy Consumption:**

Electricity Usage: Warehouses consume substantial amounts of electricity for lighting, heating, cooling, and operating machinery, contributing to overall energy consumption and carbon emissions.

Non-renewable Sources: Often, warehouses rely on energy derived from non-renewable sources, further exacerbating environmental impact.

➤ **Water Usage:**

Cooling and Cleaning: Water is used for cooling systems and cleaning processes within warehouses, contributing to water usage and potential water waste.

➤ **Waste Generation:**

Packaging Waste: Warehouse operations often generate packaging waste from incoming shipments or during storage, contributing to landfill waste and environmental pollution.

Obsolete Inventory: Inefficient inventory management can lead to the disposal of obsolete or expired products, contributing to waste generation.

➤ **Air Pollution:**

Vehicle Emissions: Operations involving trucks, forklifts, and other machinery within warehouses emit pollutants contributing to air pollution, especially if they run on fossil fuels.

➤ **Habitat Disruption:**

Land Use: Expansion or construction of warehouses may require land clearing, potentially leading to habitat disruption and ecosystem degradation.

➤ **Lighting and Emissions:**

Light Pollution: Excessive lighting in and around warehouses can contribute to light pollution, impacting local ecosystems and wildlife.

➤ **Emission of Harmful Substances:**

Chemical Usage: Some warehouse operations involve the use of chemicals for cleaning, pest control, or preserving goods, potentially leading to soil or water contamination if not managed properly.

To mitigate these environmental impacts, warehouses can adopt several sustainable practices:

Energy Efficiency Measures: Implementing energy-efficient lighting, heating, and cooling systems, as well as utilizing renewable energy sources, can reduce energy consumption.

Water Conservation: Adopting water-saving technologies and practices for cooling systems and cleaning processes can minimize water usage.

Waste Management: Implementing recycling programs, reducing packaging waste, and proper disposal or repurposing of obsolete inventory can reduce landfill waste.

Green Infrastructure: Designing warehouses with eco-friendly materials and incorporating green spaces can mitigate environmental impact.

Emission Reduction Strategies: Using electric or low-emission vehicles within warehouse premises and implementing emission reduction strategies can minimize air pollution.

By integrating sustainable practices into warehouse operations, businesses can reduce their environmental footprint while maintaining efficient logistics operations.

https://logingreen.usal.es/wp-content/uploads/LOG-IN-GREEN_Green-Logistics-Book_EN.pdf

➤ Environmental impact of transportation/ Shipping

The transport sector, including all modes, accounts for about 25% of global CO₂ emissions, with this share being around 28% for advanced economies such as the United States.

End-to-end shipping is a complex process, included is the decision on what modes of transport that will be used for a shipment. Several factors are included in this decision-making process, such as the type of cargo, shipping location, and destination, costs, but with increasing regulation around sustainable transports, carbon footprint is a growing determining factor. Though CO₂-emissions is the measure most commonly cited, it is not the only environmental impact of transport.

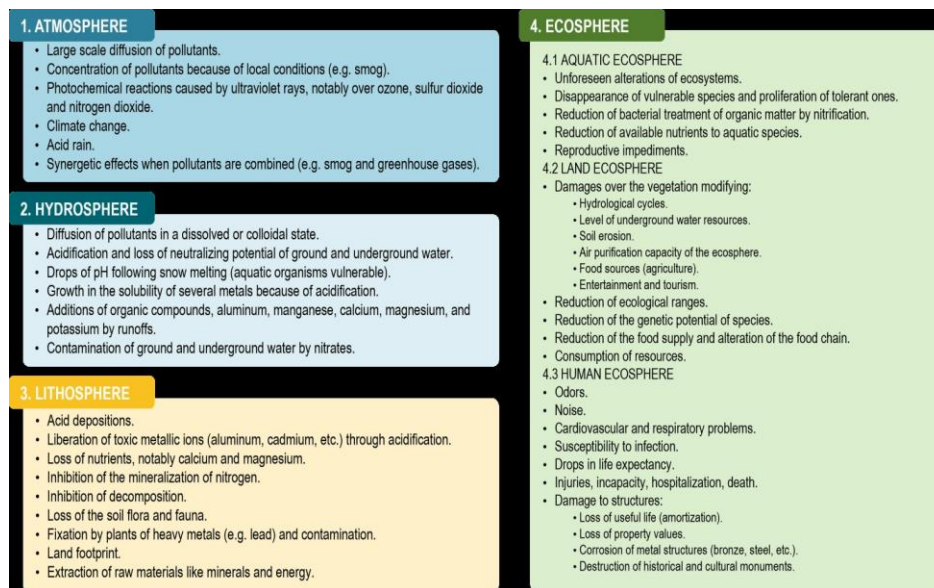


Fig. 2.2.2

Source: [*The Geography of Transport Systems*](#)

Certainly, each mode of logistics transportation bears its unique environmental impacts:

Air transportation environmental impact

Air cargo shipments offer the smallest load capacity compared to the other alternatives. Aircraft also release the highest number of CO₂. This results in the highest levels of CO₂-emissions per load transported, on average 500 grams of CO₂ per metric ton of freight per km of transportation.

Air transports largely contribute to air pollution, but also incidental water, soil pollution, noise pollution, and waste generation with high amounts of dangerous waste. Air pollution causes damage to humans, the World Health Organization estimates 4.2 million people die yearly from outdoor air pollution, but it also gravely affects our biosphere, soil, water, and buildings.

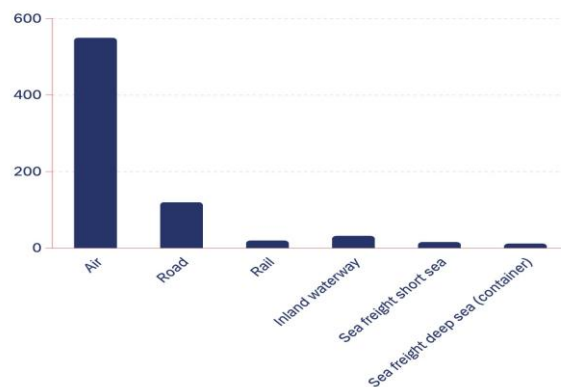


Fig. 2.2.2

Source: [rouvia](#) (CO₂ emissions in grams per metric ton of freight per km of transportation)

Road freight environmental impact

Road transport relies on fossil fuels and produces fine particles that pollute the outside air, which is dangerous to humans and ecosystems.

Emissions: Road logistics heavily rely on trucks, vans, and other vehicles powered by fossil fuels, emitting CO₂, NO_x, PM, and VOCs, contributing to air pollution and climate change.

A truck emits 60 to 150g of CO₂ per ton and a km of transportation. But it is not only about the truck itself, but contributing to road traffic congestion, which increases fuel consumption of the affected cars and trucks and in return, also CO₂ emissions and air pollution. According to a European Commission Statistical Pocket 2021 report, road freight transport accounted for 27.2 % of greenhouse gas emissions in 2019.

Congestion Impact: Traffic congestion in logistics leads to increased idling time, exacerbating emissions and air quality issues.

Diesel trucks are known as air pollutants first and foremost, but they also produce notable noise pollution. Noise pollution is not only bothering people, but also local wildlife. Studies show noise pollution makes it harder for animals to survive because it affects their ability to use sound for navigation, find food, mate, and avoid predators.

Rail freight environmental impact

Lower Emissions: Rail logistics generally emit fewer greenhouse gases per ton-mile compared to road transport.

The CO₂ emission emitted by rail freight varies depending on whether electric or diesel locomotives are being used. If the rail network is electrified, the CO₂ emissions are lower, and there is potential for a green energy source. On average, a modern freight train emits 30 to 100g of CO₂ per ton and a km of transportation.

Noise pollution and the vibrations from the moving train are of the bigger effects of rail freight transport, especially if high-speed rail is in use. Train lines often run through both countryside and communities that

are disturbed greatly by the loud noise that ensues. Noise pollution can lead to habitat loss if the local wildlife no longer uses the area on the other side of the tracks, and behavioural changes if the noise is perceived as a risk.

Building new railways is necessary for rail transport to be able to manage higher volumes of cargo and reach more destinations. Expanding the railway network does cause both physical and behavioural barriers to wildlife, like habitat fragmentation.

Energy Efficiency: Trains are more energy-efficient for cargo transport, contributing less to resource depletion and emissions.

Rail freight is still more efficient in terms of tons of goods moved per liter of fuel compared to air and road freight options. Especially comparing to road freight, rail freight offers more efficient land use, less air pollution, and less congestion and wear and tear on roads.

Water transportation environmental impact

Both cargo ships and barges can hold very large volumes of goods. One barge holds approximately 15 times more than one rail car and 60 times more than one semi-trailer. They also require less fuel to transport their cargo. This means that they are most efficient in terms of cargo transported per litre of fuel.

Ballast Water and Spills: Maritime logistics pose risks of oil spills and introduce invasive species through ballast water, impacting marine ecosystems.

Ballast water is water brought on to manage the ship's weight and level. This is a very old practice to steady the ship, but it poses a real environmental threat. Releasing ballast water that has been collected in a very different environment carries many risks for the destination port's microbiome. In the ballast water, there could be foreign bacteria, microbes, organisms, larvae, etc., of various species that may now become invasive species and cause significant damage to the local ecosystem.

To battle the ballast water problem, many countries have adopted regulations on how to treat ballast water before disposal. One type of treatment is chemical, to kill the organisms effectively and quickly strong chemicals are used, but they may create a toxic byproduct that is released instead. The best option environmentally would be to treat the ballast water using a mechanical treatment or using UV radiation to kill the organisms. Another possibility is to look at ships that do not release ballast water all by using alternative systems.

Sinking ships and oil spill disasters can have irreversible consequences on the ecosystem of oceans and sea coasts. A smaller oil spill can happen when a ship re-fuels, but even a smaller spill may still cause serious environmental harm by contaminating water and soil. Larger oil spills may also cause explosions and fires, harm wildlife, and contaminate seafood at large.

When ships are abandoned in the ocean or sink, they inevitably impact the surrounding environment. When large vessels sink, they scrape the seafloor, causing damage to a sizeable ocean habitat. Additional environmental damage is caused if the vessel is then left abandoned, as metals, chemicals, and oil will leak into the water.

Emissions:

Looking at CO₂-emissions, container ships emit on average 10 to 40 grams of CO₂ per km. The European Commission established that international shipping produces less greenhouse gases for every ton transported per km than road or air transport.

However, when it comes to container ships the main problems relate to their release of emissions either into or directly over the ocean, and those can contribute directly to ocean acidification.

The environmental impact of inland waterways transportation

Transportation via inland waterways has been on the decline since the modernization of trucks due to its lack of flexibility in routes and unreliable conditions. However, the known environmental impact, in terms of for example noise pollution, is low.

Traditional barges are without engines or motors, meaning the barge itself did not use any fuel for transportation; however, the tug vessel would, of course, require fuel. These days self-propelling barges have become increasingly common, meaning they would consume fuel, but they are still very fuel-efficient and can move cargo more than four times as far as trucks on the same amount of fuel.

So unfortunately, there is no one golden solution to more sustainable transport as there is not one option that will suit all goods and all transport needs. However, integrating rail or barge more efficiently in freight transports can lead to 60-90% reduction in CO₂-emissions compared to a truck only option.

The best way forward is to look at intermodal transportation, combining different modes, to find the most efficient route at a reasonable cost

2.2.3 Components of green supply chain performance metrics

Green supply chain performance refers to the measurement, evaluation, and optimization of supply chain activities with a focus on environmental sustainability. It involves assessing the environmental impact of supply chain operations, implementing strategies to reduce that impact, and achieving sustainable outcomes while maintaining or improving overall supply chain performance. Green supply chain performance aims to integrate environmental considerations into all aspects of supply chain management, from sourcing raw materials to delivering products to customers, with the goal of minimizing environmental harm and promoting long-term sustainability.

Below are 25 key components of metrics to measure supply chain performance and management which helps an organization, work toward a bright and prosperous future.

1. Cash-to-Cash Time Cycle

This priceless supply chain metric calculates the length of time required to transform the resources into Bonafide cash flows. Working with three core ratios - the days of inventory (DOI), the days of payables (DOP), and the days of receivables (DOR) - the cash-to-cash time cycle KPI visualizes the period required between the moment a business pays cash to its suppliers and the moment it receives cash from its customers. The shorter the conversion cycle the better.

(Cash-to-cash cycle time = receivable days + inventory days – payable days)

2. Freight Bill Accuracy

Freight bill accuracy measures the precision of your orders and shipping bills. Shipping the wrong items or the wrong amounts is wasteful and lowers customer satisfaction.

formula to calculate freight bill accuracy:

Freight bill accuracy = (number of correct freight bills / total freight bills) x 100

3. Perfect order index or perfect order delivery rate:

The perfect order index measures the percentage of your orders that are error-free from beginning to end. That means the order was recorded correctly, shipped on time and in the right quantities and arrived without damage.

Perfect order = [(total orders – errors) / total orders] x 100

4. On-time delivery:

The percentage of orders that arrive as scheduled.

On-time delivery = [(total orders – orders that do not arrive on time) / total orders] x 100

5. In-full delivery:

The percentage of sales orders that are delivered completely in the first shipment

In-full delivery = [(total orders – orders that aren't complete or are incorrect in first shipment) / total orders] x 100

6. Damage-free delivery:

The percentage of orders that are delivered without any damages

Damage-free delivery = [(total orders – orders that arrive damaged) / total orders] x 100

7. Accurately documented order:

Percentage of orders in which all documents relating to the order are accurate

Accurately documented order = [(total orders – orders without accurate documentation) / total orders] x 100

Businesses can track their perfect order indexes by using the simple formulas above. But companies may also measure the index by multiplying their perfect order rates in various categories. For example, say you have a 92% rate of on-time delivery, a 95% rate for in-full delivery, a 97% rate in damage-free delivery and a 98% rate in accurately documented delivery.

8. Gross margin return on investment (GMROI):

The gross margin returns on investment measures how much money a company makes on a specific inventory investment.

Gross margin return on investment = gross profit / [(opening inventory in the period – closing inventory in the period) / 2] x 100

9. Total supply chain management cost as percentage of sales:

Total supply chain management cost as percentage of sales is fairly self-descriptive. It measures the total cost of your supply chain operations compared with your overall sales.

Total supply chain management cost as percentage of sales = (total supply chain costs / total sales) x 100

10. Supply chain cost per unit sold:

Supply chain cost per unit sold measures your supply chain costs compared with how many of a given item your company sells.

Supply chain cost per unit sold = supply chain costs for a product over period / number of units sold in that period

11. Day sales outstanding:

Day sales outstanding measures how quickly you collect revenue from your customers. A low day sales outstanding number means you are generating revenue more quickly, which ultimately improves your cash flow.

Day sales outstanding = (receivables / sales) x number of days in a period

12. Inventory days of supply (IDS):

IDS represents the number of days it would take a company to run out of inventory if it didn't add to its supply.

Inventory days of supply helps your company understand and maintain its par level

Inventory days of supply = (average inventory in a month, in dollars / monthly product demand, in dollars) x 30

13. Days sales of inventory (DSI):

Days sales of inventory calculates the average number of days that inventory remained in stock over a certain period. It's a measurement of how long it takes a business to sell the items it makes or buys.

Days sales of inventory = (ending inventory / cost of goods sold) x number of days in period

14. Inventory-to-sales ratio (ISR):

The inventory-to-sales ratio compares the inventory you carry to your overall sales. It's a measurement of the financial stability of your company. The ratio is closely related to your inventory turnover ratio.

Inventory-to-sales ratio = inventory value in dollars / sales value in dollars

15. Inventory turnover ratio (ITR):

The inventory turnover ratio measures how often a company's entire inventory is sold in a specific period. What comprises a "good" inventory turnover ratio depends on the industry. But in general, a lower inventory turnover ratio means a company may have excess inventory due to lagging sales.

Inventory turnover ratio = cost of goods sold in period / [(opening stock in period – closing stock in period)/2]

16. Turn-earn index (TEI):

The turn-earn index combines inventory turnover and gross margin to evaluate your company's profits and use of inventory. In essence, the index recognizes that your company can get by with less inventory turnover if it makes a lot of money on that inventory. But it needs more inventory turnover if it realizes a lower margin on the inventory. Generally, most businesses will want a turn-earn index of at least 150.

Turn-earn index = (inventory turnover ratio x gross profit percentage) x 100

17. Inventory velocity (IV):

Inventory velocity is the portion of inventory that your company projects it will exhaust within the next specified period. The metric helps your company set optimum inventory levels so that you don't carry too much inventory but retain enough to satisfy projected sales in the coming period.

Inventory velocity = opening stock / upcoming period's sales forecast

18. Months on hand:

Months on hand indicates how many months of inventory you have at your disposal if you purchase no more stock while your sales continue as forecast.

Months on hand = (average inventory for year / cost of goods sold for year) x 12

19. Stock rotations:

Stock rotations, also called *stock life* or *stock coverage*, refers to the number of days, on average, that it takes for you to run out of your inventory stock. It is best to use the past 52 weeks to figure this calculation. That allows for enough time to show seasonal and other variations

Stock rotations = (average stock in a period / total sales in a period) x number of days in the period

20. Fill rate:

Fill rate is the percentage of customers' orders that are filled on their first shipments—an important contributor to customer satisfaction. It's also a measure of your company's efficiency.

It can also track specific aspects of fill rate, including:

Order fill rate: The percentage of orders delivered on the first attempt.

Line fill rate: The percentage of order lines delivered on the first attempt.

Unit fill rate: The percentage of individual items delivered on the first attempt.

21. Freight cost per unit:

Freight cost per unit measures how economically you ship your products. It measures your total freight costs divided by how many units you've shipped

Freight cost per unit = total freight cost / number of units

Fill rate = $(1 - [(total\ items - shipped\ items) / total\ items]) \times 100$

22. On-time shipping rate:

The on-time shipping rate is the percentage of times your customer receives product within the promised shipping window. Tracking this metric helps assess the efficiency of your supply chain processes.

On-time shipping rate = (number of items delivered on time in a period / total items shipped in the period) x 100

23. Average payment period for production materials:

The average payment period for production materials is the average timespan between when you receive materials and when you pay for them. By tracking this metric you can favour suppliers that offer more favourable billing terms and thus improve your cash flow.

Average payment period for production materials = (materials payables / total cost of materials) x days in period

24. Supplier on-time delivery:

Supplier on-time delivery measures the percentage of time your suppliers deliver products to you in the agreed-on timeframe.

Supplier on-time delivery = (number of products supplier delivers on time in a period / total items supplier ships in the period) x 100

Return reason:

The return reason measurement shows the top issues that cause customers to send items back. This can help you assess weaknesses in your product lines or operations and is important to track given the high cost of reverse logistics operations.

25. Order to cash:

While order to cash covers some processes outside of the supply chain, the performance of the supply chain is an important component. Order to cash is a measure of the company's entire system to receive, process and complete orders. It covers the time from when a customer places an order to when the company receives and records the customer's payment.

Order to cash = date of customer order – date of receipt and recording of customer payment for order, after receiving order

Unit 2.3 Introduction to Carbon Footprint

Unit objectives

After reading this unit, you will be able to understand the following points:

1. Meaning of carbon footprint
2. Components of measuring carbon footprint of a shipping/ transport agency

2.3.1 Meaning of carbon footprint

The carbon footprint is defined as the total trace of greenhouse gases produced by daily and economic activities that people carry out. This factor is part of the environmental footprint and is expressed in tonnes emitted during a given period of time (hours, days, weeks, months, years).

Since 1961, the human carbon footprint has increased eleven-fold. It now accounts for 60% of humanity's total impact on the environment.

However, it is not only people and companies that leave a carbon footprint. Services and products do too, as they emit greenhouse gases before (during the extraction of raw materials, their manufacture and transport), during (e.g. when driving a car) and after the end of its useful life (when it needs to be recycled or disposed of).

Carbon footprint calculation is based on two variables: activity and emission factor:

Activity data: Its function is to define the volume of CO₂ emissions generated by the activity. This would be the case, for example, of the kWh of natural gas used for heating.

ACTIVITY VARIABLE = ANY ACTIVITY THAT GENERATES EMISSIONS

The value of the activity can be stated, for example, in kilograms, tonne-kilometres or kilowatt-hours.

Emission factor: This represents the amount of greenhouse gases emitted by each unit.

EMISSION FACTOR = ACTIVITY INTENSITY

The amount of greenhouse gas emissions is expressed in carbon dioxide equivalents, i.e. as a total measure of emissions, which can be used to add up the effects of different greenhouse gases.

For example:

A corporate carbon footprint is calculated on the basis of measured or estimated consumption data (natural gas, oil, biomass, electricity, fuel, paper, materials, etc.) and emissions factors known as CO₂ equivalents. These consist of CO₂ (the most widely emitted greenhouse gas by volume) as well methane and nitrous oxide, which are converted into CO₂ equivalents. Calculating a corporate carbon footprint is not just a one-time task. Instead, this data is updated every year with the goal of progressively reducing emissions through emission abatement measures, with constant monitoring of actual emissions. The entire process of calculating a carbon footprint is a cycle of continuous improvement, making it an optimal tool for long-term management of operational climate protection measures.

A corporate carbon footprint is composed of all direct and indirect emissions generated by the entire organization. It therefore requires comprehensive recording and analysis of the company's emissions within clearly defined system boundaries. The analysis calculates emissions using available data such as consumption figures (e.g. annual electricity consumption, annual fleet mileage) and allocates them to organizational units (e.g. headquarters, production sites, external service providers), processes, products and services. The foundation for carbon footprint calculations is the international ISO 14064 standard, while the three scopes of the GHG Protocol are used to make distinctions between different emissions sources:

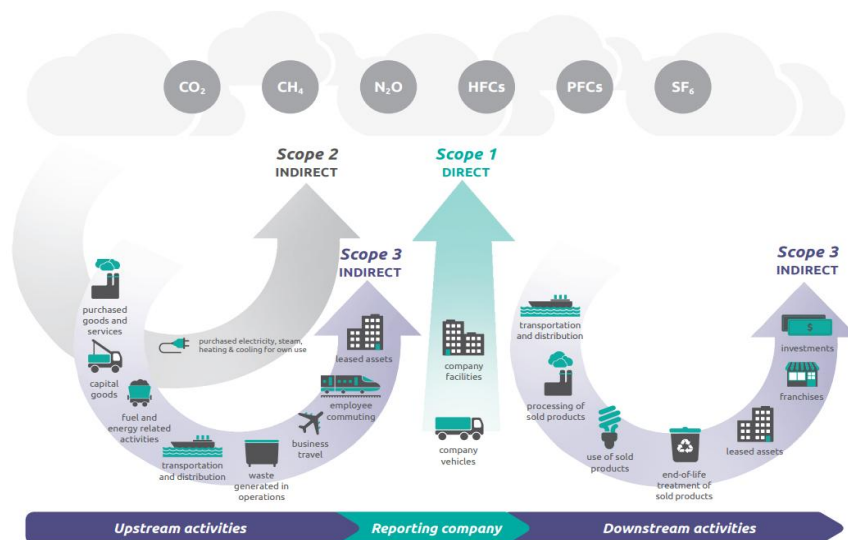


Fig. 2.3.1

Source: <https://www.ghgprotocol.org/>

The 3 scopes in the GHG Protocol

✓ Scope 1 emissions (direct):

These are emissions that occur from sources within the examined system boundaries. This includes emissions associated with the company's consumption of energy (e.g. gas) or vehicle fuels (diesel), as well as emissions generated by buildings and plant.

✓ Scope 2 emissions (indirect, upstream):

These are emissions that occur in the generation of energy purchased by the company, usually in the form of electricity or heating from energy service providers (e.g. district heating).

✓ Scope 3 emissions (indirect, upstream and downstream):

These are all gray emissions that are caused by the company's activities but are not within the company's control, such as production materials, consumable materials, packaging, waste disposal, logistics and employee commuting. These emissions occur in both, the upstream and downstream of the company.

Fig. 2.3.1

Source: [Quentic](#)

2.3.2 Components of measuring carbon footprint of a shipping/ transport agency

Measuring the carbon footprint of transportation involves assessing various components that contribute to greenhouse gas emissions within the transportation sector. Here are the key components typically considered when measuring the carbon footprint of transportation:

1. Vehicle Operations:

Fuel Consumption: Assess emissions from burning fossil fuels, such as gasoline, diesel, or aviation fuel, in various modes of transportation, including cars, trucks, airplanes, ships, or trains.

Engine Efficiency: Consider the efficiency of engines and propulsion systems, which affects fuel consumption and emissions.

2. Transport Distance and Routes:

Distance Travelled: Measure emissions based on the distance covered by vehicles or modes of transport, considering different routes and travel distances.

Emission Control Areas (ECAs): Account for emissions in designated ECAs or areas with stricter emission regulations.

3. Vehicle Types and Technology:

Vehicle Type: Consider the type of vehicles used, such as heavy-duty trucks, buses, airplanes, ships, or trains, as each has different emission profiles.

Technology and Emission Standards: Analyse the impact of vehicle technology, age, and compliance with emission standards on emissions

4. Fuel Types and Efficiency:

Fuel Quality: Assess emissions based on the type and quality of fuel used, including conventional fuels, biofuels, hydrogen, or electric power.

Fuel Efficiency: Consider the efficiency of vehicles or modes of transport, including factors like aerodynamics, weight, tire pressure, and driving behaviour.

5. Infrastructure and Traffic Management:

Traffic Congestion: Account for emissions resulting from traffic congestion and idling, which affect fuel consumption and emissions in urban areas.

Infrastructure Impact: Assess emissions associated with infrastructure development, maintenance, and improvements in the transportation sector.

6. Intermodal Transport:

Last-Mile Transport: Consider emissions from onshore transportation, such as trucks or trains, for the transport of goods from ports or terminals.

7. Reporting and Analysis:

Data Collection and Emissions Calculation: Gather comprehensive data on fuel consumption, travel distances, vehicle specifications, and use appropriate emission factors to calculate the carbon footprint.

Carbon Footprint Calculation: Utilize standardized methodologies or specific carbon accounting tools designed for transportation emissions.

Carbon Footprint Reporting: Present the findings in a detailed report outlining emission sources, mode-specific emissions, and opportunities for emission reduction strategies.

By evaluating these components within transportation activities, stakeholders can understand the emissions associated with different modes of transport, identify areas for improvement, and implement targeted strategies to reduce the carbon footprint of transportation operations.

Exercise

1. Explain the meaning of Green Logistics?

5. List down the sustainable logistics practices?

3. Identify the areas of difference between conventional and green logistics?

4. What is green supply chain management?

5. List the components of green supply chain performance metrics?

6. What is carbon footprint?

7. Explain the components to measure carbon footprint of a shipping/ transport agency?

8. What are the steps required for greening the logistics sector?

3. Steps required for greening the logistics sector



Unit 3.1 Best practices and strategies to make supply chain management green

Unit 3.2 Implementation of practices under energy/ electricity/ material/ water conservation

Unit 3.3 The use of renewable energy in the logistics sector

Unit 3.4 Introduction to circular economy in different areas

Unit 3.5 Implementation of efficient green logistics



Chapter Highlights



After reading this chapter, you will be able to: %

1. Know about the green practices and Strategies to Green the Supply Chain Management
2. Know about Sustainable materials in the sector
3. Identify how to Streamline inventory management for a greener supply chain
4. Understand about Resource Conservation (e.g., material, energy, water usage)
5. Identify how to Implement Energy/ Electricity/material/water Conservation Practices
6. Identify the use of Renewable Energy / solar energy in the sector
7. Identify how to manage different types of Waste effectively
8. Understand how to review internal operations with the point of energy, material and waste management
9. Understand the concept of circular economy and where it can be introduced
10. Know about the concept of green logistics and how efficiently it can be implemented
11. Understand the significance of optimizing logistics operations for sustainability

Unit 3.1 Best practices and strategies to make supply chain management green

Unit objectives



After reading this unit, you will be able to understand the following points:

1. Introduction to green supply chain management
2. Best practices to make supply chain green
3. Strategies to make supply chain green
4. Sustainable materials in the sector
5. Streamline inventory management for greener supply chain
6. Suggestions to make work place operations/ activities environment friendly

3.1.1 Introduction to green supply chain management

Supply chain management (SCM) is the end-to-end coordination and optimization of the flow of goods, information, and finances across a network of interconnected entities involved in producing and delivering products or services to end customers. It encompasses the planning, execution, control, and monitoring of all supply chain activities.



Fig. 3.1.1

Typical supply chain (Source: [Studentproject](#))

<https://www.mbaknol.com/logistics-management/introduction-to-supply-chain-management-scm/>

What is green supply chain management?

Green Supply Chain Management (GSCM) is an approach that integrates environmental considerations into every stage of the supply chain's lifecycle. It aims to minimize the environmental impact of product sourcing, manufacturing, transportation, and distribution while maximizing efficiency and value throughout the process.

The term 'Green supply chain management' (GSCM) refers to the concept of integrating sustainable environmental processes into the traditional supply chain. This can include processes such as product design, material sourcing and selection, manufacturing and production, operation and end-of-life management.



Fig. 3.1.1

Source: [Linkedin](#)

Benefits of Creating a Green Supply Chain

Apart from impacting the environment positively, a green supply chain has a lot of benefits for a business:



Fig. 3.1.1

Source: [edureka](#)

https://www.designingbuildings.co.uk/wiki/Green_supply_chain_management, [edureka](#)

- **Reduced Greenhouse Gas (GHG) Emissions**

Greenhouse gases like carbon dioxide (CO₂) and methane (CH₄) are released when fossil fuels are burned. Green SCM can help reduce these gases' emissions by promoting the use of green energy sources, green transportation methods, and energy-efficient manufacturing processes. Regularly adopting green SCM practices makes it possible to achieve a significant reduction in GHG emissions over time, which will help mitigate climate change.

- **Improved Environmental Sustainability**

Green SCM practices can help to improve the environmental sustainability of a company's operations. It is because green SCM practices often lead to reduced resource consumption, waste generation, and improved energy efficiency.

- **Decreased Energy Consumption**

It is evident that green SCM practices would lead to decreased energy consumption. It is due to the fact that green practices often revolve around being more efficient with resources. In other words, green practices aim to do more with less energy consumption.

- **Improved Resource Efficiency**

It goes without saying that greening your supply chain will make it more resource-efficient. After all, one of the main goals of green operations is to minimise waste and optimise processes. By working with green supply chain partners, you can be sure they are also committed to resource efficiency and waste reduction.

- **Increased Competitiveness**

In today's business environment, being green is no longer just an option – it's a necessity. A green supply chain can give the company a competitive edge by helping to improve its bottom line and increase its customer base. The business that is seen as green and sustainable are the ones that will thrive in the future.

- **Reduced Waste and Pollution**

Every business should look for ways to cut waste and pollution, and green supply chain management can help. Working with green suppliers can reduce the amount of waste produced throughout the supply chain. It is not only beneficial for the environment but can also save the business money.

- **Strengthened Supply Continuity**

Many green initiatives improve supply chain efficiency and reduce waste, resulting in increased reliability and cost savings. For example, a company might choose to source locally produced materials to reduce transportation costs and emissions. It will increase the continuation of the supply chain without making any big changes.

- **Enhanced Customer Satisfaction and Loyalty**

In many cases, green initiatives can lead to increased customer satisfaction. For example, a product manufactured using sustainable practices can be seen as more trustworthy and of higher quality. It can lead to increased customer loyalty and repeat business.

- **Better Public Relations**

Going green can also improve a company's public image and reputation. It is especially true if the company is in a highly-polluting industry. Implementing green initiatives can show the public that the company is committed to reducing its environmental impact. As a result, companies may find it easier to attract and retain customers.

- **Improved Employee Morale and Motivation**

Not just customer satisfaction but green SCM is also linked with employees' morale. Increased motivation has a direct link to employee productivity. And green SCM policies can increase employee motivation by providing opportunities for employees to work on green initiatives and be recognised for their efforts.

- **Compliance With Environmental Regulations**

Many countries have environmental regulations in place that businesses must comply with. It can be a challenge for companies, primarily if they are operating in multiple countries with different regulations. Green SCM can help companies meet these requirements while reducing their environmental impact. With green SCM, businesses can make sure they use environmentally-friendly practices throughout their supply chains.

- **Contribution to CSR**

Green SCM is not only good for the environment but also for a company's reputation. It can contribute to a company's CSR (Corporate Social Responsibility) initiatives. A green supply chain can help a company to be perceived as environmentally responsible and committed to sustainable practices

3.1.2 Best practices to make supply chain management green

Green supply chains use ethical and environmentally sound practices at every stage, with the goal of reducing air, water, and waste pollution. Below are key best practices:

- **Green purchasing:**

Green purchasing, which means finding suppliers with environmentally sustainable products and services, is just as important as greening your own operations

➤ **Green manufacturing:**

Green manufacturing focuses on using fewer non-renewable natural resources, reducing pollution and waste, and keeping emissions to a minimum, among other green practices. The most critical component of going green at this stage is reducing energy use.

➤ **Green packaging:**

Green packaging considers every phase of a package's life cycle. That includes everything from how the supplier sources materials to how consumers dispose of the packaging. Using boxes and packing materials made of postconsumer recycled materials is a good start.

➤ **Green warehousing:**

Green warehouses are facilities that prioritize environmentally sustainable and eco-friendly practices in their design, construction, and operations within logistics and supply chain management. These warehouses aim to minimize their ecological footprint by implementing strategies that reduce energy consumption, decrease waste generation, and promote overall environmental responsibility.

➤ **Green transportation:**

Greening transportation refers to the methods a business adopts to ensure that its vehicles and other modes of transportation are environmentally friendly. To go green, transportation choices made by businesses range from replacing their existing fleet with electric vehicles and introducing biofuels to using technology to optimise routes. Green transportation comes under the larger purview of green logistics that extends to other carbon-neutral practices - such as eco-conscious procurement, investing in green technology, optimising energy consumption, and other measures implemented - to reduce the ecological impact of storage, transportation, and distribution of goods and services.

➤ **Energy Efficiency and Waste Reduction:**

Efficient Operations: Embrace lean manufacturing techniques to minimize waste, reduce energy consumption, and improve resource efficiency.

Waste Management: Implement recycling programs and circular economy practices to minimize waste generation and promote resource recovery.

➤ **Eco-friendly Product Design:**

Design for Environment (DfE): Incorporate eco-design principles, creating products that minimize environmental impact throughout their lifecycle.

Materials Selection: Opt for recyclable, biodegradable, or renewable materials to reduce the ecological footprint of products.

➤ **Lifecycle Assessments and Reporting:**

Environmental Impact Assessment: Conduct regular lifecycle assessments to measure the environmental footprint of products and processes.

Reporting and Certification: Publish sustainability reports, adhere to environmental certifications, and showcase commitment to eco-friendly practices.

[Planettogether](https://www.planettogether.com/en/blog/what-are-green-supply-chain-practices-your-guide-to-supply-chain-sustainability/), <https://www.agility.com/en/blog/what-are-green-supply-chain-practices-your-guide-to-supply-chain-sustainability/>

3.1.3 Strategies to make supply chain management green

1. **Remove Waste** - Ensure that all of your resources and materials are used at peak efficiency through end-to-end visibility. Poor planning can lead to having excess materials and waste that has to be discarded.
2. **Reduce Packaging Materials** - Re-designing your packaging to use less material can lower the costs associated with packaging and transport as the total weight of the product is reduced. In addition, designing packaging that can be recycled or returned can further reduce the costs associated with packaging.
3. **Maintain Compliance** - Ensure that your supply chain meets environmental regulations for sourcing, clean air, and emissions.
4. **Source From Sustainable Suppliers** - Collaborate with all levels of the company to find suppliers that have long-term sustainability goals that are in line with your company's goals.
5. **Reduce Inventory** - Eliminate costly stock-outs and wasteful overstocks by precise prediction and balancing your supply and demand. Excess inventory can be costly to maintain and may become obsolete if products are not sold.
6. **Cut Carbon Emissions** - By identifying the most efficient route and delivery options, you can reduce the carbon footprint of the company. In addition, shortening the supply chain by sourcing raw materials closer to the end customer can further reduce carbon emissions associated with transportation.

3.1.4 Sustainable materials in the sector

Sustainable materials in a green supply chain refer to resources and substances that are sourced, produced, and utilized in a manner that minimizes environmental impact while promoting ecological balance and social responsibility. These materials are chosen for their reduced carbon footprint, renewable nature, recyclability, and overall eco-friendly characteristics. Here are examples of sustainable materials commonly utilized in green supply chains:

➤ **Recycled Materials:**

Recycled Plastics: Post-consumer or post-industrial recycled plastics reduce the need for new plastic production, lowering resource consumption and waste.

Recycled Metals: Utilizing recycled aluminium, steel, or copper reduces mining activities and energy-intensive extraction processes.

➤ **Bio-based Materials:**

Bioplastics: Derived from renewable sources like cornstarch, sugarcane, or algae, bioplastics decompose naturally, reducing reliance on fossil fuels.

Bio-based Fibers: Fabrics made from materials like organic cotton, hemp, or bamboo offer alternatives to traditional, resource-intensive textiles.

➤ **Sustainable Wood and Paper:**

FSC-Certified Wood: Wood and wood products certified by the Forest Stewardship Council (FSC) ensure responsible forest management and conservation practices.

Recycled Paper: Using recycled paper reduces deforestation and promotes a circular economy for paper products.

➤ **Natural and Renewable Materials:**

Cork: Renewable and biodegradable, cork is used in packaging, flooring, and insulation due to its sustainability and versatility.

Bamboo: Fast-growing, renewable, and versatile, bamboo is used in various applications from construction to textiles.

➤ **Eco-friendly Textiles:**

Organic Cotton: Grown without synthetic pesticides or fertilizers, organic cotton minimizes environmental impact and reduces health risks for workers.

Recycled Polyester: Repurposed polyester from PET bottles or textile waste reduces the need for new polyester production.

➤ **Green Building Materials:**

Low VOC Paints: Volatile organic compounds (VOCs) in paints can harm air quality; low VOC alternatives reduce environmental impact.

Sustainable Insulation: Materials like recycled denim or cellulose insulation offer eco-friendly alternatives for construction.

➤ **Renewable Plastics:**

PHA (Polyhydroxyalkanoates): Biodegradable plastics derived from renewable resources like plant oils or sugar. They break down in various environments, reducing plastic pollution.

PLA (Polylactic Acid): Bioplastic made from fermented plant starch (usually corn). It's compostable and offers an alternative to traditional plastics.

➤ **Green Concrete Alternatives:**

Fly Ash Concrete: Using fly ash, a byproduct of coal combustion, as a substitute for cement in concrete production reduces CO2 emissions and waste.

Recycled Aggregates: Crushed concrete or recycled aggregates from construction waste used in concrete, reducing the need for new natural resources.

➤ **Sustainable Packaging:**

Mycelium Packaging: Using mycelium (mushroom roots) as a biodegradable and compostable alternative to traditional packaging materials.

Edible Packaging: Creating packaging from edible materials like seaweed or starches, reducing waste and environmental impact.

3.1.5 Streamline inventory management for greener supply chain

Efficient inventory management plays a vital role in creating a sustainable and environmentally conscious supply chain. By optimizing inventory practices, businesses can minimize waste, reduce costs, and contribute to a greener future. Some practical tips to streamline inventory management and foster a more sustainable supply chain are:

1. Demand forecasting

Accurate demand forecasting techniques are key to aligning inventory levels with customer needs. By analyzing historical data, market trends, and customer insights, businesses can optimize inventory levels and minimize the risk of overstocking.

2. Embracing the just-in-time (JIT) approach

Adopting the JIT approach helps reduce excess inventory and minimize waste. With JIT, stock is replenished only when needed, ensuring inventory is available at the right time while avoiding unnecessary stock accumulation.

3. Supplier collaboration

Strong relationships with suppliers are crucial for efficient inventory management. Collaborate closely with suppliers, sharing demand forecasts and optimizing lead times. Improved communication and coordination contribute to inventory availability when required, reducing the need for excess stock.

4. Efficient order management

Implementing efficient order management processes minimizes errors, avoids duplicate orders, and prevents unnecessary stock accumulation. Streamlining order processes reduces waste and optimizes inventory levels, leading to a more sustainable supply chain.

5. Safety stock optimization

Analyze historical data, lead times, and demand variability to optimize safety stock levels. Having enough buffer stock is important, but excessive safety stock can lead to waste. By accurately assessing risk and demand fluctuations, businesses can strike the right balance.

[Linkedin](#)

6. Evaluating just-in-case inventory

Evaluate the necessity of "just-in-case" inventory and explore alternative solutions. Improved quality control, proactive maintenance, or flexible production capabilities may eliminate the need for excessive safety stock, reducing waste and optimizing resources.

7. Inventory tracking and visibility

Leverage technology solutions and real-time data to track inventory levels, monitor expiration dates, and identify slow-moving or obsolete items. Enhanced visibility allows for timely action, reducing waste and minimizing the environmental impact of excess stock.

3.1.6 Suggestions to make work place operations/ activities environment friendly

From a business standpoint, consumers are more inclined to buy products and invest in services from, and lend their support to, companies with a proven record of social responsibility—particularly when it comes to reducing their environmental impact. Research has shown that 80% of consumers are more inclined to support companies and brands with a history of sustainability.

Developing an eco-conscious organization has to start on the ground level, with your workforce.

There are various accessible and cost-effective measures that a company can take to reduce its carbon footprint and environmental impact.

Here are some examples of sustainable initiatives you can implement to encourage an environmentally conscious workplace for your employees, day in and day out.

1. Energy Conservation:

Efficient Lighting: Switch to LED bulbs and install motion sensors to regulate lighting usage.

Appliance Management: Turn off computers, printers, and other devices when not in use to save energy.

Natural Light Utilization: Maximize natural light to reduce reliance on artificial lighting.

2. Waste Reduction and Recycling:

Recycling Program: Implement a comprehensive recycling system for paper, plastics, and other recyclable materials.

Reduce Paper Usage: Encourage digital documentation and printing on both sides of the paper to minimize waste.

Compost Organic Waste: Set up composting bins for food scraps and organic materials.

3. Sustainable Procurement:

Eco-Friendly Supplies: Purchase products made from recycled materials and opt for biodegradable or eco-certified office supplies.

Environmentally Responsible Vendors: Work with suppliers and vendors who follow sustainable practices.

4. Transportation and Commuting:

Promote Green Commuting: Encourage carpooling, biking, or public transportation by offering incentives or flexible work arrangements.

Remote Work Options: Allow telecommuting or remote work to reduce commuting-related emissions.

5. Digitalization and Paperless Operations:

Digital Workflow: Transition to digital document management and encourage e-signatures to minimize paper usage.

Email Communications: Opt for email communications over printed memos or notices to save paper.

6. Water Conservation:

Efficient Fixtures: Install low-flow faucets and toilets to conserve water in restrooms and kitchens.

Raise Awareness: Educate employees about water-saving techniques and encourage responsible usage.

7. Employee Engagement:

Training and Education: Conduct workshops or training sessions on sustainability practices and encourage employee involvement.

Green Teams: Form teams dedicated to implementing and maintaining eco-friendly initiatives.

8. Green Spaces and Initiatives:

Indoor Plants: Introduce plants to improve indoor air quality and create a more pleasant workspace.

Outdoor Green Areas: Create outdoor spaces or gardens to promote connection with nature.

9. Setting and Tracking Goals:

Establish Targets: Set achievable sustainability goals and regularly track progress to stay accountable.

Celebrate Milestones: Recognize and reward achievements in adopting environmentally friendly practices.



Sustainable practices at
workplace

Unit 3.2 Implementation of practices under energy/ electricity/ material/ water conservation

Unit objectives



After reading this unit, you will be able to understand the following points:

1. Meaning of resource conservation (e.g., material, energy, water usage)
2. Implementation of energy/ electricity/ material/ water conservation practices
3. Management of different types of waste
4. Steps in reviewing internal operations with the point of energy, material and waste management

3.2.1 Meaning of resource conservation

Resource conservation refers to the responsible and sustainable management of natural resources to ensure their long-term availability and minimize environmental impact. It involves adopting practices that optimize resource use, reduce waste generation, and protect ecosystems. Resource conservation aims to balance human needs with the preservation of the environment, promoting the efficient utilization of resources for present and future generations. By conserving resources such as water, energy, minerals, and biodiversity, we can mitigate resource depletion, minimize pollution, and promote ecological balance, contributing to a more sustainable and resilient world.

Components of Resource Conservation

Resource conservation involves managing various types of resources.

1. Renewable Resources

Renewable resources are those that can be naturally replenished or regenerated over time.

Examples include:

- Solar energy
- Wind energy
- Water resources
- Forests

Resource conservation for renewable resources focuses on sustainable harvesting and utilization without depleting their availability. Strategies include:

- Promoting the use of renewable energy sources
- Implementing sustainable forestry practices
- Conserving water resources through responsible management

2. Non-Renewable Resources

Non-renewable resources exist in limited quantities and cannot be replenished once they are depleted. Examples include:

- Fossil fuels (coal, oil, natural gas)
- Minerals

Resource conservation for non-renewable resources involves reducing consumption, finding alternative sources, and promoting recycling and reuse. Strategies include:

- Promoting energy efficiency and reducing reliance on fossil fuels
- Exploring and investing in alternative energy sources
- Encouraging recycling and responsible waste management

Conservation in Various Resource Categories:

- **Water Conservation:** Involves reducing water wastage, optimizing usage in agriculture, industry, and households, and safeguarding water sources from pollution and depletion.

Water is a finite resource essential for all forms of life. Conserving water involves reducing waste, improving water-use efficiency, and protecting water quality. Strategies for water conservation include:

- **Water-Efficient Technologies:** Installing water-saving fixtures and appliances, such as low-flow toilets and efficient irrigation systems.
- **Water Recycling:** Treating and reusing wastewater for non-potable purposes, such as irrigation or industrial processes.
- **Awareness and Education:** Promoting water conservation practices among individuals, businesses, and communities.
- **Energy Conservation:** Focuses on reducing energy consumption through efficient technologies, promoting renewable energy sources, and minimizing energy wastage across industries and homes.
- **Material and Waste Management:** Encourages recycling, reusing, and reducing material consumption to minimize waste generation, decrease landfill burden, and conserve resources.

Importance of Resource Conservation:

- **Sustainability:** Promotes long-term viability by ensuring resources are available for future generations.
- **Environmental Protection:** Reduces environmental degradation, pollution, and habitat loss associated with resource extraction and use.

- **Economic Benefits:** Improves efficiency, reduces costs, and enhances competitiveness by minimizing waste and optimizing resource use.
- **Climate Resilience:** Helps mitigate the impact of climate change by reducing greenhouse gas emissions and promoting sustainable practices.

3.2.2 Implementation of energy/ electricity/ material/ water conservation practices

Implementing practices for conservation in energy, electricity, materials, and water within a supply chain involves adopting strategies to reduce consumption, optimize usage, and promote sustainability. Here are specific actions for each conservation area:

Energy Conservation:

- **Efficient Equipment Use:**

Upgrade to energy-efficient machinery, lighting, and appliances across facilities and warehouses to reduce energy consumption.

Implement smart technology and automated systems to optimize energy usage based on real-time demand.

- **Renewable Energy Adoption:**

Invest in renewable energy sources like solar panels or wind turbines to generate clean energy and reduce reliance on non-renewable sources.

- **Energy Audits and Monitoring:**

Conduct regular energy audits to identify areas of inefficiency and implement measures to reduce wastage.

Install energy monitoring systems to track usage patterns and identify opportunities for conservation.

Electricity Conservation:

Timely identification of electrical faults is of paramount importance at workplace. The common types of issues that arise are as follows:

1. **Electrical Surge** - It can damage equipment and needs to be checked soon. It can occur due to some wiring problem, faulty appliances or a damaged power line.
2. **Overloading** – This occurs when the light fixture has a bulb or other fitting with higher watts than the designed fixture. The risk level is quite high and can create smoke and melt the sockets as well as the insulation. It could cause sparks or a fire.
3. **Missing or Defective Residual Current Circuit Breaker (RCCB)**- It is a must to have device as it reduces the risk of shocks and equipment damage. It is also useful for detecting and preventing low voltage circuit if there is a current leak.
4. **Tripping of Circuit Breaker** - If a circuit breaker keeps on tripping, chances are that it has detected some issues in the current circuit.

5. **Frequent Bulb Burnouts** - There are a lot of reasons why bulb lights burn out much frequently. It can be due to high voltage, tightly fixed bulb or improper air circulation.
6. **Wires Sticking Out from Outlets** - Wires fall off from outlets due to excess length or inefficient wiring. If the wires are taped securely, they reduce potential damages. But this is just a temporary solution. Making sure that the sockets are rewired will reduce the chances of fire sparks.
7. **Backstabbed Outlets/Wires** - Backstabbed outlets/wires are simply wires that have cuts or holes on them and hence, expose live wires. This makes electrocution possible when someone comes in direct contact with them. Tape based safety is just temporary. It is important to get a new wire fixed.
8. **No GFCIs** - A GFCI which is also known as a ground fault circuit interrupter is another device that prevents electric shocks. The device detects human contact with electricity and shuts down the power immediately. It is important to have this installed.
9. **Ungrounded Outlets** - Ungrounded connections are ones which do not have a grounding system as an added failsafe. Not having a ground connection poses a risk to your appliances as well as someone using them.
10. **Dysfunctional Switches** - If you find that dimming or ON/OFF switches are not functioning as you expect them to, there is a chance that the switch may have been overridden. Another cause might be improper wiring and circuit faults.
11. **High Electricity Bills** - Are the electricity bills rising even when less power is consumed? The best thing that you can do from your side is to keep a check on how you utilise electricity. Follow the maintenance plan to track usage of devices.
12. **Too Bright or Too Dim Lighting** - When you notice that some lights burn brighter while some seem to be too dim, then you are experiencing either one of the following two problems:

Some bulbs consume a different wattage of power.

The main power panel is having technical problems.

13. **Uncovered Junction Box** - Junction boxes are made to separate one section of wires from another. They also protect the wires from external forces that can damage them. In large buildings, the use of junction boxes is extremely important. This is because when there is an electrical problem occurring in one part of a building, the wires to that building can easily be located in the junction box. It is important to install a junction box.

- **Lighting Efficiency:** Switch to energy-efficient LED lighting and install sensors or timers to control usage, reducing unnecessary electricity consumption.
- **Power Management Practices:** Implement power-saving modes on electronic devices, encourage unplugging unused equipment, and use power strips to manage standby power.
- **Employee Training and Awareness:** Educate employees about energy-saving practices, encouraging them to switch off lights and equipment when not in use.

Material Conservation:

- **Waste reduction and recycling:** Implement a waste reduction strategy, focusing on reducing packaging, reusing materials, and promoting recycling programs. _Source materials that are recyclable, biodegradable, or made from recycled content to reduce environmental impact.
- **Lifecycle Assessments and Eco-design:** Conduct lifecycle assessments to understand the environmental impact of products and adopt eco-design principles for sustainable product development.

Water Conservation:

- **Water-efficient practices:** Install water-saving fixtures, such as low-flow faucets and toilets, to reduce water consumption in facilities. Implement water recycling and rainwater harvesting systems to reuse water for non-potable purposes.
- **Monitoring and leak detection:** Pipe leakages are one of the main reasons why materials (water, oil, etc.) are wasted. Unseen or unfixed leaks can waste large quantities of materials over time.

Adding leak detection to toilets and sinks can also ensure that the problems are found early so that they can be rectified before a higher water bill is generated.

Monitor water usage level regularly to identify leaks. These leaks can be visible or hidden. Where installed, water meters need to be regularly read to identify irregular usage, which might indicate a leak.

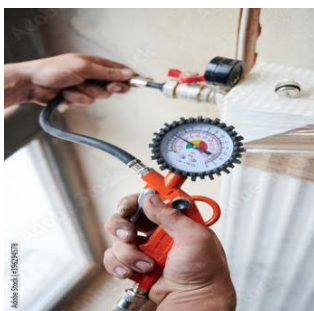


Fig. 3.2.2

- **Employee training and water awareness:** Educate employees about water conservation practices, promoting responsible usage and minimizing waste.

List of green practices at workplace

Switch all lighting to LEDs

A simple and effective way for businesses to become more sustainable is by switching their lighting over to LED (light-emitting diode) lights. These bulbs require less energy, last longer, and provide a brighter light than traditional bulbs, making it an ideal choice for businesses wanting to go green.

Motion sensors for lighting

Fitting motion sensors into your office ensures lights are only used when needed. This can help to reduce unnecessary energy consumption in the workplace and also save the business money on energy bills.

Recycle

making a concerted effort to recycle as much waste as possible in the workplace can help to reduce the business' carbon footprint.

Go paperless

The best way to reduce office waste is by not creating any at all. Many businesses are making the jump to going paperless. By implementing a paperless workflow, companies can help to save trees, cut costs associated with document storage and printing, as well as reduce energy consumption from printing equipment.

Eco-friendly printing practices

Investing in new energy efficient printers, buying remanufactured ink cartridges and introducing double-sided printing policies can go a long way to reducing the impact your use of paper has on the environment.

Water-saving plumbing fixtures

Low flow toilets, water-saving taps and pressure-reducing valves help minimise water consumption in your building. Businesses can significantly decrease the amount of water they waste by improving the efficiency of their plumbing systems by replacing traditional fixtures with these eco-friendly options, without sacrificing performance or comfort.

Rainwater harvesting system

Rainwater harvesting systems capture and store rainwater in tanks or cisterns and are a great way to support sustainability in the workplace. This captured water can be used for a variety of purposes such as watering plants, flushing toilets, and cleaning the workplace.

3.2.3 Management of different types of waste

Our activities generate a lot of waste material. It could be part of the production stage or later when we consume goods. Now offices, factories, public places and even our homes are expected to separate this waste into different types, but it is best to avoid the waste in the first place.

Separating different types of waste can help in sending different materials for different uses. For instance, paper can be recycled, wood can be reused, electronics can be repaired, organic waste can be made into manure and hazardous waste can be incinerated. Thus, segregating waste into different types is a very important task. Segregated waste is placed into different types of colour coded containers.

As a first step, you need to identify recyclable, non-recyclable and hazardous waste.

1. Dry Waste - All dry waste is recyclable. It also includes the production wastes from the industries. Some examples are:

Plastic	Plastic bags, used cups and plates, juice or Pet bottles
Paper	Newspapers, printed papers, stationary, cardboard boxes
Metal	Metal packs, offcuts, containers
Foam	Thermocol and related items
Wood and Glass	Boxes, containers or utensils

2. Organic Waste - This can be recycled into compost.

Food	Horticulture Waste
 This is typically leftover food and vegetables and can be sent for composting. Leftover food could also be used for generating biogas.	 This is the waste generated in buildings or factories and includes materials such as grass trimmings, plants, leaves and even bouquet flowers.

3. Sanitary or Biowaste - This is hazardous waste and has to be clearly identified. There is a special handling process to be carried out for this. The biowaste includes:

Sanitary items such as diapers, hair and used tissue from toilets

Used material cloth, badly soiled paper and waste from cleaning operations

Biomedical waste from medical facilities

4. Agri Waste- This kind of waste requires proper collection and disposal to avoid burning.

Agriculture waste management is an important issue. With the right kind of practices, we can avoid issues like crop burning and also save energy. Some simple steps and a disposal plan can go a long way in resolving this issue.

These include the following measures:

- Do not burn garbage or farm waste. It should always be recycled, reused, or disposed of to a landfill.
- Safely remove surface materials like pvc pipes, etc. and separately pack them.
- Check the wood condition, if it requires salvaging, chipping or excavation.
- Check if pruning can be used as it can be helpful.
- Compost nursery waste as plant waste is a valuable resource.
- Recycle agricultural plastic. Separate them and identify local programme details for this. Livestock feed plastics have specific bags as well. Similarly, greenhouse and nursery film plastics, when clean, can be made into pellets or smaller pieces for other uses.
- Recycle agricultural chemical containers. They are recycled differently from other agricultural waste and would follow the chemical waste management process.

5. E-waste - These are products that are used in electronic devices and need a special recycling process. These include:



Computer Accessories

Laptops, computers, drives and peripherals come under this category. They may be reusable for some other purpose.



Other Waste

These include mobiles, printers, circuit boards, cables and so on.



Hazardous Waste

These could include batteries, tubes and so on.

6. Chemical Waste - This could be waste from manufacturing or industrial process. This has a special treatment process for each type. These include:

Spent Solvents

Chemicals

Wasted Oils

Filters for Air and Oil

Effluent Treatment Plant(ETP) Sludge

Resins and Paints

Segregating Waste as per their Type

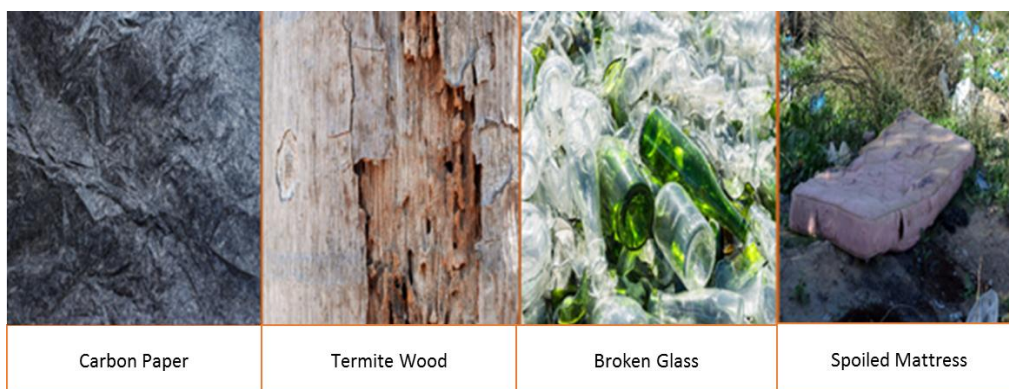
Offices and factories have various kinds of containers placed in common areas. Typically, these could be of the following types:



Everyone is encouraged to follow these rules strictly. Each location will also have stickers informing everyone about the correct use of the different types of bins.

Non-recyclable Waste

Some materials are not readily recyclable and are appropriate for disposal today. But with research and advancements in technology, they could become resources tomorrow.



For now, the Red Bin material needs to go to the landfill.

At an office or a factory site, the red bin containers are stored in a designated area. They are cleared by the service provider at a designated time and periodicity. This is dependent on the amount of waste generated per day.

Difference Between Recyclable and Reusable

The major differences between recyclable and reusable material is shown in the following table:

Recyclable	Reusable
It refers to a process of converting an item into a new product. Its aim is to reduce waste of the original material.	It refers to using the same item for a different use.
It changes the original form of the product.	It does not change the original form of the product.
For example, Preparing compost from fruit and vegetable peels.	For example, using newspaper to make a paper bag.

If the office is generating a bulk quantity of biodegradable waste, there may be a composting solution in place on location. This may be either an Organic Waste Converter (bigger offices) or slightly simpler solution for smaller locations. All waste material will need to be go to this location.

This is the recyclable material that will be taken away by a service provider from the office location. Each office has a designated location for this dry recyclable material. All material will be transferred

from different parts of the office complex to this location. These materials might be typically taken by service provider on designated days. So, keeping a track of the process is important.

The Government of India drafted National Resource Efficiency Policy in 2019, which was proposed by the environment ministry. The policy outlines a three-year resource management action plan, which will be implemented across seven sectors. The policy proposes a waste management plan that includes the following:

- Improved Waste Management Plan
- Reduced Landfill Use
- Use of materials drawn from the natural sources
- Efficiently reduce, recycle and reuse the waste
- Improved process for depositing recyclable and reused material

To deposit the recyclable material, it is essential to label it properly there are some symbols that are been proposed for recyclable material. The following figure represents the symbols for reusable material:



Fig. 3.2.3

Symbols for Recyclable Material

Hazardous Waste Disposal

Hazardous chemical waste is defined as any substance that poses a health threat to humans or could cause damage to the environment, if not managed correctly. Chemical waste can be generated by many sources, ranging from chemical laboratories and industrial manufacturing to the disposal of household batteries or cleaning fluids.

If this waste is allowed to go into our water streams, it can impact both the human life and the water animals.

Some measures that need to be taken are as follows:

- **Reduce the use of hazardous chemicals** - Make sure you only use the minimum amounts required, re-use as many chemicals as possible, recycle all recoverable materials and then dispose of any remaining hazardous substances.
- **Maintain record of all chemicals** - Inventory limits will always be set according to emergency response capacity. Maintain records in software to manage chemical stocks. This ensures that the right amount of material is used.
- **Preventing spills and maximising material usage** - Efficient production scheduling to ensure chemicals are only taken from storage when they are going to be used will also help to mitigate the risk of leaks and spills. Minimising the need to transfer materials into different containers and using pre-weighed packages to reduce handling requirements will also help reduce the risk.
- **Chemical Waste Recycling** - When chemicals are recycled, the useful materials that cannot be directly re-used, and would otherwise be discarded, are recovered, collected and separated from the chemical waste, significantly reducing the amount of waste that needs to be treated and discarded. Hazardous waste recycling can be carried out on-site, with the recycled materials being returned to a manufacturing process within the same facility, or waste can be collected and the recycling then carried out off-site by a licensed third-party.
- **Chemical Waste Disposal** - While lots of chemical waste can be safely and effectively recycled, other chemical waste will need to be treated and disposed of correctly. Hazardous chemical waste treatment is carried out at dedicated Treatment Storage and Disposal Facilities (TSDFs).

It is advisable to prevent chemical waste from accumulating at the premises and take regular advantage of local hazardous waste collections. By properly segregating your waste in your on-site chemical waste storage area, you can remove the risk of cross-contamination.

HAZARDOUS WASTE खतरनाक सामग्री	
Waste Category and Characteristics	Incompatible Waste and Substances
Total Quantity:	Date of Storage:
Physical State of the Waste (Solid/Semi- Solid/Liquid):	
Sender's name and address	Receiver's name and address:
Phone:	Phone:
E-mail:	E-mail:
Contact Person:	Contact Person:
In case of emergency please Contact:	
HANDLE WITH CARE	

Fig. 3.2.3

Sample Label for Hazardous Waste

3.2.4 Steps to review the internal operations in energy, material and waste management

When it comes to running a successful construction business, there are many factors that come into play. But one of the most important? Internal operations management. The back office, yard and warehouse teams are the backbone of any construction business. They analyse, manage, monitor and plan every project from start to finish. Every aspect of a project is controlled and managed by your internal operations staff- from preconstruction through project completion

But as times change, businesses cannot solely rely on the talent and hard-work of their internal staff. They must provide their teams with technology solutions so employees can optimize internal processes and maximize their valuable time.

Reviewing internal operations with a focus on energy, material, and waste management involves a comprehensive evaluation of how an organization consumes resources, manages materials, and handles waste.

Here's a step-by-step guide on how to review internal operations with the point of energy, material, and waste management:

1. Energy Review:

Energy Audits: Conduct audits to assess energy consumption patterns across facilities, machinery, and operations.

Identify Inefficiencies: Pinpoint areas of high energy usage and inefficiencies through data analysis and assessment.

2. Material Management Assessment:

Inventory and Procurement Analysis: Review inventory practices to minimize waste and optimize material usage.

Sustainable Sourcing: Evaluate suppliers' sustainability practices and prioritize those offering eco-friendly materials.

3. Waste Management Evaluation:

Waste Audits: Conduct waste audits to understand waste generation patterns and identify opportunities for reduction or recycling.

Recycling Programs: Review and strengthen existing recycling programs for various materials used in operations.

4. Integrated Approach:

Cross-functional Assessment: Assess how energy, material, and waste management practices intersect within different departments.

Resource Efficiency Metrics: Establish metrics to measure resource efficiency and waste reduction across operations.

5. Technology and Innovation:

Efficiency Technologies: Evaluate the use of energy-efficient machinery, smart sensors, or automation for material handling.

Innovative Waste Solutions: Explore innovative waste management technologies or circular economy approaches.

6. Employee Engagement and Training:

Educational Programs: Conduct training sessions to educate employees about energy-saving practices, efficient material usage, and waste reduction.

Behavioural Changes: Encourage a culture of resource efficiency by involving employees in identifying and implementing improvements.

7. Sustainability Reporting and Goal Setting:

Benchmarking: Compare current practices with industry benchmarks and best practices in energy, material, and waste management.

Goal Setting: Set achievable goals for energy reduction, material optimization, and waste minimization, aligning with sustainability objectives.

8. Continuous Improvement:

Feedback and Adaptation: Gather feedback from employees, stakeholders, and customers for ongoing improvements.

Regular Review: Continuously monitor performance, reassess strategies, and make necessary adjustments for continual improvement.

Unit 3.3 The use of renewable energy/ solar energy in the sector

Unit objectives



After reading this unit, you will be able to understand the following points:

1. Introduction to renewable energy in supply chain management
2. The use of renewable energy/ solar energy in supply chain management

3.3.1 Introduction to renewable energy in supply chain management

The renewable energy supply chain is mainly consisting on five phases namely procurement, generation, transmission, distribution, and demand. These phases cover all processes along the supply chain of renewable energy, from raw materials (input) to the final product (output) Alternatively, the RESC can be divided into three processes as upstream, production, and downstream. Its main objectives are to provide a regular and consistent supply of raw materials and to encourage and promote the use of renewable energy technologies

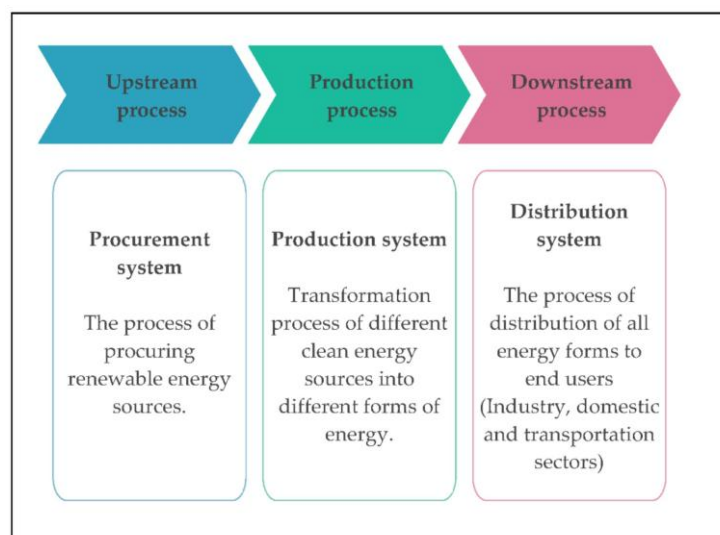


Fig. 3.3.1

Renewable energy supply chain

The renewable energy supply chain differs following the renewable energy source (biomass, wind, solar, hydropower, geothermal). A brief overview of the specific supply chain for each renewable energy source is given below.

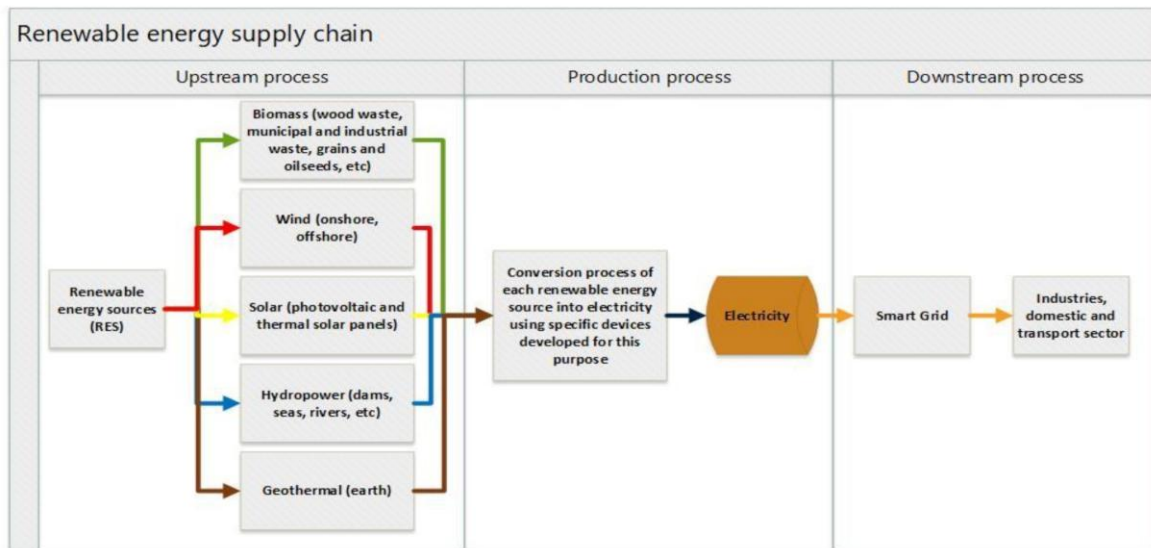


Fig. 3.3.1

Supply chain by renewable energy sources

(a) The energy supply chain for biomass resources:

The energy supply chain for biomass includes three processes [5] as follows:

Firstly, the upstream process includes all operations (collection, storage, pre-processing) performed from the source as an input. Secondly, the production process aims to explore biomass energy sources to produce electricity via conversion devices. Lastly, the downstream process includes specific operations such as securing the availability of electricity stored for end-users.

(b) The energy supply chain for wind resources:

Similarly, the energy supply chain from wind contains three processes [5][13][14] as follows:

In the upstream process, energy sources from wind are characterized by the generation of electricity through the wind turbine. The latter contains certain components, including the mast, propeller, nacelle, blade, and rotor. It is also important to choose an installation site with high wind intensity.

(c) The energy supply chain for solar resources:

Solar energy has recently been characterized by strong growth in technology investment and a large number of innovative initiatives [17]. Solar energy can be harnessed by two means, including solar photovoltaic and solar thermal.

The upstream process is based on the implementation of solar photovoltaic or solar thermal power generation systems. The production process focuses mainly on producing electricity either by PV modules or thermodynamic cycles driven by solar concentrating collectors. Finally, the downstream process in the supply chain consists of a continuous supply of electricity to end-users.

(d) The energy supply chain for water resources:

The hydropower supply chain exploits dams and seas to generate electricity.

Electric power is generated by releasing water through a turbine, which is connected to generators to create an alternating current that must be regulated to the voltage level of the power grid. The downstream process deals with the distribution of electricity to end-users based on their energy needs.

(e) The energy supply chain for geothermal resources:

The geothermal energy supply chain is based on three main processes [5] as follows:

The upstream process consists of extracting thermal resources from the earth as an input. In production process, water is infiltrated, heated until it is vaporized. Vapor drives a turbine to produce mechanical energy, which is then converted into electricity via a generator. This energy has to be adjusted to the voltage requirements of the electricity grid. Finally, the downstream process concerns the distribution of electricity to end-users.

3.3.2 The use of renewable energy/ solar energy in supply chain management

The integration of renewable energy into the supply chain involves utilizing sustainable energy sources across various stages of the supply chain process. Here's how renewable energy can be utilized within the supply chain:

1. Manufacturing and Production:

Renewable-Powered Facilities: Utilize renewable energy sources like solar, wind, or biomass to power manufacturing plants, reducing carbon footprint.

Energy-Efficient Technologies: Implement energy-saving machinery and processes to optimize energy usage during production.

2. Transportation and Logistics:

Green Vehicles: Adopt electric, hybrid, or vehicles running on alternative fuels (biofuels, hydrogen) to reduce emissions in transportation.

Renewable-Powered Fleet: Install solar panels or use renewable energy sources to power logistics vehicles and warehouses.

3. Warehousing and Distribution:

Renewable-Powered Warehouses: Implement solar panels, wind turbines, or geothermal systems to power storage facilities.

Efficient Systems: Use energy-efficient systems and smart technologies to optimize energy consumption within warehouses.

4. Procurement and Sourcing:

Sustainable Suppliers: Partner with suppliers using renewable energy or eco-friendly practices in their production processes.

Circular Economy Practices: Embrace recycling and repurposing materials to reduce reliance on virgin resources.

5. Solar-Powered Technology and Equipment:

Mobile Devices and Equipment: Use solar-powered chargers for mobile devices, laptops, and equipment used in inventory management or on-the-go operations.

Data Collection and Tracking: Implement solar-powered sensors and RFID devices for real-time tracking of inventory, reducing reliance on traditional power sources.

6. Off-Grid Solutions in Remote Areas:

Remote Warehousing: Deploy solar-powered containers or modular warehouses in remote areas lacking reliable power sources for temporary storage needs.

Off-Grid Operations: Use solar energy for operations in regions with limited access to electricity, allowing continuous functionality without grid dependence.

As renewable energy technologies advance and become more cost-effective, integrating them into various facets of the supply chain can significantly contribute to a more sustainable and environmentally friendly operation.



Example of a Company Nike for
its sustainable practices

Unit 3.4 Introduction to circular economy in different areas

Unit objectives



After reading this unit, you will be able to understand the following points:

1. Overview of circular economy
2. Areas where the concept of circular economy can be introduced

3.4.1 Overview of circular economy

A circular economy is an industrial system that is restorative or regenerative by intention and design. It replaces the 'end-of-life' concept with restoration, shifts towards the use of renewable energy, eliminates the use of toxic chemicals, which impair reuse, and aims for the elimination of waste through the superior design of materials, products, systems, and, within this, business models

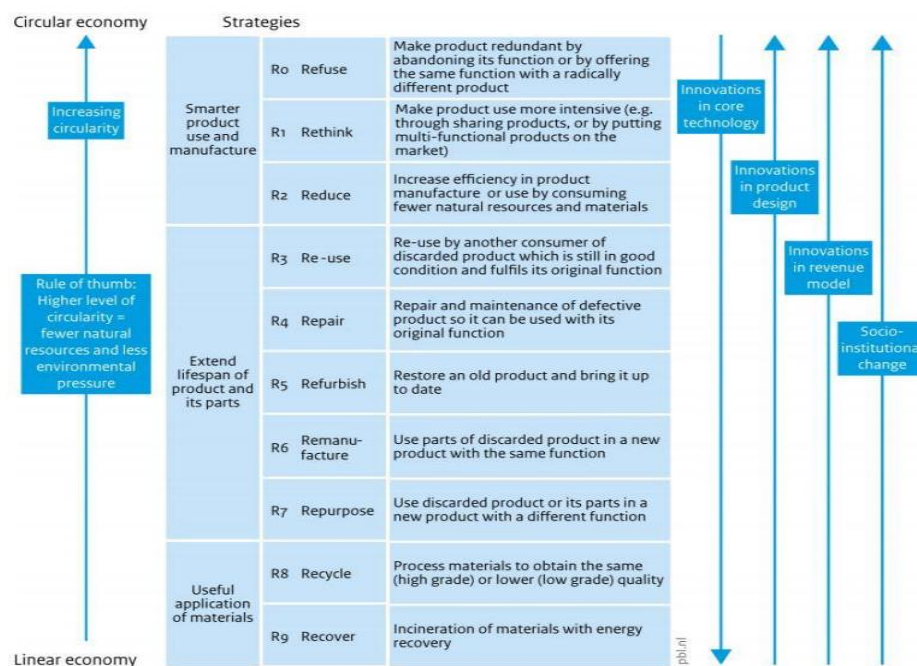


Fig. 3.4.1

Circularity strategies within the production chain

It can be seen from Figure 7.2.1 that recycling was a key element in circular economy thinking from the start. As time has gone on the number of R's has multiplied. The Japanese Government's '3R Initiative'

(reduce, reuse, recycle) dates from 2004. The European Union's waste hierarchy in its 2008 Waste Framework Directive⁴ has four Rs (reduce, reuse, recycle, recover). By 2017 nine separate Rs contributing to circularity had been identified (Figure 7.2.1).

Categories and characteristics of circular economy concepts

Categories	Characteristics	Concepts									
		Circular economy	Cradle to cradle	Closed supply chains	Regenerative design	Blue economy	Industrial ecology	Reverse logistics	Performance economy	Natural capitalism	Bio-mimicry
Motivation(s)	Focus on environment	•	•	•	•	•	•	•	•	•	•
	Focus on profitability			•				•			
	Including social aspects	•	•		•	•			•	•	
Proposition for waste management	Efficiency and waste reduction	•	•	•	•	•	•	•	•	•	•
	Zero waste	•	•		•						
	Technological/biological substances	•	•								
Guidelines and tools	Business model perspective	•	•			•				•	
	Focus on operations		•	•	•		•	•			
	Measurability	•	•	•		•	•	•	•	•	
Economic sectors covered	Policy	•		•			•				
	Primary sector	•	•		•	•	•			•	
	Secondary sector	•	•	•	•	•	•	•	•	•	•
Economic scope	Tertiary sector	•	•		•	•			•	•	
	Macro-economic perspective	•				•			•	•	
	Meso-economic perspective		•	•			•				
Activities during life cycle stages: Circular design of ...	Micro: company level	•	•	•	•		•	•			
	Micro: product level	•	•		•			•			•
	Product development	•	•	•	•	•			•	•	•
	Raw material sourcing	•	•	•							•
	Production processes	•	•	•		•	•	•			•
	Use	•	•						•		•
	CRM							•	•		
	End of life/disposal	•	•	•	•	•	•	•	•	•	•
	Transportation	•	•	•	•	•	•				

Fig. 3.4.1

It can be seen that only four characteristics are common to all the concepts: focus on the environment (which presumably also includes natural resources), efficiency and waste reduction, coverage of the secondary (manufacturing) sector, and end-of-life disposal. A definition that would therefore cover all these concepts, at least in part, would seem to be: The circular economy is one that has low environmental impacts and that makes good use of natural resources, through high resource efficiency and waste prevention, especially in the manufacturing sector, and minimal end-of-life disposal of materials

Relationship between circular economy and sustainability

Relationship types between the Circular Economy and sustainability.

General direction	Type of relationship	Short description
Conditional	Conditional relation	Circularity/closed loop systems are seen as ... One of the conditions for a sustainable system
	Strong conditional relation	The main solution for a transformation to a sustainable system
	Necessary but not sufficient conditional relation	A necessary but not sufficient condition for a sustainable system
Beneficial	Beneficial relationship	Beneficial in terms of sustainability, without referring to condition-ality or alternative approaches
	Subset relation (structured and unstructured)	One among several solutions for fostering a sustainable system
Trade-off	Degree relation	Yielding a degree of sustainability with other concepts being more and/or less sustainable
	Cost-benefit/trade-off relation	Having costs and benefits in regard to sustainability, which can also lead to negative outcomes
	Selective relation	Fostering certain aspects of sustainability but lacking others

Fig. 3.4.1

3.4.2 Areas where the concept of circular economy can be introduced

The concept of a circular economy, aimed at minimizing waste and maximizing resource efficiency by promoting reuse, recycling, and regeneration of materials, can be introduced and implemented across various sectors and areas. Here are key areas where the principles of a circular economy can be applied:

1. Manufacturing and Production:

Product Design: Design products with recyclability and reusability in mind, using eco-friendly materials.

Resource Optimization: Embrace remanufacturing and refurbishment to extend product lifecycles.

Sharing Platforms: Promote product-as-a-service models and sharing platforms to maximize product use.

2. Waste Management and Recycling:

Material Recovery: Enhance recycling infrastructure and processes to recover valuable materials from waste.

Closed-loop Systems: Establish closed-loop systems where materials are recycled and reused within the same cycle.

E-waste Management: Implement efficient e-waste recycling programs to recover valuable components.

3. Supply Chain and Logistics:

Reverse Logistics: Develop systems for efficient product returns, refurbishment, or recycling at the end of their life.

Packaging Innovation: Adopt reusable or biodegradable packaging materials to reduce waste in the supply chain.

Optimized Transportation: Implement efficient logistics to minimize resource use and emissions.

4. Construction and Infrastructure:

Circular Building Design: Use materials that are easy to disassemble and recycle for construction.

Resource Recovery: Explore methods to recover and reuse materials from demolished structures.

Adaptive Reuse: Promote adaptive reuse of buildings and infrastructure rather than demolition.

5. Consumer Goods and Services:

Product Life Extension: Encourage repair and maintenance services to extend the lifespan of consumer goods.

Product-as-a-Service: Offer subscription-based services for products, encouraging reuse and responsible disposal.

Consumer Education: Educate consumers about responsible consumption and disposal practices.

6. Agriculture and Food Systems:

Food Waste Reduction: Implement strategies to minimize food waste and promote food redistribution.

Circular Farming Practices: Embrace regenerative agriculture and closed-loop farming systems.

Bio-based Materials: Explore the use of biodegradable materials in packaging and agricultural products.

7. Energy and Renewable Resources:

Circular Economy Approach: Emphasizing renewable energy sources, optimizing energy production and consumption, and promoting energy efficiency to minimize waste and reliance on finite resources.

Implementation Focus: Investing in renewable energy infrastructure, promoting energy-efficient technologies, and exploring energy storage and grid optimization.

Applying circular economy principles across these diverse areas involves reimagining how resources are used, reused, and conserved to create a more sustainable and resource-efficient system.

Unit 3.5 Implementation of efficient green logistics

Unit objectives



After reading this unit, you will be able to understand the following points:

1. Implementation of an efficient green logistics
2. Successful examples of companies taking initiatives to make logistics activities green
3. Significance of optimizing logistics operations for sustainability

3.5.1 Implementation of an efficient green logistics

Green logistics involves considering the environmental footprint of a company when organising its logistics. It is applied to the entire supply chain and its objective is to reduce the company's impact on the environment, without affecting its economic activity.

This is achieved through more or less substantial changes in a company's logistics activity, either by the choice of suppliers, infrastructures, optimisation of storage space or automation. There is a wide variety of solutions to combine efficient logistics and sustainable development.

Although the implementation of green logistics implies changes in the operation of the company and hence time and money expenditure, these changes can be very beneficial for the company.

Compared to traditional logistics, the field of green logistics defines more objectives, more specifically speaks about the economic and environmental objectives that organisation should set for its implementation and which determine the process of management of material and information flows. Thus, in order to facilitate the application of environmentally friendly solutions in practice, the four stages in the implementation of green logistics in business operations are identified:

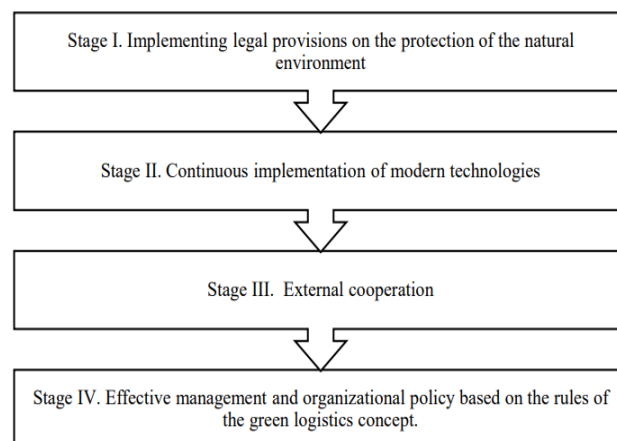


Fig. 3.5.1 (Stages of implementation of green logistics in the organization)

- ✓ **In the first stage**, organisations implement environmental provisions laid down by the management of the country in which they operate or the organization's management.
- ✓ **In the second step**, the actions are carried out related to the introduction of modern technologies in the organization, which increase the "degree of greening". These are, for example, IT solutions that allow better use of the company's transport fleet, more efficient management of warehouse premises, etc. These solutions include modern vehicles as well as the purchase of less polluting cars that meet the highest standards. Thus, it should be noted that the development of green logistics in business operations is impossible without investment in modern technologies.
- ✓ **The third step** is defined by the provision that a high level of greening of the enterprise is unattainable without cooperation with external entities. In practice, many logistics processes are carried out involving another participant in the supply/distribution chain. It is argued that the implementation of "green" logistics is only insufficient at the company level, as there is a need to review the processes of the entire supply chain – greening of raw materials, sustainable packaging, efficient warehousing, etc
- ✓ **The last step** is performed when the organization's management is based on the principles of green logistics and the 10 most applicable principles of green logistics in an organization are:
 - Use of recyclable packaging
 - Use of alternative fuels
 - Improving fleet facilities for greening
 - Route optimization
 - Use of intermodal transport
 - Eco-driving
 - Upgrading warehouse machinery and technologies for efficient use of energy
 - Optimisation of warehouse area
 - Reduction of paper documents (digitisation)
 - Adaptation of "green" criteria while choosing suppliers / business partners.

Areas of green logistics	Impact of green logistics principles
Green packaging	Reduces negative environmental impact, greenhouse effect; reduces layers of packaging used; Recyclable packaging; reduction of packaging costs due to materials used.
Green warehousing	Optimization and efficient exploitation of the place reduce the movement in the warehouse, increase the productivity of the enterprise by reducing costs and increasing profits. However, due to more sustainable technologies in warehouse, logistics services will be relatively more expensive.
Green transportation	Route optimization maximizes fuel consumption, reduces mileage, empty kilometres. In the future the use of alternative fuels (e.g., natural gas instead of gasoline or diesel) will help reduce transportation costs. In the field of transportation, it helps to reduce costs, increases competitiveness, increases speed/ reduces transportation time, reduces pollution, increases safety of workers.
Green logistics management system	The commitment of top managers to implement the environmental sustainability strategy and to involve all employees of the organisation strengthens staff's morale to practice and facilitate work and application of green logistics principles. Economic indicators are important in the development of the enterprise - this is the value of money, assets, debts, and investments. To improve these indicators, green logistics gives the opportunity to reduce costs associated with a sustainable environment, for example, reduction of energy consumption, minimization of waste, processing, environmental taxes. Environmental efficiency is measured in terms of energy savings and emissions of pollutants, reduction of waste reducing negative environmental impacts, e.g., sulphur dioxide concentration, nitrogen oxides concentration, greenhouse gas emissions and fossil fuel consumption.
Green logistics data collection and management	The organization implements information technology, transport or warehousing management systems, innovative management, use of electronic documents instead of paper.

Fig. 3.5.1

The principles of green logistics include all attempts to measure and reduce negative environmental impact of logistics operations. Green packaging, transportation, warehousing, management system itself are among the most important processes in the logistics system portfolio. The main objective of green logistics practices is to avoid any activity that causes unnecessary carbon dioxide emissions, to look constantly for opportunities to reduce the environmental impact of your business, such as trying to use office stationery made of recycled materials, optimising fuel use, reducing waste, optimising routes, and applying sustainable distribution practices.

The implementation of green logistics in an organization is a complex of all logistics processes that are carried out in accordance with the principles of green logistics and thus achieve economic and ecological objectives of the organization, which increase the competitive advantage in the market.

There are various ways to implement sustainability into logistics services. Here are some examples of best practice when it comes to sustainable logistics:

- **Sustainability assessment** - assess your overall logistics strategy to see where more sustainable practices could be implemented.

- **Optimise your travel routes** - minimise unnecessary travel and waiting times to reduce your fuel consumption.
- **Renewable energy or technology** - review the potential to incorporate more green or renewable energy or technologies into your logistics strategy.
- **Avoid wasted energy on partial capacity**- minimise waste space and energy by ensuring your containers and transport are packed to maximum capacity.
- **Sustainable packaging**- introduce recyclable or sustainable packaging into your logistics strategy.
- **Use integrated logistics**- align your supply chain across all departments to maximise your overall efficiency.
- **Greener transport options**- consider greener alternatives to your transport where possible, e.g. electric-powered vehicles.

Advantages of implementing green logistics

When it comes to tackling climate change, the logistics and transport sector can have a strong influence and impact. Large logistics companies are some of the key players that can make a positive change by implementing more sustainable logistics services and advocating this as the norm.

There are numerous benefits of sustainable logistics and supply chain management, for both the environment and for businesses. These include:

- Reduced emissions and energy usage.
- Reducing waste creates a more efficient and cost-effective supply chain.
- Protects against reputational damage.
- Improved continuity of supply by diversifying your supply chain.
- More attractive to customers.
- Increasing environmental awareness.
- Alignment with government goals and commitments.

As awareness and concern around sustainability continues to rise, it's likely sustainable logistics and supply chain management is what the future of logistics will look like.

3.4.2 Examples of companies taking initiatives to make logistics activities green

[maersk](#)

Example of companies taking initiatives to make logistics activities green

IKEA, a multinational company that designs and sells furniture, kitchen appliances and home accessories, carries out many projects for the GSC process. Climate roadmaps prepared for the company to reduce the climate footprint of materials are integrated into every IKEA process. According to the 2021 sustainability report published by the company, it is stated that the climate footprint is 26.2 million tons of CO₂ equivalent, and 1.6 million tons of CO₂ equivalent reduction was achieved compared to 2016. However, it is estimated that there will be a 5.7% increase in climate footprint in 2021 compared to 2020, due to the diminishing effects of the pandemic. The main objective of the company is to reduce greenhouse gas emissions from the value chain by at least 15% in absolute terms compared to fiscal year 2016 by 2030. To this end, the company sets a new target for the phasing out of remaining plastics in consumer packaging by 2028 and takes supporting actions. Circular pathways are developed to ensure that the life of their product is extended. Products that have become unusable are recycled into secondary raw materials for new products and are designed to last as long as possible. For example, A KLIPPAN sofa cover from recycled denim has been developed in partnership with MUD Jeans. Each cover contains 40% recycled denim (approximately two jeans). This reduces the carbon footprint of the cover by 67% and saves 27,000 litres of water during production. In 2021 55.8% of procured materials are renewable and 17.3% are recycled materials. It works to achieve 100% renewable electricity for all IKEA factories and component packaging and distribution units. It is thought that all these actions also affect customer preferences. The company's sales increased by 5.8% in 2021 compared to the previous year (IKEA, 2021).

BOEING is a company that conducts business processes with a global network of approximately 12,000 suppliers every day. It cooperates with suppliers that comply with sustainable environmental policies in supplier review, selection and support. In 2018, the company carried out the world's first commercial airplane flight using 100% sustainable fuels as part of the eco Demonstrator program. In 2020, the target of net zero carbon emissions at its factories and construction sites has been achieved. The company contributes to reducing waste by reusing and recertifying aircraft parts. Parts that meet quality standards are recertified. These parts are obtained by dismantling decommissioned aircraft. According to the company's 2021 sustainability report, it has reduced energy consumption by 12%, water use by 23%, solid waste by 44% and hazardous waste by 34%. The company also partners with community organizations to advance racial equality in terms of social responsibility, promote academic achievement, and train workforces for economic stability (BOEING, 2021).

<https://dergipark.org.tr/en/download/article-file/2251369>

More examples of companies taking initiatives to make logistics activities green:

1. Unilever

Unilever, a multinational consumer goods company, aimed to reduce the environmental impact of its logistics and supply chain operations.

Actions Taken:

- **Optimized Transport:** Utilized more efficient transportation methods, including consolidation and optimization of delivery routes.
- **Packaging Innovation:** Focused on lightweight packaging designs and sustainable materials to minimize waste.
- **Eco-friendly Warehousing:** Transitioned to energy-efficient warehouses and distribution centers.

Impact: Reduced carbon emissions and waste through optimized transportation, packaging innovation, and energy-efficient facilities.

2. UPS

UPS, a global package delivery and logistics company, has been actively pursuing sustainability in its operations.

Actions Taken:

- **Alternative Fuel Vehicles:** Invested in alternative fuel vehicles, including electric and hybrid delivery trucks, reducing emissions.
- **Route Optimization:** Employed advanced routing software to optimize delivery routes for fuel efficiency.
- **Eco-friendly Packaging:** Implemented eco-friendly packaging solutions to reduce waste.

Impact: Lowered carbon emissions and fuel consumption through the adoption of alternative fuel vehicles and optimized delivery routes.

3. Maersk

Maersk, a leading container shipping company, aimed to reduce emissions and improve the sustainability of its maritime logistics.

Actions Taken:

- **Fuel Efficiency Measures:** Adopted fuel-efficient vessels and technologies to optimize fuel consumption during shipping.
- **Low Emission Goals:** Invested in research for alternative fuels and technology to achieve carbon-neutral shipping.
- **Data-driven Efficiency:** Used data analytics to optimize vessel speed and route planning for fuel savings.

Impact: Improved fuel efficiency and reduced emissions by implementing technology and strategies focused on energy optimization.

4. Amazon:

Amazon, a major e-commerce and logistics company, is working on several initiatives to make its operations greener.

Actions Taken:

- Committing to operate on 100% renewable energy by 2025 and net-zero carbon emissions by 2040.
- Investing in electric delivery vans and deploying solar panels on fulfillment centers.
- Encouraging sustainable practices among suppliers and customers through initiatives like Frustration-Free Packaging.

5. DHL:

DHL, a global logistics company, has implemented various sustainability measures in its operations.

Actions Taken:

- Introducing electric and hybrid delivery vehicles in urban areas to reduce emissions.
- Implementing innovative packaging solutions to optimize space and reduce waste.
- Investing in renewable energy sources and energy-efficient facilities.

These companies demonstrate diverse approaches to making logistics greener, including investments in alternative fuel vehicles, renewable energy adoption, waste reduction strategies, and sustainable packaging practices. Through such initiatives, they aim to reduce their environmental footprint while maintaining efficient and effective logistics operations.

3.5.3 Significance of optimizing logistics operations for sustainability

At its core, a sustainable supply chain is centered around achieving greater results with less resource usage. With that goal in mind, optimizing your operations is a necessity. Fortunately, making the business case for doing so is easy, given the benefits of optimization.

Optimizing logistics operations for sustainability holds substantial significance for various reasons:

1. Environmental Impact Reduction:

Emissions Reduction: Sustainable logistics practices, such as efficient routing and vehicle optimization, can significantly reduce carbon emissions, mitigating the impact of transportation on climate change.

Resource Conservation: Implementing sustainable measures minimizes resource consumption, such as fuel and energy, while reducing pollution and waste generation.

2. Cost Savings and Efficiency:

Fuel Efficiency: Optimized routes and vehicle loads reduce fuel consumption, leading to cost savings and increased operational efficiency.

Resource Optimization: Streamlining operations minimizes unnecessary resource usage, leading to more efficient supply chain management and reduced overhead costs.

3. Regulatory Compliance and Reputation:

Compliance: Adhering to sustainability standards and regulations ensures compliance with environmental laws, reducing legal risks.

Reputation Management: Embracing sustainability bolsters brand reputation, attracting environmentally conscious customers and stakeholders.

4. Resilience and Adaptability:

Future-Proofing: Implementing sustainable logistics practices prepares businesses for future regulatory changes and market demands.

Supply Chain Resilience: Resilient and sustainable supply chains are better equipped to withstand disruptions, such as resource scarcity or regulatory shifts.

5. Social Responsibility:

Stakeholder Expectations: Meeting societal expectations for sustainability enhances corporate social responsibility.

Employee Engagement: Commitment to sustainability attracts and retains talent, fostering a sense of purpose among employees.

6. Long-Term Viability:

Business Continuity: Sustainable logistics strategies contribute to the long-term viability and profitability of businesses by minimizing risks associated with climate change and resource scarcity.

Innovation and Adaptation: Encourages innovation and fosters a culture of continuous improvement to adapt to evolving sustainability challenges.

7. Global Impact:

Global Supply Chains: Given the interconnected nature of supply chains, adopting sustainable logistics practices has a broader positive impact on global environmental health.

Collective Responsibility: Contributing to global sustainability efforts demonstrates shared responsibility in addressing environmental challenges.

Optimizing logistics operations for sustainability not only reduces environmental impact and operational costs but also aligns businesses with evolving societal expectations. It fosters resilience, innovation, and a positive global impact, positioning companies for long-term success in an increasingly sustainability-focused world.

Exercise

1. What are the best practices and strategies to make supply chain green?

2. What are sustainable materials in the logistics sector?

3. Explain how to streamline inventory management for greener supply chain?

4. Give suggestions to make workplace operations/ activities environment friendly?

5. What is resource conservation?

6. Explain how to implement energy, electricity, material and water conservation practices?

7. Explain how to manage different types of waste?

8. Explain how to review the internal operations in energy, material and waste management?

9. Explain how renewable energy can be used in supply chain?

10. Identify the areas where the concept of circular economy can be introduced?

11. Identify the stages involved in implementing green logistics in an organisation?

12. Explain the need to optimize logistics operations for sustainability?

4. Identifying green supply and demand networks



Unit 4.1 Concept of green purchasing in supply chain management

Unit 4.2 Selection of green logistics partner and transporter



Chapter Highlights



After reading this chapter, you will be able to: %

1. Understand the meaning of green purchasing and its importance
2. Identify the ways for Green Purchasing in Supply Chain Management
3. Identify the best Practices for Green Purchasing
4. Identify the criteria for selecting a supplier with green attributes
5. Understand how to evaluate and select supplier based on green approach in Supply Chain Management
6. Know how to choose a green logistics partner
7. Identify how to select transporter which are following green practices
8. Understand the Map of entire supply chain, identifying suppliers, transportation routes, manufacturing processes, and end consumers.
9. Know how to prepare an assessment criterion to evaluate suppliers based on their sustainability practices, covering aspects like energy efficiency, waste management, and ethical sourcing

Unit 4.1 Concept of green purchasing in supply chain management

Unit objectives

After reading this unit, you will be able to understand the following points:

1. Meaning of green purchasing and its importance
2. Ways for Green Purchasing in Supply Chain Management
3. The best Practices for Green Purchasing

4.1.1 Meaning of green purchasing and its importance

Green Purchasing refers to the procurement of products and services that have a reduced effect on human health and the environment when compared with competing products or services that serve the same purpose. This comparison can consider raw materials acquisition, production, manufacturing, packaging, distribution, reuse, operation, maintenance, and disposal of the product or service. Green purchasing is also known as environmentally preferred purchasing (EPP), environmentally responsible purchasing, green procurement, affirmative procurement, eco-procurement, and environmentally responsible purchasing.

Principles of Green Purchasing

- **Reduce, reuse, recycle:** Consider ordering products that are reusable (as opposed to single-use or disposable) or allow sharing with multiple users.
- **Quality & durability:** Consider purchasing quality products that are designed to last. Durable products may cost more at the onset, but will require less frequent replacement.
- **Energy efficiency:** Purchase energy-efficient products and appliances.
- **Water conservation:** Purchase products and appliances that conserve water.
- **Sustainable forestry:** Purchase products made from sustainably harvested wood.
- **Fair trade:** Purchase products that are made in fair trade conditions.

Importance of green purchasing

1. Environmental Impact Reduction:

Resource Conservation: Green purchasing prioritizes products made from recycled materials or those with minimal environmental impact, reducing resource depletion.

https://financial.ucsc.edu/pages/purchasing_greenpurchasing.aspx,
<http://www.igpn.org/workshop/singapore/pdf/GPN%20Sato.pdf>

Emissions Reduction: It encourages the procurement of goods and services that have lower carbon footprints or reduced greenhouse gas emissions.

2. Promotes Sustainability and Responsibility:

Supports Sustainable Practices: Choosing environmentally friendly products supports companies that embrace sustainable production and supply chain practices.

Encourages Innovation: Demand for green products encourages innovation and development of more eco-friendly technologies and materials.

3. Health and Safety Considerations:

Reduced Toxicity: Green purchasing focuses on goods and materials that are less toxic and pose fewer health risks to users and the environment.

Eco-Certifications: It emphasizes products that comply with eco-labels or certifications indicating reduced environmental impact.

4. Cost Savings and Efficiency:

Long-Term Savings: While initial costs of green products might be higher, they often result in long-term savings due to reduced energy consumption or lower maintenance costs.

Efficient Use of Resources: By choosing more durable or reusable products, organizations can reduce waste and improve resource utilization.

5. Corporate Social Responsibility (CSR):

Enhances Reputation: Embracing green procurement demonstrates an organization's commitment to environmental stewardship and social responsibility, enhancing its reputation.

Meets Stakeholder Expectations: It aligns with the expectations of stakeholders, including customers, investors, and employees, who increasingly value sustainability.

6. Regulatory Compliance:

Meets Environmental Standards: Green procurement helps organizations comply with environmental regulations and standards, minimizing legal risks and penalties.

7. Global Impact:

Contributing to Global Sustainability: The collective adoption of green procurement practices contributes to global sustainability efforts by reducing environmental degradation.

By integrating green purchasing practices into procurement strategies, organizations not only contribute to environmental protection but also drive innovation, enhance reputation, and meet the growing demand for sustainable and responsible products and services.

4.1.2 Different ways for green purchasing/ green procurement in supply chain management

Green purchasing within supply chain management involves various strategies and approaches aimed at procuring goods and services with minimized environmental impact. Here are different ways to implement green purchasing:



Fig. 4.1.2

(Source: <https://www.sketchbubble.com/en/presentation-green-purchasing.html>)

1. Supplier Selection and Evaluation

Supplier Assessment: Evaluate suppliers based on their commitment to sustainability, environmental certifications, and eco-friendly practices.

Green Supplier Networks: Collaborate with suppliers who prioritize sustainability and have eco-friendly offerings.

2. Sustainable Product Specifications:

Environmental Criteria: Define and incorporate environmental criteria into product specifications and procurement guidelines.

Eco-labels and Certifications: Prioritize products with recognized eco-labels or certifications indicating reduced environmental impact.

3. Lifecycle Assessment:

Consider Full Lifecycle: Evaluate the environmental impact of products throughout their entire lifecycle, including extraction, production, distribution, and disposal.

Environmental Impact Analysis: Assess the total environmental footprint of products before procurement decisions.

4. Material and Packaging:

Recycled or Renewable Materials: Opt for products made from recycled, biodegradable, or renewable materials to reduce environmental impact.

Minimize Packaging: Encourage suppliers to minimize excessive packaging or adopt eco-friendly packaging materials.

5. Energy Efficiency and Emissions Reduction:

Energy-efficient Products: Procure equipment and technologies with energy-efficient features to reduce power consumption.

Low Emission Criteria: Set criteria for vehicles and machinery to minimize emissions and fuel consumption.

6. Longevity and Durability:

Durable Products: Prioritize products known for durability and longevity to reduce replacement frequency and waste.

Warranty and Maintenance: Consider warranties and maintenance agreements that prolong product lifespan.

7. Circular Economy Practices:

Support Circular Products: Procure goods designed for easy repair, refurbishment, or recycling to support circular economy principles.

Reverse Logistics: Encourage suppliers who facilitate product take-back programs or recycling initiatives.

9. Performance Evaluation and Improvement:

Set Green Procurement Goals: Establish specific objectives and KPIs for green procurement and regularly assess performance.

Continuous Improvement: Encourage feedback loops and continual improvement in green procurement practices.

Implementing these strategies fosters a more sustainable supply chain by promoting eco-friendly practices, reducing environmental impact, and supporting the development and adoption of greener products and services.

4.1.3 Best practices for green purchasing

Green purchasing involves adopting sustainable practices when procuring goods and services. Here are some best practices to consider:

- Procurement of energy-efficient appliances to reduce energy consumption.

- Using recycled or recyclable building materials such as recycled plastic and concrete blocks, reclaimed wood and steel, precast concrete, and zero-VOC paint.
- Locally sourcing goods and services that prioritize lowering carbon emissions and supporting the local economy.
- Utilizing eco-friendly, natural cleaning products. This includes non-toxic products that are free of harmful chemicals.
- Investing in renewable energy, such as naturally powered fixtures that rely on solar or wind energy to power facilities.
- Transitioning to digital document management to reduce paper waste.

<https://procurementtactics.com/green-procurement-practices/>

Unit 4.2 Selection of green logistics partner and transporter

Unit objectives



After reading this unit, you will be able to understand the following points:

1. Criteria for selecting a supplier with green attributes
2. Supplier evaluation and selection based on green approach in supply chain management
3. Selection of green logistics partner
4. Selection of transporter following green practices
5. Assessment criterion to evaluate suppliers based on their sustainability practices, covering aspects like energy efficiency, waste management, and ethical sourcing.
6. Overview of supply chain map
7. Companies using upgraded technology
8. Implementation of smart logistics solutions
9. Example of emerging future technologies in green logistics

4.2.1 Criteria for selecting a supplier with green attributes

Green supplier selection is based on the critical factors comprising the selection index: cost, supply quality, lead time to delivery, warranty and capacity of the supplier, green product, pollution control, and environmental management

The criteria are categorized into three groups: General criteria, corporate social responsibility related, and Environmental responsibility related criteria.

General Criteria

This category includes general criteria describing the basic characteristics of the finished goods, including Product Price, Quality, Delivery, and Reserved Capacity.

Product Price: It refers to the supplier's bidding price of the product.

Quality: It refers to the conformance and reliability of product. **Delivery:** Delivery refers to the required day that the ordered product will arrive at the door of buying organization.

Reserved Production Capacity: It refers to the supplier that reserves an amount of production capacity to respond sudden change in demand.

<https://hub.hku.hk/bitstream/10722/189926/1/Content.pdf?accept=1>

Criteria related to CSR

The pressure exerted on companies to practice corporate social responsibility comes from both internal and external stakeholders such as customers, employees, unions, shareholders, business partners, governments, NGOs and the media. Several related criteria are identified to evaluate suppliers' corporate social responsibility practices.

Discrimination: The treatment of supplier's labour should not be based on their membership in a certain group or category. **Occupational Health and Safety:** relates to the concerns the labour safety and their health including both the physical and mental health. It could include the ergonomic, safety measure, and other instruments that ensure the labour safety.

Child Labors and Rights: refers to the employment of children in any work that deprives children of their childhood, interferes with their ability to attend regular school, and that is mentally, physically, socially or morally dangerous and harmful.

Internal Training Program: refers to the training that is given inside the company, rather than trainings given by external training program or seminars.

Criteria related to Environmental Responsibility

The following environmental criteria are related to the design phase of the product life cycle

Material Selection: refers to the appropriate material that contributes to product functionality and minimizes environmental impact. Commonly used material is considered, such as Aluminium, Copper, Iron and Steel, Lead, Zinc, Plastics and so on.

Design for Process: refers to the chosen manufacturing method that generates the least pollutant emissions.

Design for Disassembly: refers to designing a product that is easier to be disassembled to component status through selecting the right material, altering the product architecture (Modular Design), and use of fastener.

Design for Recycling and Reuse: allows a used product to be disassembled into components for recycling and reuse.

Packaging: refers to the packaging used for protecting the product. Commonly used packaging material are various, for instance, Corrugated Fibreboard, Recycle and Remoulded HDPE, Chipboard, Paper Dunnage/ Wraps, Moulded Starch Peanuts in Bags, Padded 100% Packaging Paper Bags, Suspension Style Packaging, EPU (Polyurethane Foam), Plywood Crates, Wooden Pallets, EPS (Polystyrene Foam), EPE (Polyethylene Foam), EPP (Polypropylene Foam), Plastic Bubble Wrap, Stretch/ Shrink Wrap or Bags, Pressure Sensitive Tape, Corrugated Plastic, Commingled Foam/ Corrugate, Commingled Foam/ Wood, ESD Static Shielding Bags, Foam-in-place, Foam in bag, Padded bags with plastic bubble core, PVC Plastics, and Foams with CFCs/ HCFCs

The following criteria are related to the production phase of the product life cycle.

Greenhouse Gases Emission: Greenhouse Gases (GHG) Emission includes the by-product in the manufacturing process such as CO₂, CH₄, N₂O, HFCs, PFCs, and SF₆. They are unified into CO₂ equivalent.

Air Pollution: Air Pollution refers to the gas's emission such as NO_x, SO_x, and VOCs.

Waterborne Pollution: Waterborne Pollution refers to the pollutant emitted that affects the water system, e.g. Chemical oxygen demand, Ammonia, Sulphur dioxide, and Nitrogen

Waste: Waste refers to the waste that are not appropriate to be recycled and disposal is the only treatment method.

Energy Consumption: Energy Consumption refers to the consumption of energy or power, in the form of electricity. Conventional and Renewable energy source are included, e.g. fossil fuel, solar, wind, tides, and etc.

The following criteria are geared to the End-of-Life phase in the product life cycle.

Disposal: Disposal refers to the content of product that is unable to be recycled and reused in the service provider's site. This content will be ultimately transported to landfill

Recycle Rate: Recycle Rate refers to the percentage of product unit to be collected and recycled by the service provider

Reuse Rate: Reuse Rate refers to the percentage of reused product without significant repair and refurbishment.

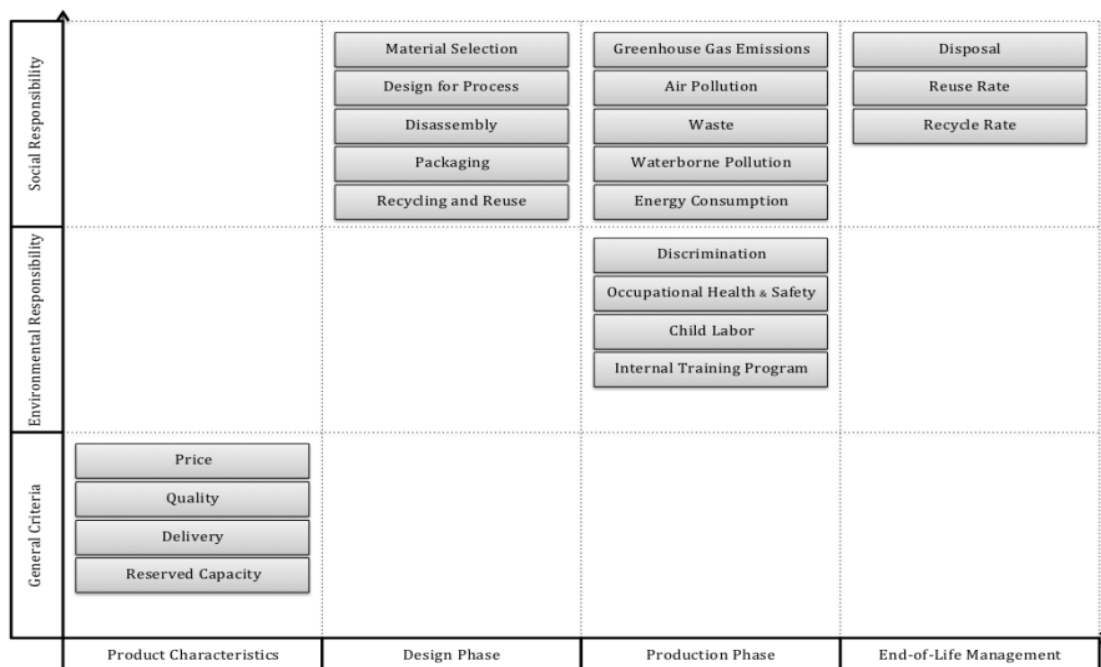


Fig. 4.2.1

The categorization of 21 defined criteria

(Source: <https://hub.hku.hk/bitstream/10722/189926/1/Content.pdf?accept=1>)

4.2.2 supplier evaluation and selection based on green approach in Supply Chain Management

Supplier evaluation and selection based on a green approach in supply chain management involves specific steps to ensure suppliers align with sustainability goals. Here's a process for evaluation and selection:

1. Define Green Criteria:

Establish Requirements: Determine specific environmental criteria that align with your organization's sustainability objectives.

Key Focus Areas: Consider aspects like energy efficiency, waste reduction, use of eco-friendly materials, certifications, and adherence to regulations.

2. Supplier Identification:

Research and Pre-screening: Identify potential suppliers through research, industry databases, or referrals.

Preliminary Assessment: Pre-screen suppliers based on their published sustainability reports, certifications, and environmental policies.

3. Request for Information (RFI) or Pre-Qualification:

Green Questionnaires: Send out detailed questionnaires or RFIs requesting information about suppliers' environmental initiatives and practices.

Pre-qualification Process: Assess suppliers based on their responses regarding sustainability, compliance, and eco-friendly practices.

4. Supplier Assessment:

Evaluate Responses: Analyze the information provided by suppliers, assessing their commitment to sustainability, environmental compliance, and performance metrics.

Site Visits or Audits: Conduct on-site visits or audits to verify claims and observe their environmental practices firsthand.

5. Performance Measurement and Rating:

Establish Metrics: Define key performance indicators (KPIs) to measure supplier performance in environmental areas.

Rating System: Develop a rating system to evaluate suppliers based on their green performance.

6. Decision-making and Selection:

Weighted Criteria: Consider a weighted evaluation where green criteria hold significant importance in supplier selection.

Supplier Scorecards: Utilize scorecards to compare and rank suppliers based on their green performance alongside other factors like cost and quality.

7. Negotiation and Contracting:

Negotiate Terms: Engage with chosen suppliers to discuss commitments to further green initiatives, including terms in the contract related to sustainability.

Clear Expectations: Set clear expectations regarding environmental goals and collaboration on improving sustainability practices.

8. Continuous Monitoring and Improvement:

Regular Evaluation: Continuously assess supplier performance against established green KPIs.

Collaborative Improvement: Collaborate with selected suppliers to implement improvement plans, fostering a culture of continuous improvement in sustainability.

9. Supplier Development:

Training and Support: Provide guidance, resources, or training to suppliers to enhance their green capabilities.

Collaborative Innovation: Encourage joint initiatives to explore innovative ways to improve environmental performance.

4.2.3 Selection of green logistics partner

Selecting a green logistics partner involves careful consideration of their commitment to sustainability, eco-friendly practices, and alignment with the organization's environmental goals.



Fig. 4.2.3

(Source: [GEP](#))

1. Environmental Considerations

- **Supply Chain Carbon Footprint:** In addition to assessing carbon emissions, delve into suppliers' efforts to minimize their carbon footprint. This can include their commitment to using renewable energy sources, optimizing transportation routes and adopting efficient packaging practices.
- **Life Cycle Assessment:** Go beyond understanding suppliers' grasp of the product life cycle. Evaluate their plans for reducing waste at each stage, optimizing resource usage and embracing circular economy principles. Consider how they handle product end-of-life, recycling or repurposing.

2. Social Responsibility

- **Ethical Labor Practices:** Scrutinize not only fair wages and safe working conditions but also the welfare of workers across their entire supply chain. Ask about their engagement with labor unions, training programs and initiatives to improve the lives of their workforce.
- **Diversity and Inclusion:** Dive deeper into suppliers' diversity initiatives. Inquire about their hiring practices, equal opportunities for advancement and programs aimed at promoting diversity and inclusion not only internally but also among their partners.

3. Governance and Compliance

- **Transparency:** Request detailed information about suppliers' transparency practices. How do they communicate their sustainability efforts? Do they openly share progress reports, metrics, and goals with their stakeholders? Transparency indicates a genuine commitment to continuous improvement.
- **Legal and Ethical Compliance:** Apart from assessing past legal infringements, inquire about the suppliers' commitment to future compliance. What measures have they put in place to ensure they adhere to changing regulations? Proactive compliance initiatives showcase a responsible approach.

4. Innovations for Sustainable Operations

- **Eco-Friendly Innovations:** Encourage suppliers to share the innovative practices they have adopted. This could involve using eco-friendly materials, implementing energy-efficient manufacturing processes or utilizing advanced technologies that minimize environmental impact.
- **Collaborative Initiatives:** Explore suppliers' participation in collaborative sustainability initiatives. This could range from partnering with other companies to reduce collective carbon emissions to sharing best practices within their industry.

5. Long-Term Sustainability Roadmaps

- **Sustainability Targets:** Ask suppliers about their short- and long-term sustainability goals. This demonstrates their commitment to ongoing improvement and aligning with your business's sustainability journey.
- **Investment in Sustainability:** Inquire about the resources they allocate for sustainability initiatives. Suppliers willing to invest in eco-friendly technologies, workforce training and sustainable practices showcase their dedication.

6. Supply Chain Transparency and Traceability

- **Supply Chain Mapping:** Request detailed information about suppliers' upstream and downstream partners. A transparent supply chain demonstrates accountability and a commitment to preventing unethical practices or environmental harm throughout the value chain.

- **Raw Material Sourcing:** Investigate the origin of suppliers' raw materials. Ethical sourcing practices, such as using responsibly harvested materials or ensuring conflict-free minerals, contribute to a more sustainable supply chain.

By embedding greening criteria into logistics partner selection, businesses actively contribute to a more ethical, equitable and environmentally conscious future. Beyond regulatory compliance, this approach fuels innovation, elevates brand reputation, and lays the foundation for a transformed business landscape.

Logistics companies known for following green logistics practices

SEROS Logistics: It is a Surat-based logistics company. It has launched several initiatives to adopt energy-efficient logistics practices. In fact, the company has a division named SEROS Green that is committed to moving cargo with cleaner fuel.

Greenline Logistics: This Mumbai-based company offers green logistics services to its customers. The company has an LNG-fuelled fleet of trucks. Switching from conventional fuel to LNG can reduce carbon emissions approximately by 30%.

DHL: A global logistics giant is committed to climate protection. It aims at electrifying 60% of last-mile deliveries and long-distance shipments with electric vehicles by 2030.

Mahindra Logistics: The company has more than 1300 electric vehicles for greening transport services. Additionally, it aims to become carbon neutral by 2040. Other initiatives of green logistics include LEED-certified facilities, solar-powered warehouses, etc.

[*TATA nexarc*](#)

4.2.4 Selection of transporter following green practices

Carrier selection refers to the process of carefully evaluating and choosing a suitable carrier or transportation service provider to handle the movement of goods or passengers from one location to another. It involves considering various factors such as service level, cost, capacity, equipment, geographic coverage, and safety and security. By making an informed decision based on these criteria, organizations can ensure the efficient and reliable transportation of their goods or passengers while balancing cost-effectiveness and service quality.

Strategic shippers realize one obvious way to “green up” their operations is to examine supply chains, which include providers, practices, and processes. Because the transportation providers they work with are such big pieces of the supply chain puzzle, it makes sense – and cents – to work with those committed to operating as environmentally soundly as possible.

Even with thousands of transportation providers out there, choosing one that is green doesn't have to be confusing or overwhelming.

Being green for the sake of being green will only get a business so far. Shippers need to think about sustainability in terms of a triple bottom line:

- ✓ financial responsibility
- ✓ social responsibility
- ✓ environmental responsibility

To be truly sustainable, any transportation provider needs to be financially stable. It should demonstrate its ability to invest in people, equipment and plans for the future. Ideally, it should have a long track record of success. A company that's been around for a while likely knows how to weather the ebbs and flows of the economy and industry-shifting changes (like the green movement).

Another element of sustainability is social responsibility. The company should be very comfortable talking about its core values and how its people live them day in and day out. Ask how the transportation providers give back to their communities too. Charitable donations and projects often give an insight into how a carrier will be when work together.

Multiple transportation mode provides sustainability options

There is more than one way to get from point A to point B. If a provider only has one mode under its umbrella, its options – and its shippers' options – are limited. What if that's not the most efficient route or mode of transport? That's the beauty of a multimodal provider. Look for a company that offers a broad portfolio of services, particularly featuring truckload, intermodal and brokerage.

Shippers can be even greener by moving freight via intermodal service, which is a combined truck and rail move.

Today, rail can move a ton of freight an average of 479 miles per gallon of fuel. In fact, moving freight by rail instead of truck lowers greenhouse gas emissions by 75%.

Another great environmental benefit of intermodal is that it also reduces highway congestion: A single train can take 280 trucks off the highway.

Evaluating energy-efficient equipment

The transportation industry has made great strides in the last decade toward the development and implementation of best practice equipment. Top performers embarked on a quest to reduce both fuel consumption and environmental footprint. The result is a simple checklist when examining a particular transportation provider.

Engines: The EPA has guided the industry to a new level when it comes to low-emitting engines. The next phase will be engineering more fuel efficiency. If a large portion of the carrier's fleet is comprised of trucks meeting the latest EPA requirements, it's not only a good sign – it's a great sign. Newer add-ons are predictive cruise control systems, which optimize shifting.

Aerodynamic features: This is a case where little things really do mean a lot. Truck design and aerodynamic additions to the tractor and trailer/container can save a fleet emissions and fuel. The most

common features on tractors include side extenders, optimized roof fairings, fuel tank skirts and aerodynamic mirrors and bumpers.

Slower speeds: Carriers who care about conservation know to place speed governors on trucks. It's proven that the slower the vehicle travels, the less fuel it consumes. In fact, a truck traveling at 75 mph consumes 27 percent more fuel than one going 65 mph.

No-idle heating/cooling systems: For decades, drivers who wanted to adjust the temperature in cabs when snoozing at a truck stop needed to idle the engine to do so. No longer. Engine-off solutions now exist.

Stackable intermodal containers: When stackable containers are an option, the amount of freight able to make that move multiplies quickly.

Safety Record: Check the carrier's safety record, including their compliance with regulations and any past incidents or accidents. Look for carriers with a strong safety culture and a good track record of safe operations.

Experience and Expertise: Consider the carrier's experience and expertise in the type of transportation you require. Carriers possess expertise in different sectors or categories of cargo, such as hazardous substances, perishable items, or bulky loads. Opt for a carrier with a well-established reputation for effectively managing your unique transportation needs.

Licensing and Insurance: Verify that the carrier holds all necessary licenses and permits required for transporting goods. Additionally, ensure that they have appropriate insurance coverage to protect your cargo in case of damage or loss during transportation.

Equipment and Fleet: Evaluate the carrier's equipment and fleet to ensure they have the capacity and capability to handle your shipments. Take into account elements like the age and state of the vehicles, their capacity, and any specialized equipment or advanced technology they possess

Network and Coverage: Determine the carrier's network and coverage area to ensure they can transport your goods to the desired destinations. If you require international transportation, check if they have experience and partnerships in the relevant countries or regions

Reliability and On-Time Delivery: Look for carriers with a reputation for reliability and on-time delivery. Late deliveries can disrupt your supply chain and impact customer satisfaction, so it's essential to choose a carrier that can meet your delivery deadlines consistently

Pricing and Cost Transparency: Compare the pricing structures and rates of different carriers to find one that aligns with your budget. Additionally, ensure that the carrier provides transparent pricing, with no hidden fees or charges

Customer Service and Communication: Evaluate the carrier's customer service and communication practices. A carrier with responsive and proactive customer service can provide updates on shipment status, address any concerns promptly, and maintain clear lines of communication throughout the transportation process

Environmental Considerations: If sustainability is a priority for your business, inquire about the carrier's environmental practices and initiatives. Some carriers may offer greener transportation options or have implemented strategies to reduce their carbon footprint.

References and Reviews: Gather references or peruse reviews from fellow businesses or individuals who have previously utilized the carrier's services. This will offer valuable insights regarding their performance, dependability, and overall level of customer satisfaction.

4.2.5 Assessment criteria to evaluate sustainable suppliers

When evaluating suppliers based on sustainable practices, consider the following comprehensive assessment criteria:

1. Environmental Management:

Environmental Policies: Review their documented environmental policies and commitment to sustainability.

Compliance and Certifications: Check adherence to environmental laws and presence of certifications (ISO 14001, EMAS).

Environmental Performance: Assess their track record in reducing environmental impact, emissions, and resource usage.

2. Energy Efficiency:

Energy Consumption: Evaluate efforts to reduce energy use in operations, facilities, and logistics.

Renewable Energy Use: Consider adoption or plans for renewable energy sources in their operations.

Energy-saving Initiatives: Assess implementation of energy-efficient equipment or practices.

3. Waste Management:

Waste Reduction: Check strategies to minimize waste generation and increase recycling rates.

Waste Disposal: Assess responsible waste disposal practices, including hazardous waste management.

Circular Economy Initiatives: Consider efforts promoting reuse, repair, or recycling of materials.

4. Sustainable Sourcing:

Sustainable Material Use: Assess the use of sustainable or ethically sourced materials in their products.

Supply Chain Transparency: Check transparency in supply chain practices and ethical sourcing.

Avoidance of Conflict Minerals: Ensure compliance with regulations avoiding conflict minerals.

5. Social Responsibility:

Labor Standards: Evaluate compliance with labour laws, fair wages, worker safety, and diversity policies

Community Engagement: Assess involvement in community programs and social initiatives.

Human Rights Commitment: Check alignment with international human rights standards.

6. Innovation and Continuous Improvement:

Innovative Solutions: Evaluate adoption of innovative technologies or practices for sustainability.

Continuous Improvement: Check commitment to setting and achieving sustainability goals.

7. Metrics and Reporting:

Data Reporting: Assess ability to measure and report environmental impact through key metrics.

Transparency: Check willingness to share environmental performance data and reports.

8. Risk Management:

Risk Assessment: Evaluate identification and mitigation strategies for environmental risks.

Emergency Preparedness: Assess readiness to handle environmental emergencies or incidents.

9. Stakeholder Engagement:

Supplier Collaboration: Consider engagement in collaborative sustainability initiatives.

Stakeholder Involvement: Assess involvement and response to stakeholder sustainability concerns.

10. Regulatory Compliance:

Legal Compliance: Ensure adherence to environmental laws, regulations, and industry standards.

By assessing suppliers against these criteria, organizations can partner with suppliers aligned with sustainable practices, fostering a more environmentally responsible supply chain.

4.2.6 Overview of supply chain map

What is a Supply Chain Map?

A supply chain map is a visual representation of the entire supply chain, from raw materials to the final product. It is a comprehensive overview of all the stages involved in the supply chain process of producing and delivering goods or services. A map can be used to identify key players, including suppliers, manufacturers, distributors, retailers, and customers, and it can help businesses understand the flow of goods and services throughout the supply chain.

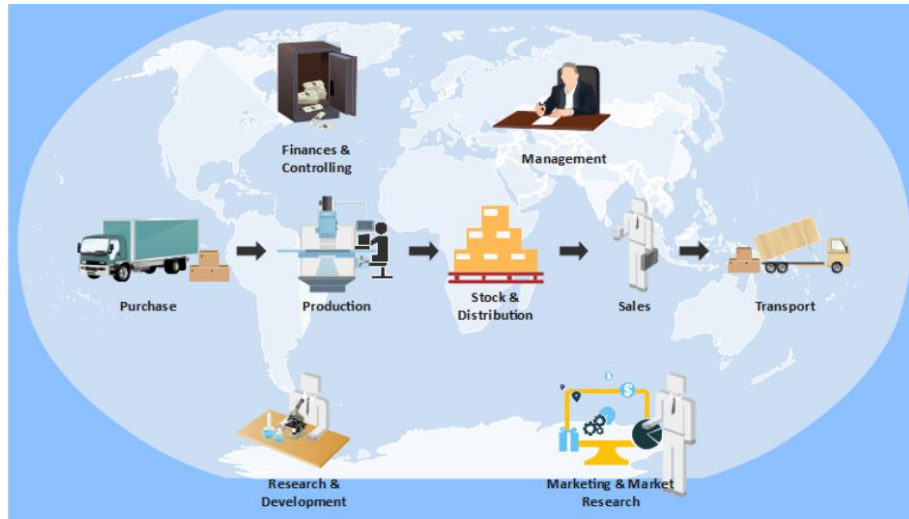


Fig. 4.2.6

(Source: [Wondershare EdrawMax](#))

A map should provide a comprehensive overview of all the stages in the supply chain

Components of a Supply Chain Map

A *supply chain process map* consists of four key components: suppliers, manufacturers, distributors, and customers. Understanding these components is essential for creating an accurate and detailed map of the supply chain

Suppliers

Suppliers are the entities that provide raw materials or components to the manufacturers. They can be local or international, and their location can have a significant impact on the lead time and cost of production.

For example, if a manufacturer relies on a supplier located in a different country, they may face longer lead times due to shipping and customs clearance processes. It is essential to identify all the suppliers and their locations to understand the complete picture of the supply chain.

Manufacturers

Manufacturers are the entities that transform raw materials or components into finished products. They can be located in different parts of the world and can have different production capacities. Understanding the manufacturing process and the location of manufacturers in the supply chain is crucial for identifying potential bottlenecks and inefficiencies.

For instance, if a manufacturer is located in a region with a high labour cost, the cost of production may be higher than in other regions, leading to increased costs and longer lead times.

Distributors

Distributors are the entities that distribute the finished products to the customers. They can be wholesalers, retailers, or online marketplaces. Distributors play a vital role in the supply chain, as they bridge the gap between the manufacturer and the end-user. Understanding the role of distributors in the supply chain is crucial for identifying potential inefficiencies and improving the overall efficiency of the supply chain.

For example, if a distributor is located far away from the end users, the lead time for delivery may be longer, affecting customer satisfaction.

Customers

Customers are the end-users who consume or use the finished products. They can be located in different geographical locations and have different buying behaviours. Understanding the needs and behaviours of customers is crucial for designing an efficient supply chain that meets their demands.

For instance, if a product is in high demand in a particular region, it may be necessary to establish a local manufacturing or distribution centre to reduce lead times and costs.

How to Create a Supply Chain Map?

Several steps to create *supply chain maps* and *supply chain mapping examples* are outlined below.

- **Step 1: Identify the Entities Involved in the Supply Chain**

The first step in supply chain process mapping is to identify all the entities involved in the supply chain. This can include suppliers, manufacturers, distributors, retailers, and end-users. It's essential to identify every entity that plays a role in the supply chain to ensure that the map is comprehensive and accurate.

- **Step 2: Map the Flow of Goods, Services, and Information**

Once all the entities involved in the supply chain are identified, the next step is to map the flow of goods, services, and information between them. This can be done using a flowchart or a diagram that shows the movement of products, information, and services from suppliers to end users. Mapping the flow of goods, services, and information is crucial in identifying potential bottlenecks and areas for optimization within the supply chain.

- **Step 3: Analyze the Supply Chain**

After mapping the flow of goods, services, and information, the next step is to analyze the supply chain. This involves identifying areas where the supply chain can be optimized, such as reducing lead times, improving quality, or lowering costs. Analyzing the supply chain can help businesses identify areas for improvement, which can lead to increased efficiency, reduced costs, and improved customer satisfaction.

- **Step 4: Develop Strategies to Improve the Supply Chain**

Based on your analysis, develop strategies to improve the supply chain. This can include renegotiating contracts with suppliers, implementing new technology, or improving communication channels. Developing strategies to improve the supply chain is crucial in ensuring that businesses can meet the demands of customers, reduce costs, and stay competitive in the market.

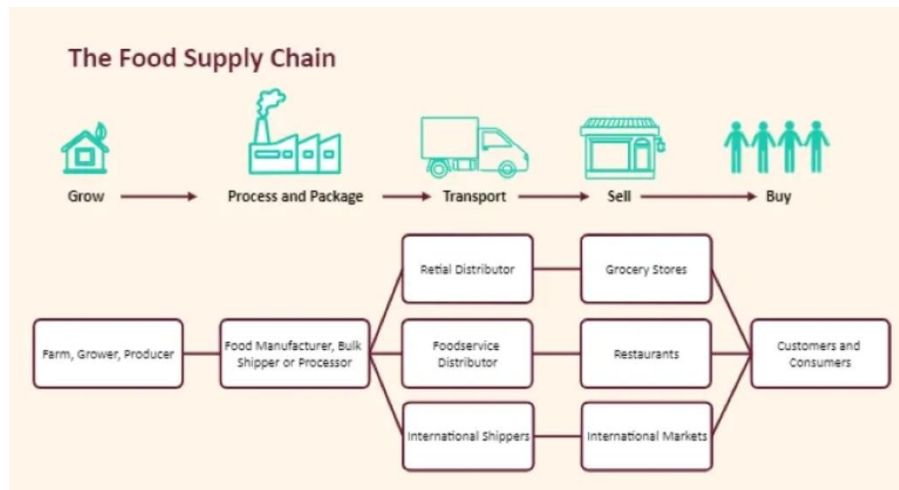


Fig. 4.2.6

Supply chain mapping template for a food & beverage company

The Environmental impact of supply chain at each stage

The environmental impact of a product varies greatly depending on its life cycle and the specific practices employed at each stage of its supply chain. However, here's a general overview of the potential environmental concerns associated with different stages:

1. Raw Material Sourcing:

- **Resource depletion:** Mining, harvesting, and extraction of raw materials can deplete non-renewable resources like fossil fuels, minerals, and timber.
- **Habitat destruction:** Clearing land for raw material production can damage ecosystems and contribute to biodiversity loss.
- **Water pollution:** Mining and processing raw materials can lead to water contamination from chemicals and waste.
- **Air pollution:** Emissions from extraction, transportation, and processing of raw materials can contribute to air pollution and climate change.

Examples:

Mining for metals (e.g., gold, copper) can result in habitat destruction, water pollution, and soil contamination.

Logging for timber can lead to deforestation, loss of biodiversity, and soil erosion.

2. Manufacturing:

- **Energy consumption:** Manufacturing processes typically require significant energy, often sourced from fossil fuels, leading to greenhouse gas emissions.
- **Water usage:** Industrial processes consume large amounts of water, potentially straining local water resources.
- **Air and water pollution:** Manufacturing often generates air and water pollution from emissions, chemicals, and waste disposal.
- **Hazardous materials:** Certain manufacturing processes use hazardous materials that can pose health and environmental risks if not handled responsibly.

Examples:

Energy-intensive manufacturing processes in industries like automotive, steel, and chemical production contribute to carbon emissions and air pollution.

Chemical manufacturing can produce hazardous waste and emissions, leading to environmental contamination.

3. Distribution and Transportation:

- **Greenhouse gas emissions:** Transportation, particularly using fossil fuels, is a major contributor to greenhouse gas emissions and climate change.
- **Air pollution:** Vehicle emissions contribute to air pollution, impacting local air quality and human health.
- **Packaging waste:** Excessive or unsustainable packaging materials can create significant waste disposal challenges.

Examples:

Freight transportation by trucks and airplanes releases greenhouse gases and pollutants into the atmosphere, contributing to air pollution and climate change.

Shipping and maritime transport can result in oil spills, marine pollution, and habitat destruction in coastal areas.

4. Warehousing and Storage:

- **Energy Consumption:** Warehouse operations require energy for heating, cooling, lighting, and equipment, contributing to carbon emissions and energy consumption.

- **Waste Generation:** Packaging waste, obsolete inventory, and discarded materials can accumulate in warehouses, contributing to landfill waste and environmental pollution.

Examples:

Refrigerated warehouses consume significant energy for temperature control, leading to higher energy consumption and carbon emissions.

Inefficient inventory management practices can result in excess inventory, leading to waste and environmental impact.

5. Retail and Consumer Use:

- **Energy consumption:** Retail stores and online platforms require energy for lighting, heating/cooling, and technological operations.
- **Waste generation:** Unsold products, expired items, and packaging materials from consumer disposal contribute to waste generation.

Examples:

Retail stores often use energy-intensive lighting and heating systems, contributing to carbon emissions and energy consumption.

Consumer products packaged in single-use plastics contribute to plastic waste and environmental pollution.

6. End-of-Life Management:

- **Landfilling:** Disposing of products in landfills creates land pollution and consumes valuable space.
- **Incineration:** Incineration of waste can release harmful air pollutants and generate toxic ash.
- **Recycling and reuse:** Improper recycling infrastructure or insufficient product reusability can limit the potential for these more sustainable end-of-life options.

Examples:

Electronic waste (e-waste) disposal can result in hazardous materials leaching into the environment if not properly managed.

Recycling of materials such as paper, plastic, and metal helps conserve resources and reduce environmental impact.

4.2.7 Companies using upgraded technologies

The logistics industry has been undergoing a massive transformation in recent years, and technology is at the forefront of this revolution. With the advancement of artificial intelligence, robotics, and automation, the supply chain is becoming more efficient and streamlined than ever before.

One of the key areas where technology is making a significant impact is in inventory management. With the help of smart sensors and IoT devices, warehouses can now monitor stock levels in real-time, allowing for better inventory planning and optimization. This not only reduces costs but also improves customer satisfaction by ensuring products are always available when needed.

Another exciting development in logistics is the use of drones and autonomous vehicles for last-mile delivery. These innovative solutions are not only faster and more cost-effective but also environmentally friendly, reducing carbon emissions associated with traditional delivery methods.

Key technologies revolutionizing logistics

- **Automation and robotics in the supply chain**

Automation in distribution centers or warehouses pertains to anything from process automation to physical automation. Some of the usual warehouse automation includes digitization of manual processes, pick-to-light systems with operators using barcodes to scan and LED lights indicating the number of items for pick-up and their destination, and autonomous mobile robots working inside warehouses.

- **Artificial intelligence and machine learning**

There's no doubt that artificial intelligence (AI) and machine learning (ML) have been playing significant roles in the transformation of the transportation and logistics sector. The large flow of data coming from the supply chain is now commonly being utilized and analyzed to spot trends in the supply chain so logistics firms can make necessary changes in their organizations.

The use of AI and ML to identify every link within the supply chain is set to continue. AI, when used for decision-making in the supply chain, reduces human error. AI in warehouse management makes the planning process much easier by speeding up analysis time.

- **Blockchain Technology**

One of the most significant advancements in logistics is the implementation of blockchain technology. Blockchain is a decentralized digital ledger that allows for secure and transparent transactions. In the logistics industry, blockchain can be used to track and verify the movement of goods throughout the supply chain.

By using blockchain, companies can create a secure and immutable record of every transaction, ensuring transparency and reducing the risk of fraud. This technology also enables real-time tracking of goods, allowing for better visibility and traceability. Additionally, blockchain can streamline the documentation

process by eliminating the need for paper-based records, reducing paperwork and human error.

- **Internet of Things (IoT)**

IoT is transforming all aspects of our lives. In logistics, expanding IoT results in the growth of various connections between goods, packaging, transportation hubs, and vehicles. It offers more data that helps in managing assets remotely, predicting risk, ensuring proper cargo handling, and forecasting traffic congestion. IoT when combined with blockchain technology may offer end-to-end visibility of packages.

- **Autonomous Vehicles and Drones**

The logistics landscape will see a significant increase in the use of autonomous vehicles for transportation. Self-driving trucks will revolutionize long-haul freight, reducing labor costs and improving safety. Drones, both ground-based and aerial, will be used for last-mile delivery in urban areas, enhancing speed and efficiency while reducing congestion and emissions.

→ **Several logistics companies have adopted upgraded technologies to improve efficiency, reduce costs, and enhance overall operations.**

Here are a few examples of logistics companies leveraging advanced technologies and the benefits they are experiencing:

1. FedEx: Autonomous Delivery Robots

Technology: FedEx introduced the SameDay Bot, an autonomous delivery robot, to handle short-distance deliveries in urban areas. The bot uses artificial intelligence, sensors, and machine learning algorithms to navigate sidewalks and deliver packages.

Benefits:

Improved Last-Mile Delivery: The use of autonomous delivery robots enables FedEx to make faster and more efficient last-mile deliveries, reducing delivery times and costs.

Enhanced Customer Experience: Customers benefit from shorter delivery times and increased flexibility, leading to higher satisfaction and loyalty.

Sustainability: By reducing the need for traditional delivery vehicles, FedEx decreases carbon emissions and environmental impact, contributing to sustainability goals.

2. Amazon: Drone Delivery

Technology: Amazon Prime Air utilizes drones for package delivery, enabling faster and more efficient deliveries to customers' doorsteps. The drones are equipped with advanced sensors, GPS, and computer vision technology to navigate and deliver packages autonomously.

Benefits:

Expedited Delivery: Drone delivery allows Amazon to expedite delivery times, particularly for urgent orders, enabling same-day or even within-the-hour delivery options.

Cost Reduction: By automating delivery processes and reducing reliance on traditional delivery methods, Amazon saves on labor and transportation costs.

Innovation Leadership: Amazon's use of drone technology demonstrates its commitment to innovation and customer-centricity, enhancing its brand reputation and market competitiveness.

3. DHL: Internet of Things (IoT) in Warehousing

Technology: DHL employs IoT devices and sensors in warehouses to track inventory levels, monitor storage conditions, and optimize warehouse operations. IoT devices are integrated with warehouse management systems (WMS) to provide real-time visibility and insights into warehouse activities.

Benefits:

- **Inventory Accuracy:** IoT sensors enable DHL to accurately track inventory levels, reduce stockouts, and minimize inventory discrepancies, improving order fulfillment and customer satisfaction.
- **Operational Visibility:** Real-time data from IoT devices provides DHL with visibility into warehouse operations, enabling proactive decision-making, resource allocation, and process optimization.
- **Cost Reduction:** By improving inventory management and operational efficiency, DHL reduces costs associated with excess inventory, stockouts, and manual processes, leading to cost savings and improved profitability.

4. Delhivery - Optimizing Last-Mile Delivery with Technology

Delhivery is one of India's leading logistics and supply chain solution providers, offering a wide range of services including express parcel delivery, freight transportation, warehousing, and fulfillment. Delhivery faced challenges in optimizing last-mile delivery operations, including inefficient route planning, delivery delays, and high operational costs. The company aimed to enhance its last-mile delivery capabilities to meet growing customer expectations for faster and more reliable service which are as follows:

- **Route Optimization Software:** The company implemented route optimization software powered by AI and machine learning algorithms to optimize delivery routes based on factors such as package volume, delivery locations, traffic patterns, and time constraints.
- **Mobile Delivery Apps:** Delhivery equipped its delivery personnel with mobile apps that provide real-time visibility into delivery assignments, route guidance, customer information, and digital proof of delivery. The apps enable drivers to efficiently manage their delivery routes, communicate with customers, and handle exceptions.
- **GPS Tracking and Telematics:** Delhivery deployed GPS tracking devices and telematics systems in its delivery vehicles to monitor vehicle location, speed, and performance in real-time. This allows the company to track the progress of deliveries, optimize route adjustments, and improve overall fleet efficiency.

5. Mahindra Logistics - Driving Digital Transformation in Supply Chain Management

Mahindra Logistics is a leading third-party logistics (3PL) service provider in India, offering end-to-end logistics solutions across various industries including automotive, consumer goods, pharmaceuticals, and e-commerce.

- **Supply Chain Visibility Platform:** The company implemented a supply chain visibility platform powered by IoT, cloud computing, and data analytics. This platform provides real-time visibility into inventory levels, shipment status, and transportation routes, enabling proactive decision-making and exception management.
- **Warehouse Automation:** Mahindra Logistics invested in warehouse automation solutions such as robotics, automated guided vehicles (AGVs), and warehouse management systems (WMS). These technologies optimize inventory storage, order picking, and fulfillment processes, improving efficiency and accuracy in warehouse operations.
- **Transportation Management System (TMS):** The company adopted a TMS to streamline transportation planning, route optimization, and load management. The TMS integrates with GPS tracking systems to monitor vehicle movements in real-time, optimize delivery routes, and provide ETAs to customers.
- **Digital Collaboration Tools:** Mahindra Logistics implemented digital collaboration tools and mobile applications for communication, collaboration, and workflow automation across its supply chain network. These tools enable seamless communication between stakeholders, automate manual tasks, and enhance transparency and accountability.

Benefits:

- **Improved Efficiency:** Mahindra Logistics achieved significant improvements in supply chain efficiency through real-time visibility, automation, and optimization of inventory and transportation processes.
- **Enhanced Customer Experience:** The company's digital transformation initiatives led to faster order processing, accurate inventory tracking, and on-time deliveries, resulting in higher customer satisfaction and loyalty.
- **Cost Reduction:** By optimizing inventory levels, streamlining transportation routes, and automating manual processes, Mahindra Logistics realized cost savings and improved overall profitability.
- **Agility and Scalability:** The digital transformation initiatives enabled Mahindra Logistics to adapt quickly to changing market dynamics, scale its operations, and meet evolving customer demands more effectively.

4.2.8 Implementation of Smart Logistics solutions

Smart logistics is the process of adopting new technologies in all operations carried out inside and outside the warehouse, from the receipt of goods at the loading docks to their distribution and delivery to the end customer as per the agreed upon conditions. In other words, this concept involves the implementation of smart devices to automate logistics operations.

Technologies such as AI (artificial intelligence), big data, and the Industrial Internet of Things (IIoT) have gained a foothold and are set to completely transform warehouses. A study by consulting company Research and Markets indicates that the Internet of Things (IoT) in the logistics market will grow by 13.2% annually until 2030, going from a turnover of \$34.5 billion in 2019 to \$100 billion by 2030.

Smart logistics aims to increase process control, optimize warehouse throughput, and minimize operating costs in the facility through the use of interconnected devices and the collection of all the data produced in it. Analysis of these data provides logistics managers with comprehensive information on what is happening in the warehouse before making a decision.

Smart logistics involves using technology and data analytics to optimize the supply chain. This can include:

- **Real-time tracking:** Using sensors and GPS technology, intelligent logistics systems can track the location and condition of goods in real-time. This allows companies to identify any issues or delays and take corrective action quickly.
- **Predictive analytics:** By analyzing data from past shipments, innovative logistics systems can predict future demand and optimise routes and schedules to ensure that goods are delivered on time.
- **Automated inventory management:** Smart logistics systems can use sensors and RFID tags to automatically track inventory levels and alert warehouse managers when supplies are running low.
- **Robotics and automation:** Automated robots can be used to move goods within warehouses, reducing the need for manual labour and increasing efficiency.

- **Blockchain technology:** Blockchain technology can create a secure and transparent supply chain, reducing the risk of fraud and ensuring that goods are delivered to the right place at the right time.

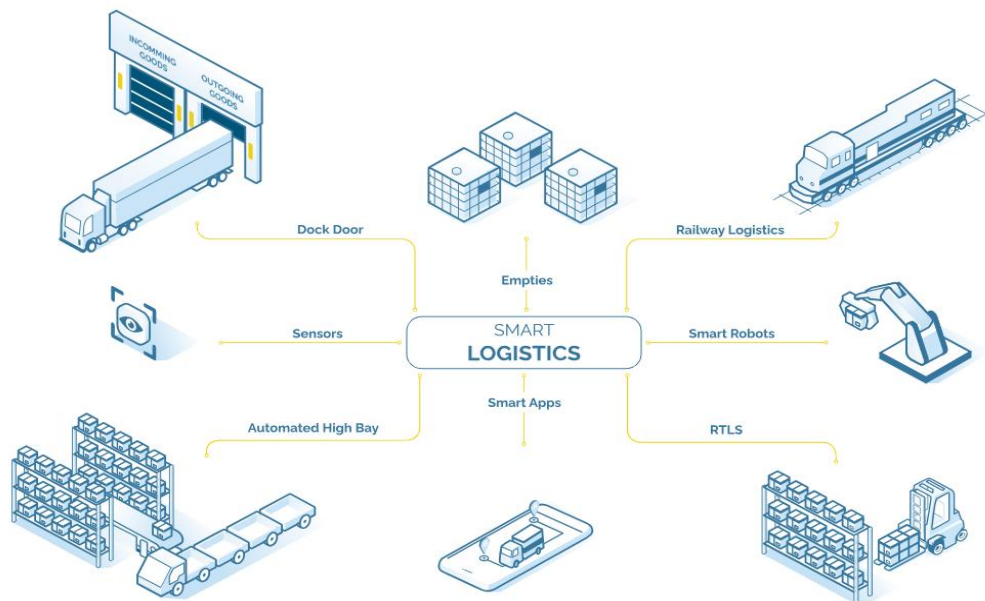


Fig. 4.2.8

Source: [leogistics](https://www.leogistics.com)

Benefits of Smart Logistics

Implementing smart logistics solutions can bring a range of benefits:

- **Reduced costs:** By optimising routes and schedules and automating manual processes, companies can save money on transportation, labour, and storage costs.
- **Improved efficiency:** Smart logistics systems can help companies streamline operations, reduce errors, and increase productivity.
- **Enhanced customer experience:** Companies can improve customer satisfaction and loyalty by providing real-time tracking and delivery updates.
- **Increased visibility:** Smart logistics systems provide real-time visibility into the entire supply chain, allowing companies to identify issues and take corrective action quickly.

Examples of Smart Logistics in Action

Some real-world examples of companies that have implemented intelligent logistics solutions:

- **Amazon:** Amazon is known for its advanced logistics capabilities, including everything from real-time tracking to robotics and automation. By leveraging data and technology, Amazon has been able to provide fast, reliable delivery to customers around the world.

- **United Parcel Service (UPS):** UPS is an American company that transports packages. UPS has invested heavily in intelligent logistics solutions, including a new system that uses artificial intelligence to optimise package delivery routes. This has allowed UPS to reduce delivery times and increase efficiency.
- **Maersk:** Maersk, the world's largest container shipping company, has implemented blockchain technology to create a secure and transparent supply chain. Using blockchain, Maersk has been able to reduce the risk of fraud and ensure that goods are delivered to the right place at the right time.
- **Walmart:** Walmart has implemented intelligent logistics solutions to improve its inventory management and reduce costs. By using sensors and data analytics, Walmart has optimised its supply chain and reduced waste.

4.2.9 Example of emerging future technologies in green logistics



Future technologies and their potential applications in green logistics

Exercise

1. What is Green Purchasing and its importance?

2. What are the different ways to implement green purchasing in supply chain management?

3. Explain the criteria for selecting a supplier and a transporter following green practices?

4. How to evaluate and select a supplier based on green approach in supply chain management?

5. Explain the steps involved in selecting green logistics partner

6. What is a supply chain map

7. Explain how to implement smart logistics solutions

8. What are the emerging future technologies in green logistics?

5. Sales and transportation of agricultural residues



Unit 5.1 Occupational health & safety standards in green logistics

Unit 5.2 Emergency situations in green logistics

Unit 5.3 Administering first aid

Unit 5.4 Usage of PPE in green logistics

Unit 5.5 Housekeeping practices



Chapter Highlights



After reading this chapter, you will be able to: %

1. Understand the occupational health & safety standards and requirements for safe work area in green logistics
2. Identify the different emergency situations in green logistics
3. Know about the fire protection in green logistics
4. Understand how to administer first aid
5. Understand the work safety procedures in green logistics
6. Identify the usage of personal protective equipment (PPE) in green logistics
7. Establish the housekeeping practices and how to prevent infection at the workplace
8. Understand the case study on practice of safety in green logistics

Unit 5.1: Occupational health & safety standards in green logistics

Unit Objectives

At the end of this unit, you will be able to:

1. Introduction to occupational health & safety standards in green logistics
2. Regulations & requirement for safe work area in green logistics

5.1.1 Occupational health and safety in green logistics

The following safety standards are required in the workplace

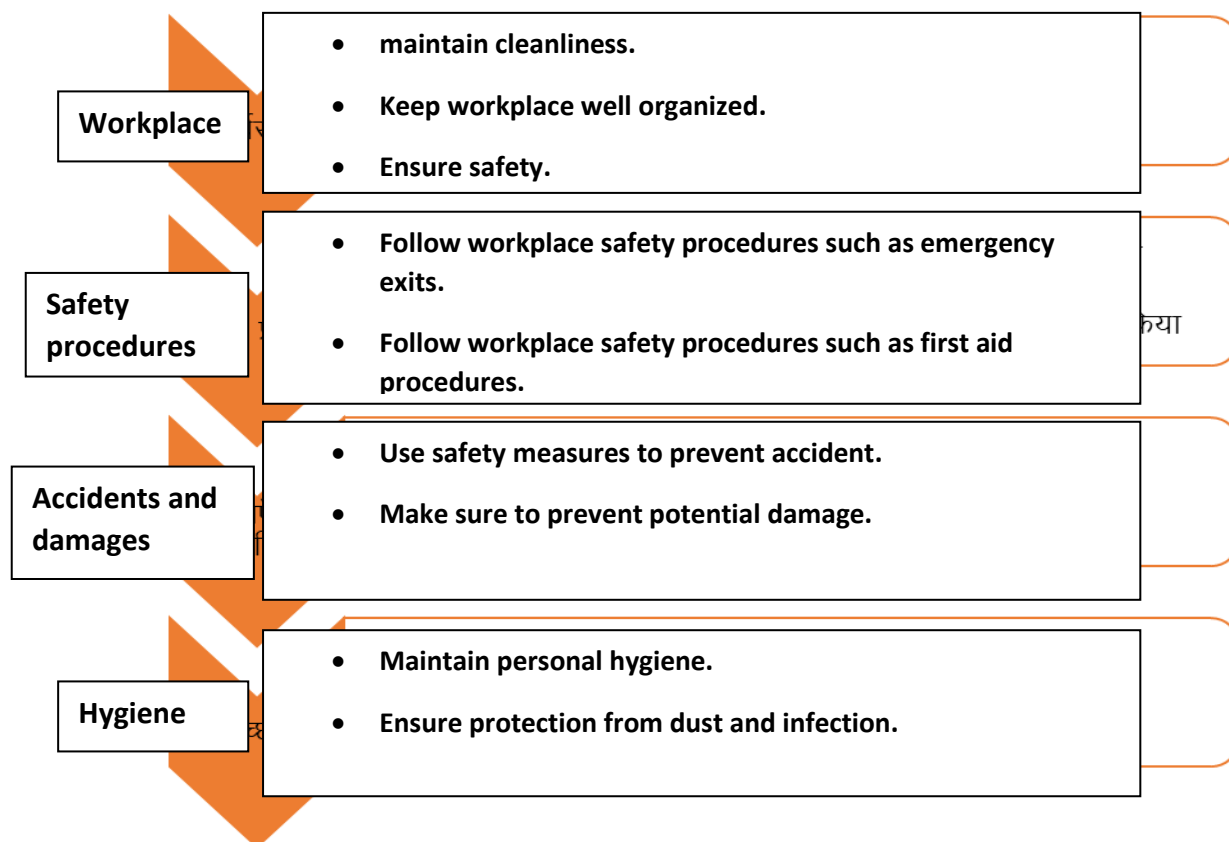


Fig. 5.1.1

Occupational health and safety standards

5.1.2 Regulations and requirement for safe work area in green logistics

Accidents can be caused by weather conditions, use of machinery, equipment, or vehicles. In fact, non-observance of safety rules or lack of knowledge can also cause injuries. The following measures are necessary for accident prevention:

- **Handle equipment with utmost care and safety.**

- **Equipment should always be usable.**

- **Harmful substances and chemicals should be used carefully.**

- **Such material should never be left unattended in the open.**

- **Personal Protective Equipment (PPE)**

- **PPE must always be in working condition and available at all times.**
- **Must have the necessary information to operate the PPE.**
- **Demonstrate use of safe and essential practices for personal**

Item description	Equipment Name	Pictures of equipment
1	Safety helmet or head protection: The worker should use safety helmet while working at the site. Safety helmets help to avoid any head injury.	
2	Safety Goggles: Safety goggles should be worn to protect your eyes.	
3	Nasal Mask: Nasal mask will help protect against bacteria and virus etc.	
4	PVC rubber hand gloves: Gloves will help protect against direct contact with mud.	
5	Acid /base proof suit: Acid /Base proof suit provides protection from direct contact with mud.	
6	Gum Boots: The worker should wear gum boots while performing the activities.	

7	Safety Light / Torch: Safety torch can be used properly for inspection.	
8	First Aid Box: First aid box can be used in case of any injury.	
9	Fire Extinguishers: Fire extinguishers can be used in case of fire on site.	

Fig. 5.1.2 Use of personal protective equipment

UNIT 5.2: Emergency situations in green logistics

Unit Objectives

At the end of this unit, you will be able to:

1. Handle the multiple emergencies like **operational emergency**, fire emergency, security emergency and medical emergency.

5.2.1 Handle the multiple emergencies in green logistics

Given the number of hazardous materials at the workplace, a contingency plan should be prepared beforehand to handle such situation. Broadly the emergency measures at the workplace can be divided into operational emergencies, fire emergency, security emergency and medical emergency.

This unit focuses on operational emergencies, fire emergencies, medical emergency and theft or vandalism.

General steps to be taken in order to be able to handle of any emergency at the workplace:

- Prepare a list of emergency contact numbers.
 - Number of the police station in whose jurisdiction the workplace falls
 - Number of the nearest hospital
 - Number of the fire brigade under whose jurisdiction the workplace falls.
- This contact list should be posted at multiple locations around the plant and should be easily visible from all the staff and visitors.

At all workplace, but especially those in remote areas, first aid materials, supplies and equipment must be provided. A typical emergency procedure consists of the following actions:

Call your nearest hospital or local emergency number or take the person there.



Contacting the appropriate emergency personnel: this can be a fire fighter in case of a fire and an ambulance in case of a medical emergency.	
Contacting the plant manager if not already on site; and providing support to affected personnel until emergency personnel arrives and takes control of the emergency situation.	

Table 5.2.1: Emergency procedure

Emergencies must be documented on an emergency report form that is then sent to management for investigation. All emergencies must also be fully detailed in the operators log book.

Operational emergency

In Case of Spillage from the Truck

Cause: Failure of outlet valve of desludging vehicle or wrong operation of outlet valve of the desludging vehicle.



How could this happen?

- Damage of the desludging vehicle's outlet valve during feeding into the receiving station
- Desludging vehicle outlet valve stuck in open position during feeding into the receiving station
- Spillage from the hose pipe used for feeding of faecal sludge into the receiving station

Emergency response measures to be taken:

- Desludging vehicle driver should close the outlet valve according to their standard operation.
- To clean the spilled sludge, pour soil over the sludge and let it dry for at least 2 hours and then clean it with water



In Case of Spillage from the Valves

Cause: Failure of valve due to blockages or wrong operation of valves.

How could this happen?

- Failure of valve may happen due to solid waste/debris stuck at the valve's opening.

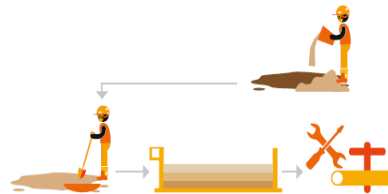


- Damage to the valve may happen due to wrong operations of the valve by the operator and turning the valves in the wrong direction forcefully.

Emergency response measures to be taken:

If sludge has spilled near the valve, clean the sludge performing the following steps.

- Pour soil over the sludge. Leave it for at least 2 hours.
- Using the shovel collect all the soil mixed with sludge in a plastic bowl.
- Dispose this sludge in the Sludge Drying bed SDB.
- Report the problem to the Urban Local Body.
- Repair or replace the valve if necessary.



Overflow from any Treatment Modules

Cause: The module outlet or the inlet of the next downstream module is clogged.



How could this happen?

- This can happen due to excessive accumulated scum or sludge as well as debris blocking the pipes or modules.
- Crushed or frozen modules or damage in the pipes connecting the various modules or excessive inflow of water into the module due to flooding may also be responsible for this kind of issue



Emergency response measures to be taken:

- Stop the flow into the module immediately if any.
- Clear the blockage in the pipes by inserting an iron bar or as instructed by the manufacturer of the treatment module and force pump water.
- Check if any debris is stuck in between outlet of the module and inlet of downstream module. If found, try to push it to the next module using the iron bar or as instructed by the designer of the treatment module and collect the debris from the inlet of downstream module. If debris cannot be moved from its place, immediately report it to the plant manager
- Check for damage/crushing of pipe. If found, immediately report it to the plant manager.
- Repair or replace the valve if necessary.



Gas Leakage

How could this happen?

This can happen due to external physical damage to a pipe or a treatment module



Emergency response measures to be taken:

- Locate where the smell is coming from;
- If damage is found, inform the plant manager
- Do not light any flammable objects near the area of gas leakage.



Fire emergency: fire protection in green logistics

Fire Emergency

Due to the number of chemicals and the flammable nature of the **multiple gases** present in **the unit/ storage area/ warehouse**, fire is a real hazard at the workplace. The following are the measures which should be taken in the event of a fire.

Learn the cause

It is crucial to learn that there are four classes of fires (A, B, C, D) depending on the source of fire. Fires can be set off by different factors: chemical, physical, mechanical or electrical. They can occur because of facilities or equipment, or unground electrical connections (wiring), accumulation, storage and improper handling of combustible materials, tank or fuel spillage and for each of these kinds of fires.

Fire extinguishers

Different substances are used to extinguish fire depending on the source of origin. The equipments containing these substances are called EXTINGUISHERS. Fire extinguishers should be located in an area which is visible and easily accessible. It is portable and can be easily transferred to the site of incipient fire.

Remember

When fire is past its incipient stage, it is usually too big to fight with fire extinguisher and in that case safe evacuation is the best strategy.

Classification of fire and extinguishing agent

Type of Fire Class A



Source of origin

Fires involving solid combustible materials of organic nature such as wood, paper, rubber, plastics, etc, where the cooling effect of water is essential for extinction of fires

Extinguishing agent : Water, foam, ABC dry powder and halocarbons.



Type of Fire Class B



Source of origin

Fires involving flammable liquids or liquefiable solids or the like where a blanketing effect is essential.

Extinguishing agent : Foam, dry powder, clean agent and carbon dioxide extinguishers.



Type of Fire Class C



Source of origin

Fires involving flammable gases under pressure including liquefied gases, where it is necessary to inhibit the burning gas at fast rate with an inert gas, powder or vaporising liquid for extinguishment.

Extinguishing agent : Dry powder, clean agent and carbon dioxide extinguishers.



Type of Fire Class D



Source of origin

Fires involving combustible metals, such as magnesium, aluminium, zinc, sodium, potassium, etc, when the burning metals are reactive to water and water containing agents and in certain cases carbon dioxide, halogenated hydrocarbons and ordinary dry powders. These fires require special media and techniques to extinguish.



Extinguishing agent : Extinguishers with special dry powder for metal fires.

Type of Fire Electrical Fire



Source of origin

Where energized electrical equipment is involved in a fire, non-conductivity of the extinguishing media is of utmost importance

Extinguishing agent : Extinguishers expelling dry powder, carbon dioxide (without metal horn) or clean agent should be used. Once the electrical equipment is de-energized, extinguishers suitable for the class of the fire risk involved can be used safely.



Type of Fire Sensitive items

Source of origin

Equipment sensitive to dirt, contamination or whose control systems are likely to be affected are categorised as sensitive items.

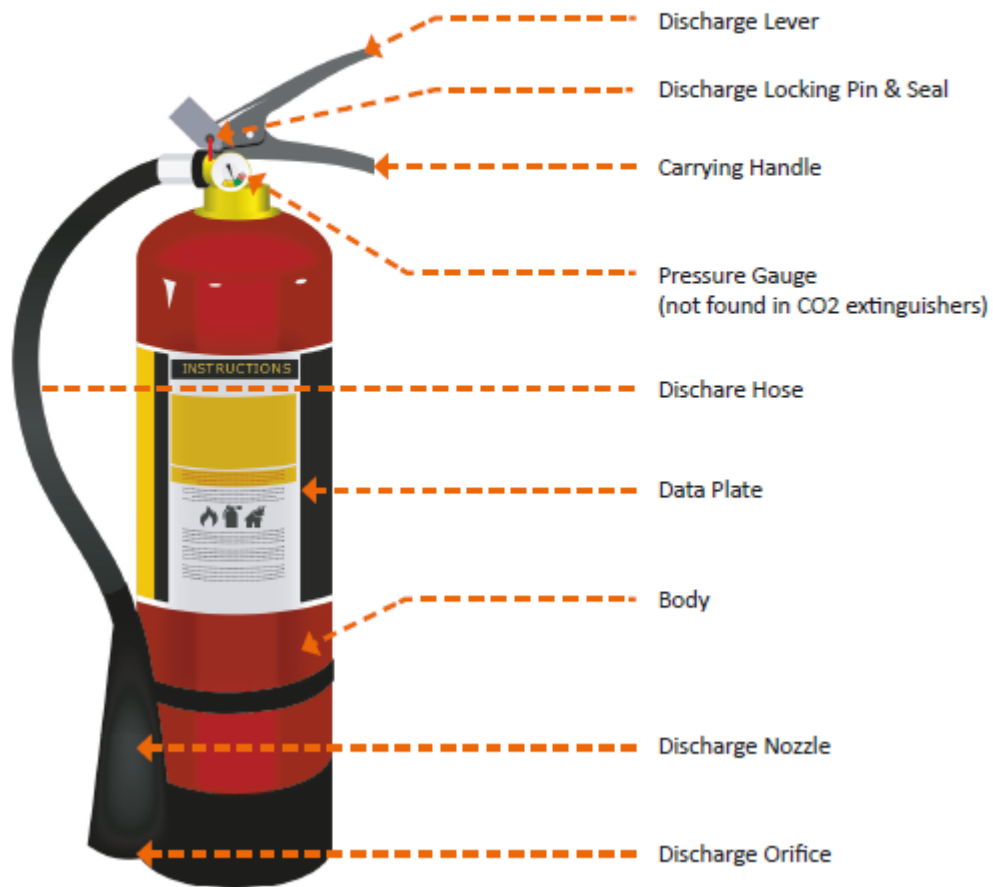


Extinguishing agent : carbon dioxide or clean agent type extinguishers

(Bureau of Indian Standards, 2010)

Anatomy of a fire-extinguishers

The following first illustrates different parts of a fire extinguisher.



Using fire extinguisher

The steps to use a fire extinguisher is summed up using the acronym

"PASS"

P

PULL

Pull the pin of the fire extinguisher.



A

AIM

Aim the nozzle at the base of the fire. Do not hit the top of the flame.



S

SQUEEZES

Squeeze the trigger. In a controlled manner, squeeze the trigger of the extinguisher to release the agent.



S

WEEP

Sweep from side to side. Sweep the nozzle of the extinguisher from side to side until the fire is put out. Keep aiming at the base while you do so. Most extinguishers will give you about 10-20 seconds of discharge time.



Slowly walk backwards, away from the fire. Even if the fire appears to be extinguished, don't turn your back on it. There might be unseen hot spots or hidden fires that can ignite into a large flame at any moment. You should be cautious about it. Once a fire extinguisher has been used, it must be recharged or replaced even if the entire contents were not used.

Do's and don'ts

Keep only limited people around the place of fire while extinguishing the fire. Ask the rest of the people to evacuate calmly.



Keep the fire extinguishers in places away from any corrosive fluids.



Preventive measures

- To identify the chemical properties of the products being stored
- Do not overload electric lines.
- Switch off electronic equipments when not in use
- Avoid connecting more than one appliance in each outlet.
- Redistribute equipment or install additional circuits.
- No smoking in the plant.
- Do not throw matches or any other lighted things indiscriminately. Extinguish them completely and then dispose them.
- Do not place oily rags around electric plugs, treatment modules or any other flammable items.
- Report the presence of gas or fuel leaks or flammable liquid spills.
- Identify fire exits and nearby phones to call external support groups.
- Adapt specific sites for fuel supply and storage.
- Place fire extinguishers according to the combustible material and in the appropriate places to provide emergency care.



Action in case of emergency

- Immediately warn the workplace manager
- Remove people to a secure site
- If the threat is not controlled, get help from the Firefighters
- Ensure that all staff are in the safe meeting place



- After inspecting the fire investigation to proceed Start immediate evacuation, with the support of the brigade.
- Remove people at least at a distance of 10 m.
- Try to find out the source of fire and use the appropriate fire extinguisher to control the fire, if possible. If the outbreak of fire is not controlled immediately notify the firefighters.
- Help those who are injured, give first aid and take the person to the nearest hospital or call the emergency number of the hospital for sending an ambulance.

(General Guidelines on Occupational Health and Safety: Emergency and contingency plan, n.d.)

Security Emergency: theft/ vandalism

Security Emergency: Theft or Vandalism

Theft or vandalism of equipments and tools from the plant should be dealt as a security threat. Take the following steps in such event:

- Do not touch or move anything at the scene
- If you are not certain that the intruder has left the scene, leave.
- If people are around at the workplace, raise an alarm
- Notify the police or call 100
- Report:
 - Location
 - Nature of the event
 - Your name and the number you are calling from
- Request immediate assistance
- Take action: Follow the instructions of the police personnel. If you are witnessing the crime, move to a safe location and call the police. Provided it is safe to do so, take a picture or record a video



(Yale University, n.d.)

Medical Emergency:

Medical Emergency

A sudden onset of medical condition characterised by acute symptoms of sufficient severity such that the absence of medical attention could reasonably be expected to result in:

- placing the patient's health in serious jeopardy,
- serious impairment to bodily functions or serious dysfunction or any bodily organ or part.

Steps to be followed while administering first aid during a medical emergency.

Keep the victim lying down. Examine the victim- look for serious bleeding 	Lack of breathing and poisoning. 	Keep the victim warm. 
Send someone to call a physician or ambulance. 	Remain calm. Do not be rushed into moving the victim unless absolutely necessary. 	Never give an unconscious victim anything to eat or drink. 
If there is a crowd, keep it away from the victim. 	Ensure the victim is comfortable. 	Don't allow the victim see his injury. 
Give CPR, if required. 		

Source: (Operation and Maintenance Manual FSTP, Devanahalli)

Emergency contact or communication in danger

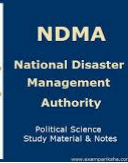
serial number	Agency Name	contact number	pictures or sign boards
1	Police: In an emergency, the police will help enforce the law and protect the public.	100	
2	Ambulance: In case of any emergency, the ambulance will help in taking the injured people to the hospital.	102	
3	Fire: In an emergency, the fire brigade will help rescue people.	101	
4	Disaster Management: Disaster Management organization helps people in the event of any natural or man-made disaster.	108	 
5	Urban Local Body: ULB is a local government which helps its citizens in case of emergency.	ULB helpline no.	

Table 5.2.1 Emergency Contact

Unit 5.3 Administering First Aid

Unit Objectives

At the end of this unit, you will be able to:

1. Follow the safety precautions at the workplace.
2. Illustrate the First-aid procedure.

5.3.1 Safety precautions to be taken at the workplace

In a logistics warehouse, any bio-hazardous material can affect your health. Similarly, fall and accidents at the workplace are things to be watched out for. Some of the common sickness, experienced by people working at warehouse/ unit/ facility are:

- Stomach infection
- Diarrhoea
- Jaundice
- Typhoid
- Worm infestation
- Falls
- Cuts
- Burns
- Bleeding
- Fractures
- Electric shocks

Precautions to be Undertaken

The following precautions can help in avoiding serious health hazards while working at the workplace:

Wear personal protective equipment	
Perform hand hygiene	
Always be alert at work	
Take vaccination: Ask your employer to provide vaccines against: • hepatitis A= 2 doses at least 6 months apart • typhoid= One dose provides protection. It should be given at least 2 weeks before travel to allow the vaccine to work. A booster dose is needed every 2 years for people who remain at risk.	
Deworming tablet: once in every 6 months	

Tetanus injection: to be taken following an injury. It is not required only if the last dose of the primary series, or of subsequent booster injections, was given less than 5 years ago for dirty wounds or less than 10 years ago for clean wounds



5.3.2 Administering First Aid for Different Kinds of Situation

First aid is the initial assistance provided to any person suffering a sudden illness or injury, care provided to preserve life, prevent the condition from worsening, or to promote recovery. Since workplaces are usually located in the periphery of the city, first aid should be available at the workplace at all times to deal with any medical emergency situation till professional medical help comes. Creating A first aid kit is the first step towards being prepared to give first aid.

List the number of the nearby hospitals and emergency medical services



Anti-bacterial soap



Anti-bacterial wipes in case of heavy bleeding



Disinfection solution/ lotions/ powders: to be used to disinfect cuts and abrasion



Cotton: rolls and balls: for applying anti-septic lotions, medicine; to stop bleeding and to dress the wounds



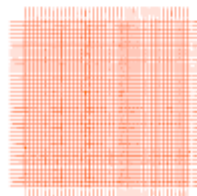
Tweezer: can be used to remove debris such as glass, dirt, or splinters from a wound.



Adhesive bandages in different sizes



Sterile gauze pads



Bandage: in case of larger cuts



Safety Pins: can be used to hold and secure wraps and bandages



Adhesive Tape: used for holding a bandage or other dressing onto a wound.



Scissors: in order to cut the bandage



Anti-septic cream/ ointment: to be applied on minor cuts, burns, grazes, scalds



Crepe Bandages: to be used in case of a sprain in joints



Oral Rehydration Solution: to be used in case someone is feeling dehydrated or having diarrhoea



Thermometer: to test temperature of the person



- Keep the first aid kit in a clean, cool and dry place.
- Keep the first aid kit in an easily accessible location and everyone in the plant should know it.
- Some items, such as solutions, may have use-by dates. Check regularly and replace when necessary.
- If an item is used from the first aid kit, promptly replace it

First Aid procedure

Diarrhoea

Diarrhoea is the most common illness among sanitation workers. This is mainly due to accidental ingestion of pathogens.

Prevention: Wearing gloves while working and taking de-worming tablets can help prevent diarrhoea.

The following are the steps to be taken to provide first aid for diarrhoea:

Make sure the person drinks more fluids than they are losing through diarrhoea. Give the person Oral Rehydration Solution regularly to replace the fluid.	
Have the person rest as needed and avoid strenuous exercise. Send the person to residence, if possible. Ask the person to continue drinking fluids and ORS.	
Medical advice should be sought if symptoms do not improve in 48 hours.	

Call the doctor immediately if any of the following symptoms are there:

- Severe abdominal or rectal pain
- Blood in stool
- Black, tarry stools
- High fever (greater than 101.3 F)
- Signs of dehydration such as:
 - Moderate dehydration
 - Thirst
 - Dry or sticky mouth
 - Not peeing very much
 - Dark yellow pee
 - Dry, cool skin
 - Headache
 - Muscle cramps
 - Signs of severe dehydration include:
 - Not peeing or having very dark yellow pee
 - Very dry skin
 - Feeling dizzy
 - Rapid heartbeat
 - Rapid breathing
 - Sunken eyes
 - Sleepiness, lack of energy, confusion or irritability
 - Fainting



Table 5.3.2: First-aid procedure for diarrhoea

In case you are the one suffering, take the above measures as applicable and call for help if needed.

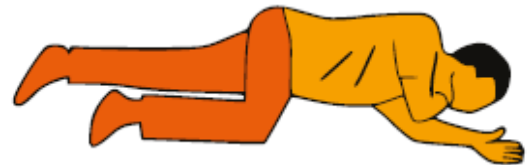
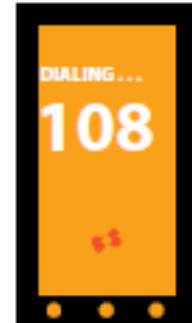
What should be done if someone is not breathing and heart beat has stopped?

You may find a person not breathing and their heart beat stopped under a number of circumstances:

- Head injury
- Heavy bleeding
- Heat stroke
- Burns
- Electric shock

The person should receive immediate medical attention. Call the nearest hospital to send an ambulance. Meanwhile, the person should be given first aid, which is as follows:

- Take the person to a safe space away from dust and debris.
- Tap the person's shoulder and shout to see if the person can hear you.
- Call the nearest hospital for an ambulance. In case you are not able to make the call, ask someone to do so.
- Loosen any tight clothing, if possible.
- First, open a person's airway to check if they are breathing (don't begin CPR if a patient is breathing normally).
- Check Airway
 - Clear any obstruction in the airway: The most common cause of airway obstruction is the tongue. To keep the airway open, perform the head-tilt, chin-lift. Place one hand on the victim's forehead and put the fingers of your other hand under the bony part of the chin. Press down on the forehead and lift out the chin so that the mouth is slightly open. If you suspect a spinal injury, do not press down on the forehead nor tilt the head back. Perform a chin lift only. The victim may start to breathe after you open the airway.
- If the victim is breathing, and no spine injury is suspected, place in the Recovery Position (on his/her side (preferably left) extending the lower arm above the head).
- If an obstruction to the airway is visible in the back of the mouth and the victim is unconscious, remove the object with your index finger. (Gloves should be worn.)
- Never place your fingers in the mouth of a conscious or semiconscious individual. If the victim is conscious, give first aid for Choking (Handal, n.d.)



CPR: Cardiopulmonary Resuscitation

Do hands-only CPR (Cardiopulmonary Resuscitation) to help circulation and get oxygen into the body.

1	Call 108/Ambulance or ask someone else to	
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2	Kneel besides victim's chest. Loosen clothing if practical.	
3	Position your hand Make sure the patient is lying on their back on a firm surface. Kneel beside the patient and place the heel of your hand on the centre of the chest.	
4	Interlock fingers. Keeping your arms straight, cover the first hand with the heel of your other hand and interlock the fingers of both hands together. Keep your fingers raised so they do not touch the patient's chest or rib cage.	
5	Give chest compressions Lean forward so that your shoulders are directly over the patient's chest and press down on the chest about two inches. Release the pressure, but not your hands, and let the chest come back up. Repeat to give 30 compressions at a rate of 100 compressions per minute. Continue until patient starts breathing normally or medical help has arrived or you are exhausted giving CPR.	

Table 5.3.2: CPR method

Choking

Choking occurs when a foreign object lodges in the throat or windpipe, blocking the flow of air. In adults, a piece of food often is the culprit. Since choking cuts off oxygen to the brain, give first aid as quickly as possible.

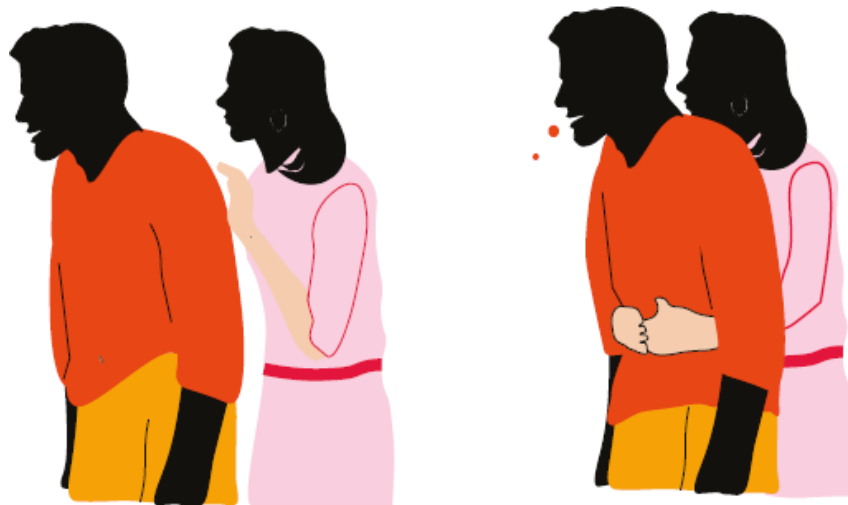
Prevention: Wearing mask while at work can help protect against choking to some extent.

The universal sign for choking is hands clutched to the throat. If the person doesn't give the signal, look for these indications:

- Inability to talk
- Difficulty breathing or noisy breathing
- Squeaky sounds when trying to breathe
- Cough, which may either be weak or forceful
- Skin, lips and nails turning blue or dusky
- Skin that is flushed, then turns pale or bluish in colour
- Loss of consciousness

If the person is able to cough forcefully, the person should keep coughing. If the person is choking and can't talk, cry or laugh forcefully, the American Red Cross recommends a "five-and-five" approach to delivering first aid:

Give 5 back blows	Give 5 abdominal thrusts	Alternate
Stand to the side and just behind a choking adult. Place one arm across the person's chest for support. Bend the person over at the waist so that the upper body is parallel with the ground. Deliver five separate back blows between the person's shoulder blades with the heel of your hand.	Perform five abdominal thrusts (also known as the Heimlich manoeuvre).	Alternate between 5 blows and 5 thrusts until the blockage is dislodged.



To perform abdominal thrusts (Heimlich manoeuvre) on someone else

The American Heart Association doesn't teach the back-blow technique, only the abdominal thrust procedures. It's OK not to use back blows if you haven't learned the technique. Both approaches are acceptable. To perform abdominal thrusts (Heimlich manoeuvre) on someone else:

- Stand behind the person. Place one foot slightly in front of the other for balance. Wrap your arms around the waist. Tip the person forward slightly. If a child is choking, kneel down behind the child.
- Make a fist with one hand. Position it slightly above the person's navel.
- Grasp the fist with the other hand. Press hard into the abdomen with a quick, upward thrust — as if trying to lift the person up.
- Perform between six and 10 abdominal thrusts until the blockage is dislodged.



If you're the only rescuer, perform back blows and abdominal thrusts before calling your local emergency number for help. If another person is available, have that person call for help while you perform first aid. If the person becomes unconscious, perform standard cardiopulmonary resuscitation (CPR) with chest compressions and rescue breaths.

If you are the one who is choking then perform the abdominal thrusts on yourself as follows:

Abdominal thrusts (Heimlich manoeuvre) on yourself: First, if you're alone and choking, call your local hospital immediately. Then, although you'll be unable to effectively deliver back blows to yourself, you can still perform abdominal thrusts to dislodge the item.

- Place a fist slightly above your navel.
- Grasp your fist with the other hand and bend over a hard surface — a countertop or chair will do.
- Shove your fist inward and upward.



To clear the airway of a pregnant woman or obese person:

- Position your hands a little bit higher than with a normal Heimlich manoeuvre, at the base of the breastbone, just above the joining of the lowest ribs.
- Proceed as with the Heimlich manoeuvre, pressing hard into the chest, with a quick thrust.
- Repeat until the food or other blockage is dislodged. If the person becomes unconscious, follow the next steps.



To clear the airway of an unconscious person:

- Lower the person on his or her back onto the floor, arms to the side.
- Clear the airway. If a blockage is visible at the back of the throat or high in the throat, reach a finger into the mouth and sweep out the cause of the blockage. Wear gloves. Don't try a finger sweep if you can't see the object. Be careful not to push the food or object deeper into the airway.

- Begin CPR if the object remains lodged and the person doesn't respond after you take the above measures. The chest compressions used in CPR may dislodge the object. Remember to recheck the mouth periodically. Refer **CPR: Cardiopulmonary Resuscitation**, (Mayo Clinic, 2017).

Heat Stroke

Staff working at the workplace work in open environment for long hours. In Indian conditions, where summer are very hot for most part of the country, there is a possibility of heat strokes. Symptoms of heat stroke:

- Core temperature is greater than 40 oC or 104oF
- Dizziness
- Throbbing heart
- Red, hot and dry skin
- Muscle cramps or weakness
- Rapid, shallow breathing
- Unconsciousness
- Lack of sweating despite of heat
- Seizures
- Nausea and vomiting
- Behavioural changes such as confusion, disorientation or staggering

Take the following steps to provide first aid for heat stroke:

Heat stroke is a medical emergency and can be fatal if immediate medical care is not provided. Call the nearest hospital for ambulance or rush the person to the hospital.

Meanwhile, take the following steps to provide first aid by the time the person gets medical help:

- Lower Body Temperature While Waiting for Emergency Services to Arrive
- Get the person into air conditioning if possible or out of the sun and into the shade.
 - Spray the person with cool water, or apply cold wet clothes or ice packs to the armpits, neck, and groin. Fan air across the person to increase cooling. These methods help cool the person more quickly.
 - Do not give the person anything to drink if the person is not alert or is vomiting.
- Treat Symptoms
 - If the person experiences seizures, keep him or her safe from injury.



- If the person vomits, turn the person on his or her side to keep the airway open.

Tilt the head backwards: This is done to avoid tongue to fall backwards and block the airway. Tilting the head backwards and pulling the tongue forward will help to clear the airways.



If you suspect a back or neck injury, it is still advisable to tilt the head slightly in order to help the patient breathe. The priority is to keep the patient breathing. Try to keep their spine in a straight line when turning them. If possible, get someone's else to help to turn them.



If someone is feeling faint, advise them to lie down on their back and raise their legs to improve blood flow to the brain. Fainting is caused by a temporary reduction in the flow of blood to the brain and can result in a brief loss of consciousness. A person who has fainted should quickly regain consciousness. If they don't, treat them as an unconscious person.



Give hands-only CPR.



(National Health Portal, India, 2019)

In case you are the one suffering, take the above measures and call for help.

Minor Cuts

Workers at the workplace can get injured and bleed under a number of circumstances such as:

- Handling broken items such as glass, metals etc
- Tripping over or falling down

Take the following measures to provide first aid for the following:

Wash your hands thoroughly before touching the wound. Wear a pair of gloves.	
Stop the Bleeding: Apply direct pressure on the area	
Clean and Protect: Clean the area with warm water and gentle anti-septic soap. Apply an anti-septic ointment to reduce chance of infection. Put a new adhesive bandage on the area.	
Call a health care provider if: • The cut is deep or over a joint • You cannot get the cut or laceration clean • The injury is a deep puncture wound or the person has not had a recent (within the last 5 to 10 years) tetanus shot or booster • The cut is from a human or animal bite (WedMD, 2018)	

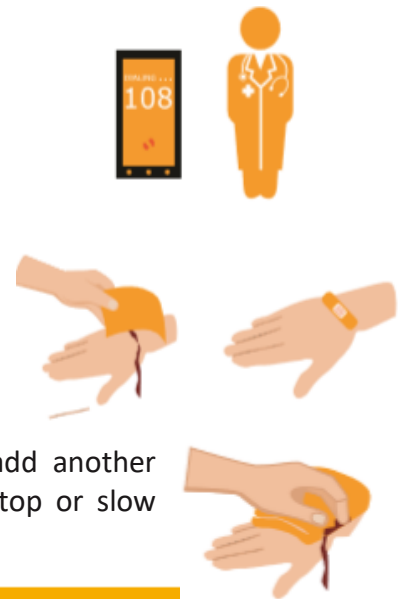
Table 5.3.2: First-aid for minor cuts

In case you are the one suffering, take the above measures as applicable and call for help if needed. If the bleeding is heavy, refer to the next section.

Heavy Bleeding

Take the following measure to provide first aid for heavy bleeding:

- Wash your hands thoroughly before touching the wound. Wear a pair of gloves.
- Clean the wound with a gauze pad or cloth in saline solution or tap water, or use an alcohol-free wipe, and gently dab or wipe the skin with it
- gently pat the area dry using a clean towel or pad of tissues, but nothing fluffy such as a cotton wool ball –strands of material can get stuck to the wound
- apply a sterile dressing, such as non-adhesive pad with a bandage, or a plaster – use a waterproof dressing if available.
- if blood soaks through the dressing, leave it in place and add another dressing, and continue to apply pressure on the wound to stop or slow down the flow of the blood.



Do's for heavy bleeding

1

Reassure victim that help is on the way.



2

Be calm.



3

Call emergency as soon as possible or take the patient to the hospital



4

Call ambulance immediately



5

Check victim's status regularly



6

Use direct pressure to stop bleeding





Source: (National Health Portal, India, 2019)

In case you are the one suffering, take the above measures and call for help.

Abdominal Wound

In case the wound is caused by tripping over a sharp object with stomach down or some form of deliberate or accidental injury to the abdomen, the patient must be taken to the hospital immediately. However, on the way to the hospital, the following first aid measures must be taken:

- Keep the patient flat on his/her back.
- Give nothing by mouth.
- Maintain warmth.
- If intestines protrude from the wound, do not attempt to touch or replace them.
- Apply sterile dressing and binder on the wound.
- Provide immediate transportation to the hospital.
- (Ministry of Labour & Employment, n.d.)



In case you are the one suffering, call the nearest hospital for help immediately.

Head Injury

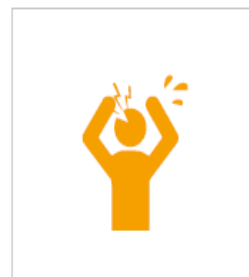
Head injuries usually happen if a person falls or trips from a height or hits a hard, blunt or sharp object with their head. This can be prevented by wearing a helmet while working at the plant machineries/ workplace.

Most head trauma involves injuries that are minor and don't require specialized attention or hospitalization. However, even minor injuries may cause persistent chronic symptoms, such as headaches or difficulty in concentrating.

Call local emergency number or take the person to the hospital if any of the following signs or symptoms are apparent, because they may indicate a more serious head injury.

Symptoms of head injury:

- Head or facial bleeding
- Bleeding or fluid leakage from the nose or ears
- Headache
- Change in level of consciousness for more than a few seconds
- Black-and-blue discoloration below the eyes or behind the ears
- Cessation of breathing
- Confusion
- Loss of balance
- Weakness or an inability to use an arm or leg
- Unequal pupil size
- Slurred speech
- Seizures



Until medical help is reached, take the following steps:

Keep the person still: Until medical help arrives, keep the injured person lying down and quiet, with the head and shoulders slightly elevated. Don't move the person unless necessary, and avoid moving the person's neck. If the person is wearing a helmet, don't remove it.	
Stop any bleeding: Apply firm pressure to the wound with sterile gauze or a clean cloth. But don't apply direct pressure to the wound if you suspect a skull fracture.	
Watch for changes in breathing and alertness: If the person shows no signs of circulation — no breathing, coughing or movement — begin CPR. (Mayo Clinic, 2018)	

Table 5.3.2: First-aid for head injury

In case you are the one suffering, call the nearest hospital for help immediately.

Burns

Do's

1

Call an ambulance for any serious burns.



2

Apply CPR if the person is not breathing normally.



3

Try to remove clothes and jewellery (from the area that has been burned) only if it is not sticking to the burned area.



4

Hold the burned area under gently running water, for about 10 minutes to half an hour.



5

To prevent corneal damage (in the case of chemical burns to the eyes), immediately but gently wash the eyes with water or a saline solution.



6

For second degree burns (affecting outer and inner surfaces of the skin) on the limbs – elevate the limbs higher than the heart.



7

To reduce shock as well as loss of body heat, place clean, dry, non-fluffy cloths lightly over the burn.



8

Cover the person with a cool, wet, lint-free cloth, while waiting for an ambulance or when transporting the person to hospital.



Do's

1

Do not apply lotions, butter, grease or oil to burned area.



2

Do not use ice, as it may cause frostbite.



3

Do not remove any piece of clothing or any other thing stuck to the burnt area but remove clothing and other items around the burnt area.



4

Do not put plaster on the burnt area.



(National Health Portal, India, 2019)

In case you are the one suffering, take the above measures as applicable and call for help if needed.

Symptoms of nose-bleeding:

- Bleeding from either or both nostrils
- Sometimes bleeding from ears/ mouth too.

Q. What are the causes of nose bleeding

- Dryness
- Blowing nose with force
- Use of medications, like aspirin
- Nose picking
- Pushing objects into nose
- Injuries / blow to the nose
- Infections of the nose
- Atherosclerosis
- Blood-clotting disorders

Q. How to manage nose bleeding?

- One should not panic and should make the patient sit in upright position with his/her head slightly forward.
- With thumb and index finger, one should apply pressure on soft part of nostrils below the nose bridge.
- Continue applying pressure until the bleeding stops.
- Ask the patient to breathe through the mouth while nostrils are pinched
- Loosen the tight clothing around the neck
- After 10 minutes, release the pressure on the nostrils and check to see if the bleeding has stopped
- If bleeding persists, seek medical aid

Note: Ask the patient not to sniff or blow their nose for at least 15 minutes



In case you are the one suffering, take the above measures as applicable and call for help (preferably the nearest hospital) immediately

Poisoning

There are chances of poisoning at a workplace if person deliberately or accidentally swallows, inhales or comes in direct contact with the chemicals present in the warehouse. Rush the person to the nearest hospital or call the ambulance, whichever is earlier.

Meanwhile professional medical help is reached, the following needs to be done:

- Swallowed poison. Remove anything remaining in the person's mouth. If the chemical is identified, follow the instructions written on the bottle for accidental poisoning.
- Poison on the skin. Remove any contaminated clothing using gloves. Rinse the skin for 15 to 20 minutes in a shower or with a hose.
- Poison in the eye. Gently flush the eye with cool or lukewarm water for at least 15 minutes or until help arrives.
- Inhaled poison. Get the person into fresh air as soon as possible.
- If the person vomits, turn his or her head to the side to prevent choking.
- Begin CPR if the person shows no signs of life, such as moving, breathing or coughing.
- Have somebody gather the bottles, packages or containers with labels, and any other information about the poison to give to the ambulance team.



In case you are the one suffering, take the above measures as applicable and call for help (preferably the nearest hospital) immediately

Snake Bite

- Keep the victim calm, restrict movement.
- Assure the victim and do not let him panic. When under panic, it will enhance heart rate and would circulate the venom faster in the body.
- Remove any rings or constricting items; the affected area may swell.
- Stop lymphatic spread of venom - bandage firmly, splint and immobilise. The limb, which has been affected by the bite, should be immobilized with splint. Victim to keep the hand as close to the level of the heart as possible - this reduces the flow of venom to major areas. A compression bandage (as firm as you would put on a sprained ankle) should cover the entire limb with the splint. The wrapping should start from the digits and extend till armpit in case of hands and groin in case of bite to the leg.
- A snakebite victim is under tremendous psychological stress. It is necessary to keep the patient warm. However, no alcohol/hot beverages should be given.



- The patient should not be allowed to exert himself/herself in any manner. Do not allow the victim to eat or to drink water in order to keep metabolism at low rate. No water No food is the golden rule.
- DO NOT COVER THE BITE AREA AND PUNCTURE OR CUT MARKS. The wound should be gently cleaned with antiseptic.
- Try to aspirate the venom out of the puncture marks with standard suction devices. It has been identified that a suction more than 270 mmHg can initiate the flow from the puncture marks. Suction instruments often are included in commercial snakebite kits. But the suction should be applied within 5 minutes of the bite.
- The only remedy for venomous snakebite is the anti-venom serum, which is available at most government hospitals and public health centres. Some private nursing homes have also started stocking it and treat snakebite cases. (Auroville, 2014)

First- Aid treatment protocol

First-aid currently recommended may be remembered by mnemonic		
" CARRY NO R.I.G.H.T"		
CARRY	Do not let victim to walk even for short distance. Transport by conveyance, especially when bite is in legs	
NO	NO- Tourniquet. NO- Cutting. NO- Electrotherapy. NO- Pressure immobilisation, nitric oxide donor (nitrogesic ointment/nitrate spray)	
R.I.G.H.T.		
R.	Reassure patient, since 70% of all snake bites are from non- venomous species. Only 50% of bites by venomous type of snakes actually envenomate (poison with venom) victims.	
I.	Immobilise limb in a fashion similar to a fractured limb, in case of bites on the limb. A bandage or cloth is used to hold the splints. Do not apply pressure and ensure that blood supply is not blocked. Compression in the form of tight ligatures does not work and may be dangerous even.	

G.H.	Get to Hospital immediately.	
T.	Tell any systemic symptoms that manifest on way to hospital.	

Table 5.3.2: First-aid treatment protocol

(National Health Portal, India, 2019)

In case you are the one suffering, take the above measures as applicable and call for help (preferably the nearest hospital) immediately.

Fractures

Symptoms of a broken bone:

- Pain and bleeding
- Swelling
- Bruising or discoloured skin around the affected area
- The patient is unable to put weight on the injured area
- The patient cannot move the affected area

(National Health Portal, India, 2015)

First-aid care for a broken bone

- Stop any bleeding: Elevate and apply pressure to the wound using a sterile bandage, a clean cloth, or a clean piece of clothing. Refer to Heavy bleeding
- Immobilize the injured area: If you suspect they've broken a bone in their neck or back, help them stay as still as possible. If you suspect they've broken a bone in one of their limbs, immobilize the area using a splint or sling.
- Apply something cold to the area: Wrap an ice pack or bag of ice cubes in a piece of cloth and apply it to the injured area for up to 10 minutes at a time.
- Treat them for shock: Help them get into a comfortable position, encourage them to rest, and reassure them. Cover them with a blanket or clothing to keep them warm.
- Get professional help: Call the nearest hospital or clinic or help them get to the emergency department for professional care.
- If the person doesn't appear to be breathing, is unconscious, or both, medical help and begin CPR.



Electric Shock

The danger from an electrical shock depends on the type of current, how high the voltage is, how the current travelled through the body, the person's overall health and how quickly the person is treated. Seek immediate medical help if the patient is showing the following symptoms:

- Severe burns
- Confusion
- Difficulty breathing
- Heart rhythm problems (arrhythmias)
- Cardiac arrest
- Muscle pain and contractions
- Seizures
- Loss of consciousness



Do's and Don't

Don't touch the injured person if he or she is still in contact with the electrical current.	
Turn off the source of electricity, if possible. If not, move the source away from you and the person, using a dry, nonconducting object made of cardboard, plastic or wood.	
Call your nearest hospital or local emergency number or take the person there.	
Don't move a person with an electrical injury unless he or she is in immediate danger.	

Take these actions immediately while waiting for medical help:

- Begin CPR if the person shows no signs of circulation, such as breathing, coughing or movement.
- Try to prevent the injured person from becoming chilled.
- Apply a bandage. Cover any burned areas with a sterile gauze bandage, if available, or a clean cloth. Don't use a blanket or towel, because loose fibres can stick to the burns.



Table 5.3.2 Dos and don'ts while electric shock

Unit 5.4 Work safety procedures and instructions in green logistics

Unit objectives



After reading this unit, you will be able to understand the following points:

1. Work safety procedures and measures

5.4.1 Work Safety procedures

The safety procedure required to avoid any mishap is given below:

Emergency exit

- Plan routes for emergency evacuation.
- Plan to turn off critical equipment.
- set-up the alarm system.
- make others aware.

Primary treatment

- First aid kit with essential items should be available at all times.



Your colleague got hurt he is bleeding profusely from his hand, help him and give him first aid.

Material:

- Cotton
- Clean water
- dettol
- Bandage
- Ointment

Suggestion:

- Wash the wound with clean water.
- Soak cotton in Dettol and squeeze it. Press the wound with the Dettol cotton and raise your hand slightly.
- Press the cotton and tie a bandage on it.
- Get medical help as soon as possible if the flow stops or slows down

First aid is the first care given to an injured person before medical help is received. Its objectives:

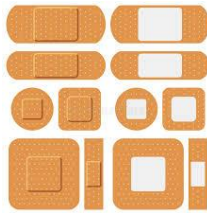
First aid at the scene of an accident

Prevent injury progression

Quick relief to injured person

Transport the injured person to the nearest hospital

components of first aid box

Serial no.	first aid box item name	pictures
1	First Aid Chart: The adjacent chart details the use of a first aid box in an emergency.	
2	Medical Ointment: Elastic bandages are used without dressing the injured area. It helps in reducing swelling etc.	
3	Medical gauge: small medium and large sterile gauge bandage	
4	Bandages: These bandages are used to hold a dressing over a wound.	

5	Safety Pin: Safety pin is used to fix the bandages.	
6	Disposable Sterile Gloves: These gloves are used in medical exams and prevent cross infection between patients and caregivers.	
7	Scissors: Scissors are used for cutting bandages, gauze and injured area etc.	
8	Alcohol-free cleaning wipe: Wipes are used for cleaning wounds.	
9	Sticky Tape: Sticky tape is used to keep the bandages in place.	
10	Thermometer: Thermometer is used to measure body temperature.	
11	Cream: (Spray antiseptic cream for relief of skin rash, insect bites and sting.	

12	Medicines: (Pain relievers cough antihistamines etc.)	
13	Distilled water for cleaning the wound.	
14	Eye washes and eye baths: This can be used to clean the eyes or soothe irritation.	

Electric safety measures

Electric safety measures refer to all the measures to be taken while performing operation and maintenance activities on powered devices. According to the Occupational Safety & Health Administration, United States of America, the following are the key things to be kept in mind while working with electricity:

1. Always be cautious while working with electricity
2. Never operate electrical equipment while you are standing in water or if you have wet hands.
3. Overhead wires: Assume that all overhead wires are energized at very high voltages. Never assume that a wire is safe to touch even if it appears to be insulated. Never touch a fallen overhead power line. Call the electric utility company to report fallen electrical lines.
4. Never repair electrical cords or equipment unless qualified and authorized.
5. Have a qualified electrician inspect electrical equipment that has got wet or are malfunctioning before energizing it

Do's and Don'ts while Working with Electric Equipment

Do's

1

Keep hands dry while working on electricity



2

Switch off the machine while performing repair and maintenance activities



3

Use electric safety gloves while working on highly energised electric machineries and equipments



4

Wear shoes while working with electrical equipments



Don't's

1

Do not work with wet hands and feet



Confined space safety measures

A confined space is defined as any place at workplace that is enclosed and has limited access, such as tanks, dry wells, pump rooms. According to OSHA Guidelines, any space that meets the following three criteria is designated as a confined space:

- Is large enough for a worker to enter it;
- Has limited means of entry or exit; and
- Is not designed for continuous occupancy.

A space may also be a permit-required confined space if it has a hazardous atmosphere, the potential for engulfment or suffocation, a layout that might trap a worker through converging walls or a sloped floor, or any other serious safety or health hazard. (OSHA Fact Sheet: Confined Spaces in Construction: Sewer Systems)

Confined spaces are potentially hazardous as the breathable atmosphere may become compromised, either by depletion of oxygen or the presence of chemical gasses, such as chlorine or hydrogen sulphide. When maintenance is required inside these areas, certain procedures can be defined in order to protect the worker. These typically include the following:

- a confined space entry permit is prepared by the worker and signed by the supervisor;
- prior to entry, the atmosphere is tested with an oxygen meter or, in the case of maintenance holes, with a hydrogen sulphide meter; and
- the work is conducted using the buddy system, with one person entering the confined space secured with a harness attached to a safety rope, and one person located outside of the confined space ready to provide assistance if needed. When the work is completed, the permit is returned to the supervisor for signature indicating the completion of the task.

Checklist for Confined Spaces

Take permission from the plant manager before entering the confined space

☐

If confined space is a maintenance holes, or any space that is below ground level or has inadequate ventilation, it can contain deadly gases. Prior to entering any confined space clean the confined space off sludge and keep the cover slab on for a minimum of 1 hour.

☐

Wear personal protective equipment

☐

Keep a co-worker located outside for rescue and support

☐

Chemical and Material storage safety

Chemical and material safety refers to the safety measures which need to be undertaken for the safety of the workers and the plant. A Material Safety Data Sheet (MSDS), a document that contains information on the potential hazardous materials present at a site, should be made available and the required safety measures should be followed accordingly. The following are the measures to be taken up:

Chemical storage	
Store the chemicals away from heat and direct sunlight	
Reduce the amount of chemicals stored by buying in limited quantities. Also, clear old stock by means of use, replacement & safe disposal.	
Store the chemicals in a segregated way based on considerations such as temperature, ignition control, ventilation, type and identification. Proper segregation is necessary to prevent incompatible materials from inadvertently coming into contact. A physical barrier and/or distance is effective for proper segregation.	
All chemical storage areas and cabinets should be inspected at least annually and any unwanted or expired chemicals should be removed.	
Flammable materials should be stored in an approved, dedicated flammable materials storage cabinet or storage room if the volume exceeds 37 litres. Keep cabinet doors closed	

Chemical storage	
Chemicals should be stored no higher than eye level and never on the top shelf of a storage unit. Do not overcrowd shelves. Each shelf should have an anti-roll lip.	

Avoid storing chemicals on the floor (even temporarily) or extending into traffic aisles.	
Liquids should be stored in unbreakable or double-contained packaging, or the storage cabinet should have the capacity to hold the contents if the container breaks.	
Volatile or highly odorous chemical shall be stored in a ventilated cabinet. Chemical fume hoods shall not be used for storage as containers block proper air flow in the hood and reduce available work space. Chemical fume hoods are chimney-like structures in a place where chemicals are used. They contain vapors, dusts, gases, and fumes generated within the hood, and remove them as air flows into the hood and then out via the laboratory exhaust system.	
All chemicals should be labelled and dated upon receipt in the lab and on opening.	
Chemicals stored in explosion-proof refrigerators or cold rooms shall be sealed and labelled with the name of the person who stored the material in addition to all other required hazard warnings.	

Chemical storage	
Look for unusual conditions in chemical storage areas, such as: • Improper storage of chemicals • Leaking or deteriorating containers • Spilled chemicals • Temperature extremes (too hot or cold in storage area) • Lack of or low lighting levels • Blocked exits or aisles • Doors blocked open, lack of security • Trash accumulation • Open lights or matches • Fire equipment blocked, broken or missing • Lack of information or warning signs ("Flammable liquids", "Acids", "Corrosives", "Poisons", etc.)	

Table 5.4.1: Conditions of safe chemical storage

Source: (Chemical Storage Guidelines from The CDC)

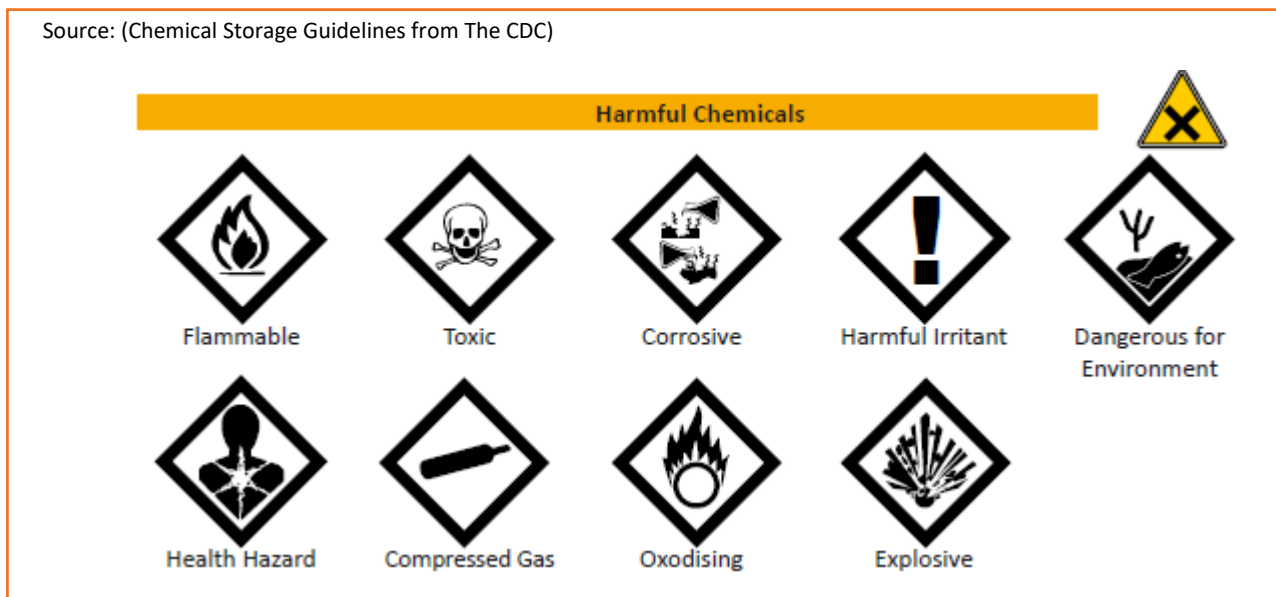


Fig. 5.4.1: Signs of harmful chemicals

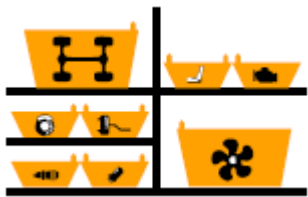
Material safety	
Materials and supplies used at a workplace should be stored in a neat and orderly manner to prevent them from falling off the shelves.	
Junk parts removed from the treatment module should be disposed off in a proper manner	
Spare parts used in the operation of the facility plant should be kept in a neat and orderly manner with the item labelled to indicate on the of equipment on which the spare part is to be used.	
Do not allow paper and other lighter combustible materials to accumulate in the treatment plant premises to prevent them from getting into the treatment modules and causing fire.	
Do not store flammable liquids such as gasoline and diesel fuel at the premises where they may cause a fire or leak onto the floor causing hazardous working conditions.	
Do not allow accumulation of oily rags and papers as they can spontaneously combust.	
Consider the size and weight of any object before attempting to lift or move the object. Do not lift any materials that cannot be handled comfortably. If necessary, take assistance or wait until assistance is available. When carrying objects near treatment modules take extra care to avoid falling in the tanks or dropping objects into the tanks.	
Use carbon dioxide or halon compressed gas extinguishers to control fires.	

Table 5.4.1: Safe storage conditions of materials

Managing movement of vehicles within the warehouse

Managing movement of vehicles within the workplace is essential to avoid accidents. This includes managing the movement of:

- Trucks
- Vehicles of the staff and visitors
- Any other vehicles

Checklist for vehicle movement within the plant	
Trucks coming to the plant for disposal or cleaning should move only on their pre-defined route within the plant	
Vehicles of staff and visitors should be parked at designated parking area	
Other vehicles coming to the plant for maintenance purpose should follow the route for trucks.	
All the personnel on the vehicle should wear gloves	

Table 5.4.1: Checklist for easy vehicle movement

Safety measures during hazards

Natural hazard

Natural hazard are natural processes and therefore, human control over it is limited to preventing/ minimizing the damage to life and property. Usually, warehouses are constructed as per the standard norms, hence, the chances for damage to life and property is minimal. However, it is good to keep a watch in the deterioration of the warehouse over time and take measures to address them. At the same time, it is possible that the warehouse can be affected if the intensity of the natural hazard reaches unprecedented levels, something for which the warehouse was



not constructed. Under such condition, it is essential to know the things to be done during and after the occurrence of such an event.

Thunderstorm

Look for early warning of thunderstorm such as vivid and frequent lightening.



Keep the emergency numbers (such as hospitals, fire brigade etc. ready)



Prepare an emergency kit with first aid, dry food items, mineral water, torch, batteries and sturdy ropes, in case of an event of thunderstorm. Periodically check the expiry date of the articles in this kit.



In case thunderstorm is forecasted, postpone outdoor activities. Keep mobile phones, laptops any other chargeable items charged.



If at the workplace during a thunderstorm:

- Stay indoors. Get inside the office of the workplace.
- Secure all the doors, ventilation and windows.
- Unplug any electronic equipments: both plant machineries and office equipments.
- Avoid contact with corded phones and devices including those plugged for recharging.
- Avoid contact with electrical equipment or cords. Unplug appliances and other electrical items such as computers and turn off air conditioners. Power surges from lightning can cause serious damage.
- Avoid contact with plumbing. Do not wash your hands, do not wash dishes. Plumbing and bathroom fixtures can conduct electricity.
- Stay away from windows and doors, and stay off porches.
- Do not lie on concrete floors and do not lean against concrete walls.
- Avoid natural lightning rods such as a tall, isolated tree in an open area.
- If in a vehicle, safely park and stay in the vehicle until the strong winds



subsides. Avoid touching metal or other surfaces that can conduct electricity in and outside the vehicle.

If at the workplace after a thunderstorm:

- Check if someone has been injured.
 - If yes, give appropriate first aid and take the person to the hospital
- Stay away from storm-damaged areas to keep from putting yourself at risk from the effects of severe thunderstorms.
- Stay away from drowned power lines and report them immediately.
- Report the damages to your plant manager



(National Disaster Management Authority, 2018)

Natural Hazard

If at the workplace during a flood:

- Secure all the doors, ventilation and windows of the FSTP office.
- Unplug any electronic equipments: both plant machineries and office equipments.
- Evacuate calmly and move to higher secure areas.
- Do not walk through moving water. Six inches of moving water can make you fall. If you have to walk in water, walk where the water is not moving. Use a stick to check the firmness of the ground in front of you.
- Do not drive or ride a bike into flooded areas. If floodwaters rise around your car, abandon the car and move to higher ground if you can do so safely. You and the vehicle can be quickly swept away.



If at the workplace after flood

- Check if someone has been injured.
- If yes, give appropriate first aid and take the person to the hospital
- Stay away from drowned power lines and report them immediately.
- Report the damages to your plant manager

(National Disaster Management Authority, India)



Earthquakes

If at the workplace during an earthquake:

- Shutdown all electrical equipments in the office.
- If indoors:
 - DROP to the ground; take COVER by getting under a sturdy table or other piece of furniture; and HOLD ON until the shaking stops. If there is no a table or desk near you, cover your face and head with your arms and crouch in an inside corner of the building.
 - Protect yourself by staying under the lintel of an inner door, in the corner of a room, under a table
 - Stay away from glass, windows, outside doors and walls, and anything that could fall, (such as lighting fixtures or furniture).
 - Move outside only if you are sure that the doorway is strong and it is safe to go outside.



- If outdoors:
 - Do not move from where you are. However, move away from buildings, trees, streetlights, and utility wires.
 - If you are in open space, stay there until the shaking stops. The greatest danger exists directly outside buildings; at exits; and alongside exterior walls. Most earthquake-related casualties result from collapsing walls, flying glass, and falling objects.
- If trapped under debris:
 - Do not light a match.
 - Do not move about or kick up dust.
 - Cover your mouth with a handkerchief or clothing.
 - Tap on a pipe or wall so rescuers can locate you. Use a whistle if one is available. Shout only as a last resort. Shouting can cause you to inhale dangerous amounts of dust.



- If in a moving vehicle
 - Stop as quickly as safety permits and stay in the vehicle. Avoid stopping near or under buildings, trees, overpasses, and utility wires.
 - Proceed cautiously once the earthquake has stopped. Avoid roads, bridges, or ramps that might have been damaged by the earthquake.
- If at the workplace after an earthquake:
 - Check if someone has been injured.
 - If yes, give appropriate first aid and take the person to the hospital
 - Report the damages to your plant manager



(National Disaster Management Authority, India)

Workplace hazards

The situation which can create threat to human life, property of workplace or environment is defined as hazard. Most of the hazards are hidden in nature which reflects theoretical risk but the fact is that if it becomes active it may lead to emergency. Situation of hazard when finish it turns to be in incident. Hazard and its possibility of getting activated act together to form a risk. Identification of hazard risks is the first step in performing a risk assessment.

Hazards are generally classified into three categories:

- **Dormant:** The circumstance project a possibility of active hazard but it does not affect people, property or environment. For instance, a hillside may be unstable, with the potential for a landslide, but there is nothing below or on the hillside that could be affected.
- **Armed:** People, property or environment is in potential harm's way.
- **Active:** A dangerous occurrence in which hazard has played its role. It is sometimes referred as accident, emergency, incident or disaster not hazard.

The common types of hazards are:

- **Physical hazards** - A situation which cause physical harm to body or create extreme stress is referred as physical hazard. A physical hazard may be natural or human made also.
- **Chemical hazards** are substances which can create harm to human health, harm body parts, property or environment. Chemical hazard may be natural or human made also.
- **Biological hazards** - In this type of hazard biological agents plays a crucial role can cause harm to human health or body parts. These biological agents can be fungus, bacteria, viruses, parasites and foreign toxins.
- **Psychological hazards** usually taken place due to stressful working environment. Here potentially a
- person could be a hazard when affected psychological disturbance through stress or shift patterns and when a person is under the influence of alcohol, illness and lack of training.

- **Electrical hazards** are dangerous condition where a worker can or does make electrical contact with equipment or a conductor. From that contact, the person may sustain an injury from shock, and there is a potential for the worker to receive burn or blast injury.

Maintaining safe workplace

The responsibilities of the employers for maintaining safe workplace are:

- Healthy workplace should be provided
- Train the worker as per their job profile
- Maintain training records (who, what and when)
- Establish and maintain a comprehensive occupational health and safety program, including a written
- Support supervisors, safety co-ordinators and workers in their health and safety activities
- During potential hazard reporting action should be taken immediately
- During incident instant investigation should be initiated
- Provide adequate first aid facilities and services
- Ensure supply of personal protective equipment as per nature of job Worker training is usually carried out by the supervisor. However, the employer has the legal responsibility to ensure that every worker receives adequate training. The employer must follow up to see that the supervisor is carrying out all the required training. Employers should also work with supervisors to help foster positive worker attitudes to safety, health & safety policy and an incident investigation procedure.

The responsibilities of the supervisors for maintaining safe workplace are:

- Instruct workers to follow safety procedures
- Train workers on tasks assigned to them, and check that their safety at work
- Make sure that only authorized, adequately trained workers operate tools and equipment or use hazardous chemicals
- Check out that equipment and materials are handled properly, stored and maintained
- Enforce health and safety requirements
- Take corrective measures on unsafe acts and conditions
- Identify and help workers with problems who could affect safety at the worksite
- Develop health and safety rules and inspect the workplace for hazards

The responsibilities of the workers for maintaining safe workplace are:

- Know and follow health and safety requirements affecting your job
- If you don't know how to do something safely, ask for training before you begin work

- Work safely, and encourage your co-workers to do the same
- Correct any unsafe conditions or immediately report them to your supervisor
- Immediately report any injury to a first aid attendant or supervisor
- Take the initiative. Make suggestions to improve health and safety

Unit 5.5 Personal Protective Equipment used in green logistics

Unit objectives



After reading this unit, you will be able to understand the following points:

1. List the personal protective equipment to be used by a technician.
2. Illustrate the use and putting on personal protective equipment.

5.5.1 List of personal protective equipment

Wearing personal protective equipments (PPEs) is the first and foremost requirement while working at a workplace, especially while handling chemicals, wastewater, plant equipments, any sharp tools and engaging in housekeeping activity. However, desk activities such as writing a report, entering logbook don't require PPE. Most of the risks and damages at the plant can be avoided by wearing personal protective equipment. These equipments provide protection against head injuries (eg. helmets), protection from contamination by avoiding direct contact; burns, abrasion and electric shock. The equipments include the following:

List of personal protective equipment	
1. Helmet Helmets to provide head protection in the event of a fall, slippage and falling objects. To be worn by everyone at the workplace at all places where there is a danger of falling, slippages, objects falling over head, near large tanks and uneven surface.	
2. Safety glasses or goggles for eye protection To be worn by operation & maintenance technician, plant workers, labourers and other staff while working with the plant machineries, tools and equipments which can throw up sparks, dust or micro-particles	

3. Face shields to protect against chemical or dust To be worn by operation & maintenance technician, plant workers, labourers and other staff while working with the plant machineries, tools and equipments which can throw up sparks, dust or micro-particles	
4. Gloves Gloves (made from rubber or latex) for hand protection from transmitting infections, pathogens, burns and abrasion. To be worn by everyone working around the plant machineries and equipments at the FSTP	

List of personal protective equipment	
5. Electric Safety Gloves Gloves for protection against electric shocks. To be worn by operation & maintenance technician, warehouse workers and other staff while working on highly energised electric equipments or repairing electric equipments.	
6. Respirators, dust masks or self-contained breathing apparatus for protection from inhaling dust, fumes etc, While masks or respirators are to be used while regular operations, self-contained breathing apparatus are to be used for special tasks such as entering a unit such as maintenance holes, wet well, dry well etc., where air is highly contaminated and there are high chances of suffocation and inhaling dangerous fumes. To be worn by operation & maintenance technician, plant workers, labourers and other staff while working with the plant machineries, tools and equipments which can throw up sparks, dust, micro-particles, fumes, other dangerous gases especially if entering any treatment unit.	
7. Gum boots For protection of foot from burns, abrasion. To be worn by operation & maintenance technician and plant workers while working in areas with high level of contamination such as sludge dryings beds, large tanks, spillages, maintenance holes, broken tools and other areas with risk of contamination through feet.	

8. Plant Uniform Separate plant uniform to be worn at the plant to avoid contamination and damage to regular clothes. To be worn by operation & maintenance technician and plant workers while working with the plant machineries, tools and equipments.	
9. Safety Harness Harness for entering maintenance holes and other treatment modules below ground. To be worn while entering the maintenance holes	

Table 5.5.1: Personal protective equipment

5.5.2 Using Personal Protective Equipment (PPE)

Correct method of putting personal protective equipment (PPE) is essential to be maintain hygiene and be comfortable wearing the PPE. 5.2.1.1 and 5.2.1.2 detail out all the steps to be followed while putting on and removing PPE.

Putting on Personal Protective Equipment

Sequence of putting on personal protective equipment:

1. Perform hand hygiene
2. Put on the uniform
3. Put on the gum boots
4. Perform hand hygiene
5. Mask, respirator or self-contained breathing apparatus
6. Put on the goggles, face shield
7. Helmet
8. Perform hand hygiene
9. Gloves
10. Harness, if needed

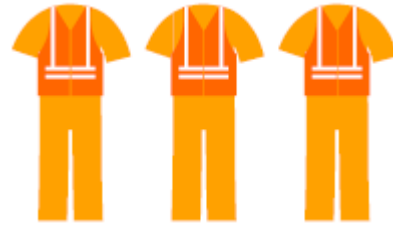
Perform hand hygiene

1. Wet your hands with clean, running water (warm or cold), turn off the tap, and apply soap.
2. Lather your hands by rubbing them together with the soap. Be sure to lather the backs of your hands, between your fingers, and under your nails.
3. Scrub your hands for at least 20 seconds.
4. Rinse your hands well under clean, running water.
5. Dry your hands using a clean towel or air dry them.
6. Source: (Centre for Disease Control and Prevention)
7. Duration of the procedure: 40 to 60 seconds



Wear uniform

- Wear the uniform as any other clothing, but it should be well fitted with no loose ends and comfortable.
- There should be at least 3 to 4 pairs of uniform. Wash each pair after each use.

**Wear gum boots**

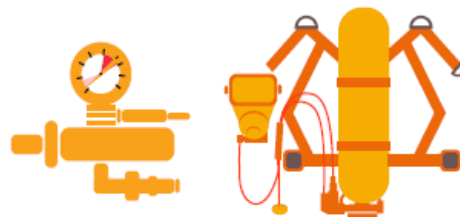
- Wear gumboots as any other footwear, but it should be well fitted and comfortable.

**Perform hand hygiene****Wear mask, respirator and self-contained breathing apparatus****Mask or respirator**

- Secure ties or elastic bands at middle of head and neck
- Fit flexible band to nose bridge
- Fit closely and tightly to face and below chin
- Fit-check respirator

**Self-contained breathing apparatus (SCBA)**

- Check if the bottle is full
- Check for leakages in hose and other parts



Check for tightness of the face mask and the whistles

- With demand valve still closed, try to hold the face mask on your face with hands and without using the straps
- If you are able to hold the face mask, it means face mask is able to hold the vacuum you created and is airtight.
- Activate the demand valve. Once demand valve is activated and air pressure is released and reaches the alarm level, whistle should sound.

**Two ways of wearing SCBA**

- Over the head: hold the back plate with both hands and lift the SCBA over your head
- Coat: hold one strap closer to the back plate and wear the SCBA like wearing a coat. After the SCBA is on the back, lean forward and pull the straps.

Wear the mask and open the cylinder valve before entering the maintenance holes, wet well etc.

Put goggles or face shield

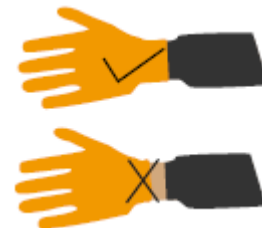
- Place over the eyes and face and adjust to fit.

**Wear helmet**

- Place the helmet over your head and adjust to fit.

**Perform hand hygiene****Wear gloves**

- Wear the gloves on both the hands. They should extend to cover wrist of the uniform and should be tightly fitted.



Wear a harness

- Hold the harness by the rear D-ring and shake it to allow the straps to fall into place
- Place your arms through the shoulder straps, ensuring the webbing remains untwisted
- Ensure the D-ring is in the middle of the back between the shoulder blades. Adjust shoulder straps so the sub-pelvic strap sits under the buttocks.
- Pull each thigh strap through the legs and fasten the buckles. Adjust the tension or release the slack as necessary
- Connect the chest strap and adjust so that the shoulder straps are centred on the shoulder and tighten
- Once all straps are buckled, tighten them so that the harness fits comfortably and does not pinch or restrict movement.



Removing Personal Protective Equipment

Sequence of removing personal protective equipment	
1. Remove the harness (if any)	
2. Remove gum boots	
3. Remove gloves	
4. Perform hand hygiene	
5. Remove helmet	
6. Remove goggles or face shield	
7. Remove safety uniform	
8. Remove mask or respirator	Remove the SCBA (Self-contained breathing apparatus) is before removing the uniform
9. Perform hand hygiene	

Table 5.5.2: Sequence of removing PPE

Remove safety harness

Unbuckle the harness and remove your arms followed by your legs one by one. Put the harness away for cleaning.

- It is best to begin with cleaning the harness on a flat surface, leaving open to visible inspection.



- Using a moist sponge, wipe down the harness to remove excess dirt and dust.
- Mix a cleaning solution using laundry detergent or dish soap. DO NOT use any cleansers that contain chlorine, bleach, or abrasives.
- Dip your sponge into the solution and thoroughly scrub each portion of the harness until a thick lather form.
- Using a sponge dipped in CLEAR water, wipe down the harness to remove the suds and soap residue.
- Let the safety harness dry in room temperature air. DO NOT use a mechanical heat dryer or expose the harness to long periods of sun-drying.
- When cleaning multiple harnesses, store each in a separate, dry compartment. Hang them in such a way that they are not crushed, worn, or creased.
- Never use gasoline or other 'drying solvents' to clean harnesses.

And the two most important cautions:

- Dampen but DO NOT SOAK the harness. The excessive expansion of the fibres by soaking (and the contraction by drying) can compromise the fabric's effectiveness and shorten the harness's life.
- NEVER put a harness in the dryer. Excessive heat and tumbling can (and will) damage the harness.



Remove safety boots

If wearing rubber boots, remove them while gloves still on your hands. Place the boots in a container with disinfectant.



Remove gloves

1. Pinch and hold the outside of the glove near the wrist area.
2. Peel downwards, away from the wrist, turning the glove inside out.
3. Pull the glove away until it is removed from the hand and hold the inside-out glove with the gloved hand.
4. With your un-gloved hand, slide your finger/s under the wrist of the remaining glove, taking care not to touch the outside of the glove.
5. Again, peel downwards, away from the wrist, turning the glove inside out.
6. Continue to pull the glove down and over the inside-out glove being held in your gloved hand.

This will ensure that both gloves are inside out, one glove enveloped inside the other, with no contaminant on the bare hands. Dispose the gloves in the waste container. The urban local body should collect the waste from the waste container daily.

**Perform hand hygiene**

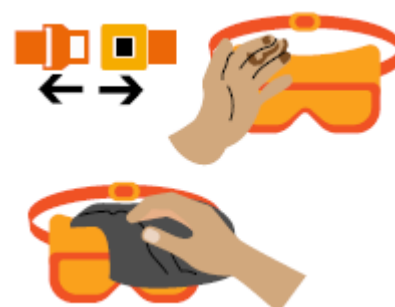
Refer to the beginning of **Perform hand hygiene**

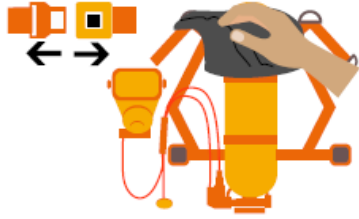
Remove helmet

- Unbuckle the helmet.
- Avoid contact with the outer surface of the helmet.
- Wipe the surface with a cloth soaked in disinfectant followed by a dry cloth or as per the manufacturer's instructions

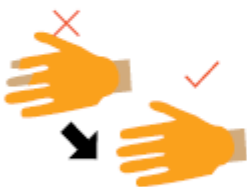
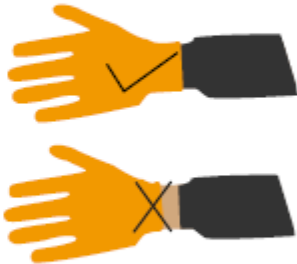
**Remove goggles or face shield**

- Remove goggles or face shield from back by lifting head band or the ear pieces.
- Avoid contact with the front of the goggle or face shield.
- Wipe the surface with a cloth soaked in disinfectant followed by a dry cloth or as per the manufacturer's instructions.



Remove SCBA (self-contained breathing apparatus) • Remove the mask from the back while avoiding contact with the front surface. • Unbuckle the SCBA. • Wipe the surface with a cloth soaked in disinfectant followed by a dry cloth or as per the manufacturer's instructions.	
Remove uniform • Remove the uniform by rolling it inside out in order to avoid contact with the outside surface. • The uniform should then be sent for washing.	
Remove mask and respirator • Remove the mask from the back while avoiding contact with the front surface. • Dispose it off if it is meant for single use. • In case it is recyclable, disinfect by wiping the with a cloth soaked in disinfectant followed by a dry cloth or as per the manufacturer's instructions.	
Perform hand hygiene Refer: 5.2.1.1 - Perform hand hygiene	

Do's and Don'ts

1 Keep hands away from mouth and face 	2 Limit surfaces touched 	3 Avoid touching surfaces such as door knobs, stationary, mobiles while working to limit chances of contamination. 
4 Change personal protective equipment immediately if damaged. 	5 Ensure rubber gloves are long enough to extend well above the wrist, leaving no gap between the glove and coat or shirt sleeve 	
6 Perform hand hygiene whenever contamination is suspected while putting on and removing personal protective equipment and in case of damage of gloves 		

Unit 5.6 Housekeeping practices and infection control

Unit objectives

After reading this unit, you will be able to understand the following points:

1. Housekeeping and prevention practices
2. Safety success stories in green logistics

5.6.1 Housekeeping practices and prevention practices

Housekeeping Practices

Housekeeping in simple words means maintaining a house on a daily or long-term basis or looking after its cleanliness, tidiness, upkeep and smooth running.

Cleanliness doesn't mean housekeeping. Housekeeping includes keeping work areas tidy and arranged; keep floors free of slip and trip accidents; clearing of waste materials (paper, cardboard) and other fire hazards.

Workplaces hazards can be eliminated by effective housekeeping and complete a job safely and properly. Poor housekeeping and hiding hazards can cause frequent accidents which can cause injuries.

Principle of Workplace Housekeeping

Housekeeping consists of the simpler aspects of care and maintenance of workplace and its associated items:

- Maintaining operational tidiness and order,
- Properly controlling waste, and
- Regulating activities such as smoking that can lead to hazardous conditions.

Good housekeeping is a vital factor in preventing accidents. The great majority of all work accidents are caused during the handling of goods or materials, and by people falling, being hit by falling objects, or striking against objects in the workplace. All these causes can be reduced by good housekeeping practices—in fact, good housekeeping is the only cure for hundreds of accidents that occur.

Here are some kinds of accidents commonly caused by poor housekeeping:

- Tripping over loose objects on floors, stairs and platforms.
- Articles dropping from above.

Good Housekeeping Checklist

Check off your housekeeping programme against this checklist. Better still, make a more comprehensive list of your own.

Buildings

1. Walls clean.
2. Windows clean.
3. Walls free of unnecessary hangings.
4. Proper light provided.
5. Platforms in good condition.
6. Stairs clean and well lit. Handrails and steps of sound construction and well maintained.



Floors

1. Good floor surface.
2. Kept clean and free of loose material. Clean in corners, behind radiators, along walls, and around pillars or columns.
3. Free of oil, grease, etc.
4. Operating floors, or work positions free of loose scrap, metal or other materials.
5. Free of unnecessary articles.
6. Bins provided for refuse



Aisles

1. Free of obstructions.
2. Safe and free passage to fire-fighting equipment and fire exits.
3. Safe and free access to work positions.
4. Clearly defined.



Stock and material

1. Properly piled and arranged
2. Kept in storage areas.

Tools

1. Properly arranged in place.
2. Free of oil and grease.
3. Inspected and maintained in good order.
4. Tool rooms and racks in clean and orderly condition.



Grounds

1. Yard and building surrounds free of refuse such as fruit peelings, scrap, wood, Iron, etc.
2. Grounds kept free of weeds and overgrown vegetation.
3. Wastes and refuse removed frequently.



Good housekeeping helps to create:

- Better working conditions
- Safer workplaces
- Greater efficiency.



Practices the regular collection, grading and sorting of scrap leads to good housekeeping. Separate the materials from waste which can be recycled and reuse. For systematic waste gathering and disposal, place the bin containers, near the place where waste is generated most.

5.6.2 Safety success stories in green logistics

The Indian chemical industry has grown 13-14% annually in the last 5 years (2007-2012). The encouraging growth of the chemical industry also brings hazardous incidents and worker safety into sharp focus. Lack of safety measures along with limited knowledge of potential hazards of chemicals often leads to several dangerous incidents and fatalities in the industry.

Aditya Birla Chemicals is a leading manufacturer of bulk and speciality chemicals and viscose filament yarn. The business is spread across multiple manufacturing locations in four countries - India, Thailand, Germany and China. The business is creating a strong footprint in the chemicals industry - It is a market leader in the chlor-alkali segment in India; at the number two position in viscose filament yarn and is the third largest manufacturer in the world in sulphites. The Chemicals business has 14 manufacturing sites - seven in India, five in Thailand, one in Germany and one in China.

Aditya Birla Chemicals has a tremendous focus on providing a safe working environment to its workforce and a consistent focus on reducing the number of accidents, injuries and illnesses. However, manufacturing operations, material handling and logistics posed numerous safety challenges to the company in improving safety and taking it to a world-class level. For example, the company's rayon unit at Veraval, Gujarat, was facing the constant challenge of recurrent bursting of spinning pots for which no readymade solution was available in the market.

Similarly, worker safety at Rehla site in Jharkhand was another area of concern primarily because of traditional operating methods and poor awareness about safety risks among local contractor workers.

With a belief in "Good Safety=Good Business", the leadership decided to make a step change in its safety performance across all operations including design, manufacturing operations and distribution.

In collaboration with dss+, the organization started moving towards world-class safety performance and making safety a part of integrated operations. Four Indian sites of the business started their journey of safety excellence with dss+ at the beginning of 2012. The journey started with Work Place Safety (WPS), wherein at-risk behaviours were addressed. This was further extended to include Process Safety Management (PSM).

Solution

With the initial assessment completed, dss+ and Aditya Birla Chemicals decided on the primary goals of the engagement, which were:

Reduce injuries and incidents and ultimately bring injuries to 'Shunya'

Involve and engage senior leadership to drive a safety culture and make it a proactive commitment

Ensure that safety is integrated into line management responsibilities

Define metrics and establish an annual progress review of safety and cultural transformation in the company

Developing a roadmap for world-class safety

Aditya Birla Chemicals, with support from dss+, developed and implemented a safety management system, focused around operations environment. This included:

Setting up of corporate objectives: Aditya Birla Chemicals Business leadership team finalized a set of seven safety principles as foundation stone for all their activities. Annual improvement goals were identified and tracked at corporate level by a steering committee.

Sub-Committees with leadership commitment: The company formed six sub-committees to cover Safety Rules & Procedures, Safety Observation & Audit, Incident Investigation, Capability Development and Communication, Safety in Transportation and Distribution, and Contractor Safety Management. Each sub-committee was led by a senior management team.

Identification of high-risk areas: The Workplace Safety (WPS) engagement framed ten high-risk focus areas, which included personal protection, working at height, materials handling, job safety analysis, work permit, lock-out tag-out, hot work, scaffolding, incident investigation and excavation. The management reinforced these standards and expectations through consistent auditing and observation.

Process Safety Management implementation: As a part of Process Safety Management (PSM), the manufacturing process is routed through a 14-element wheel to address risks qualitatively and quantitatively. In addition, the company drives control mechanisms to mitigate safety issues.

Measurement procedures: The team developed systematic procedures and measurement metrics including dss+ Safety Perception Survey, data analysis, and field visits to assess the current state.

With this comprehensive strategic process, Aditya Birla Chemicals has been successful in involving all workforce including support functions in safety activities- both on the job and off the job. Moreover, by addressing Operational and behavioural aspects of safety, the Chemicals Business was able to reduce injury rates by nearly 40 per cent at the four sites between 2012 and 2013.

Benefits

Average Loss Time Injury Rate reduced by nearly 40% at 4 Sites

Improved productivity and reduced potential hazards in logistics -In line with the safety principle of "Management is committed to providing a safe working and operating environment", the company worked to provide movable conveyors to unload salt bags, which were previously lifted manually by the workers. The introduction of conveyor belt has helped prevent the incidence of falling salt bags on the workers and has improved their condition. Similar loading/unloading conveyor belts have now been procured for other products. Automation in stabilized bleaching powder filling areas has

significantly reduced dust hazards, which will help in improving productivity and reducing non-compliance to Personal Protective Equipment (PPE).

Improved working conditions for workers - There is a visible change in the salt unloading area In the working style and people's attitude towards safety at the Rehla site. Workers are more cognizant of safety risks and demonstrate safe behaviour, which includes wearing proper coveralls while at work. Contractors openly talk about their safety concerns with senior managers at the morning toolbox talk at Nagda site.

	Personal protective equipment for safety at workplace
	Warehouse safety
	Fire Extinguisher
	Emergency Action Plan
	Occupational health & safety

Exercise

1. What are the types of emergencies at a workplace?

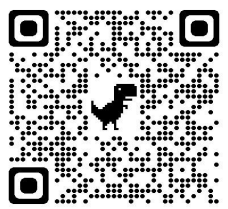
2. What are the types of fire extinguishers?

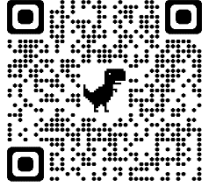
3. True or False

	T/F
1. ORS is to be given to a person suffering from diarrhoea	
2. Wet hands can be used while handling electric equipments	
3. Emergency numbers should be only available to the Personnel at warehouse	
4. A deep wound should be washed with water	

ANNEXURE



Module No.	Unit No.	Topic Name	Page No.	URL	QR Code
Module 2		Introduction to Green Logistics	65	https://www.youtube.com/watch?v=12M1r58rVSo	
				https://www.youtube.com/watch?v=f58TVB20A58	
Module 3		Steps required for greening the logistics sector	97	https://www.youtube.com/watch?v=CfMOMxBfi2g	
			116	https://www.youtube.com/watch?v=FRfB4ZEnh_A	
Module 4		Identifying green supply and demand networks	160	https://www.youtube.com/watch?v=plnQN-B9ZaM	

Module 5		Perform Health Safety Green Logistics	238	https://www.youtube.com/watch?v=vWlBgmE6oT8	
				https://www.youtube.com/watch?v=06CqZlj6yxY	
				https://www.youtube.com/watch?v=L1XS5B5D70U	
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